

AWS Certified AI Business Strategist

By Stéphane Maarek



COURSE →



EXTRA PRACTICE EXAMS →

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- Best of luck for the exam and happy learning!

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AWS Certified AI Business Strategist AIB-C01



Welcome! We're starting in 5 minutes

- Let's prepare for the **AWS AI Business Strategist exam (AIB-C01)**
- It's a Business & AI-focused certification, very little on AWS Cloud itself
- No IT knowledge needed whatsoever
- No AWS knowledge needed either
- Slides-based only course
- Very "high level / corporate / business" information
- Will try to make it as interesting as possible

Important: what this course is and isn't



- This course is not

- A course on how to use ChatGPT
- A course on how to use other AI-related tools (images, music, etc...)
- A course on the broad landscape of AI
- A deep dive on the AWS Cloud



- This course is

- Intended for business professionals who want to learn about AI
- Intended to help you pass a business certification administered by AWS

About me

- I'm Stephane!
- 12x AWS Certified
- Worked with AWS many years: built websites, apps, streaming platforms
- Veteran Instructor on AWS (Certifications, CloudFormation, Lambda, EC2...)
- You can find me on
 - LinkedIn: <https://www.linkedin.com/in/stephanemaarek>
 - Instagram: <https://Instagram.com/stephanemaarek>
 - Medium: <https://medium.com/@stephane.maarek>
 - Twitter: <https://twitter.com/stephanemaarek>
 - GitHub: <https://github.com/simplesteph>



★ 4.7 Instructor Rating
📊 1,110,468 Reviews
👤 3,636,815 Students
▶ 76 Courses

Your AWS Certification journey

Business

Validate strategic expertise in applying cloud and AI to business outcomes.

No technical experience is needed.



Associate

Role-based certifications that showcase your knowledge and skills on AWS and build your credibility as an AWS Cloud professional.

Prior cloud and/or strong on-premises IT experience recommended.



Foundational

Knowledge-based certification for foundational understanding of AWS Cloud.

No prior experience needed.



Professional

Role-based certifications that validate advanced skills and knowledge required to design secure, optimized, and modernized applications and to automate processes on AWS.

2 years of prior AWS Cloud experience recommended.



Specialty

Dive deeper and position yourself as a trusted advisor to your stakeholders and/or customers in these strategic areas.

Refer to the exam guides on the exam pages for recommended experience.



Domain I: AI Fundamentals and Literacy

Section Overview

- The question of this domain: do we all mean the same thing when we say “AI”?
- We align on all the definitions
- AI Business Strategist works in translation for the technical teams and the business
- AWS services for recognition only

Task 1.1

Core AI Concepts

algorithms, models, training, inference, data

Task 1.2

The Right Type of AI Solution

rules vs AI, agents, monitoring, shadow AI

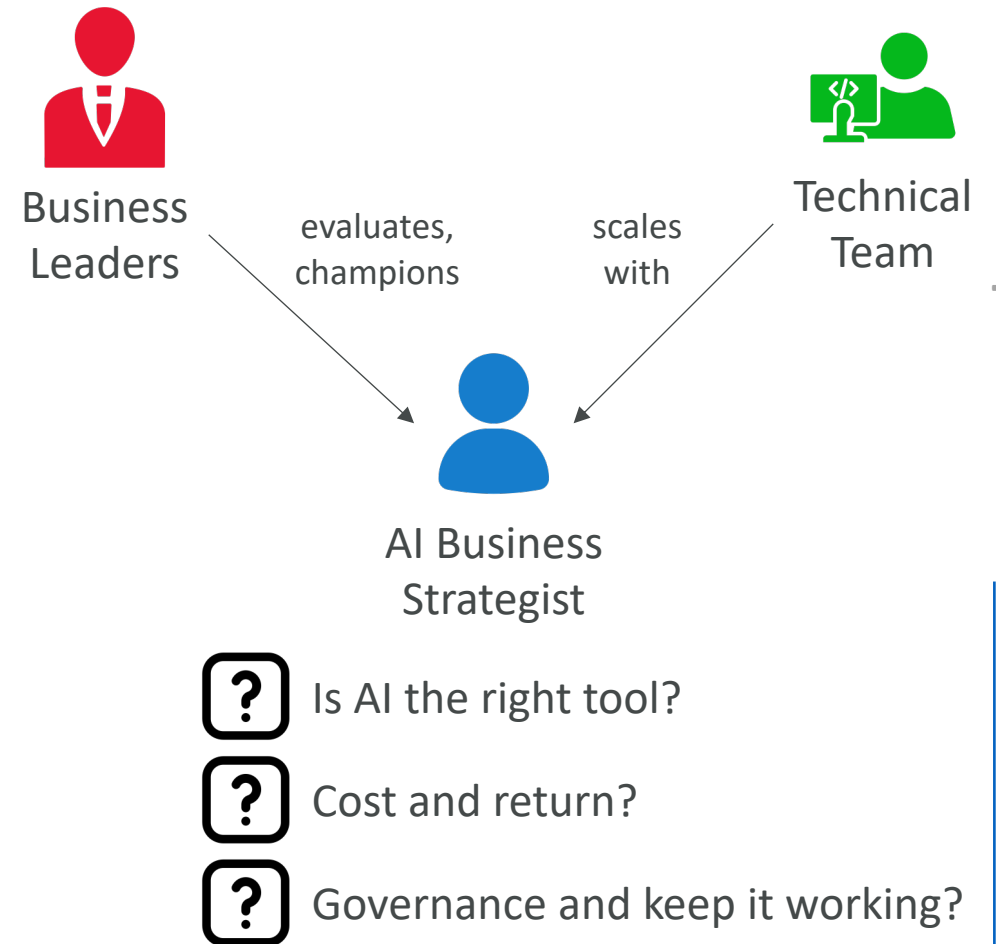
Task 1.3

GenAI Concepts and Techniques

prompts, tokens, RAG, fine-tuning

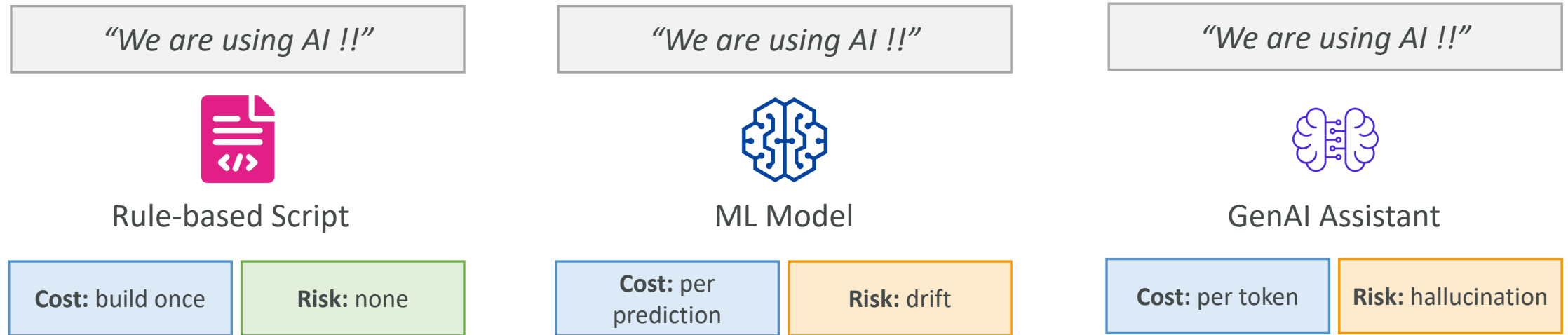
Who is the AI Business Strategist?

- Evaluates, champions and scales AI initiatives in the business
- Works with technical teams, but does not build models
- Answers 3 questions on every initiative:
 - Is AI the right tool for this problem?
 - What will it cost, and what will it return?
 - How do we govern it and keep it working?



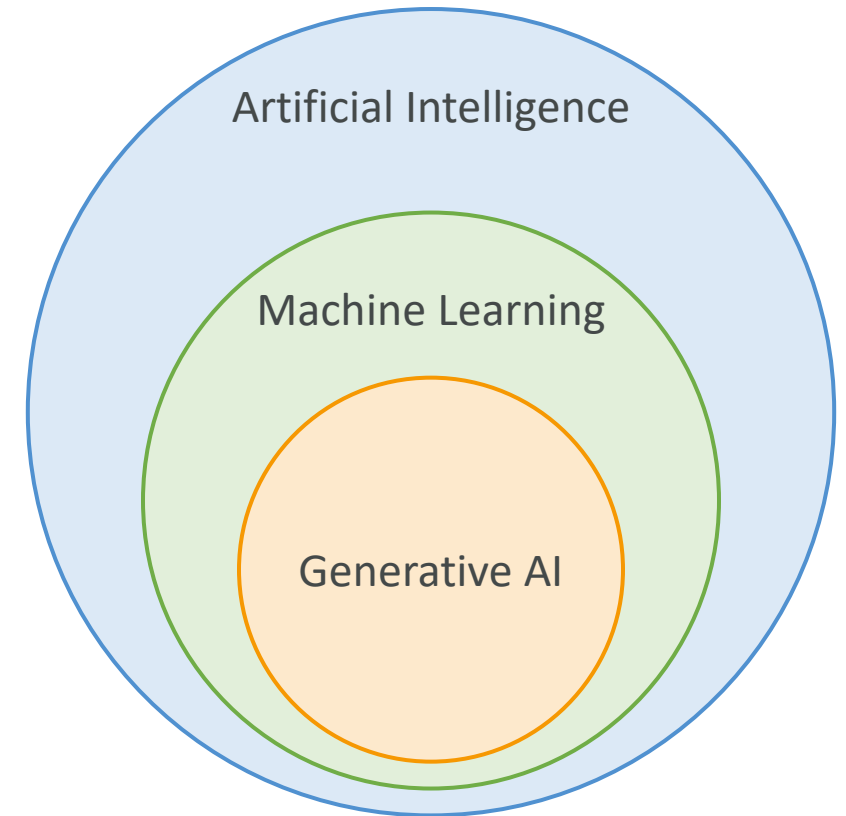
Why We Need a Shared Vocabulary?

- “We are using AI” says nothing: same words, very different things
- Without it: quiet, expensive mistakes (wrong budget, wrong governance)
- With it: initiatives become comparable across the portfolio
- Domain 1 = the vocabulary reused in Domains 2, 3 and 4

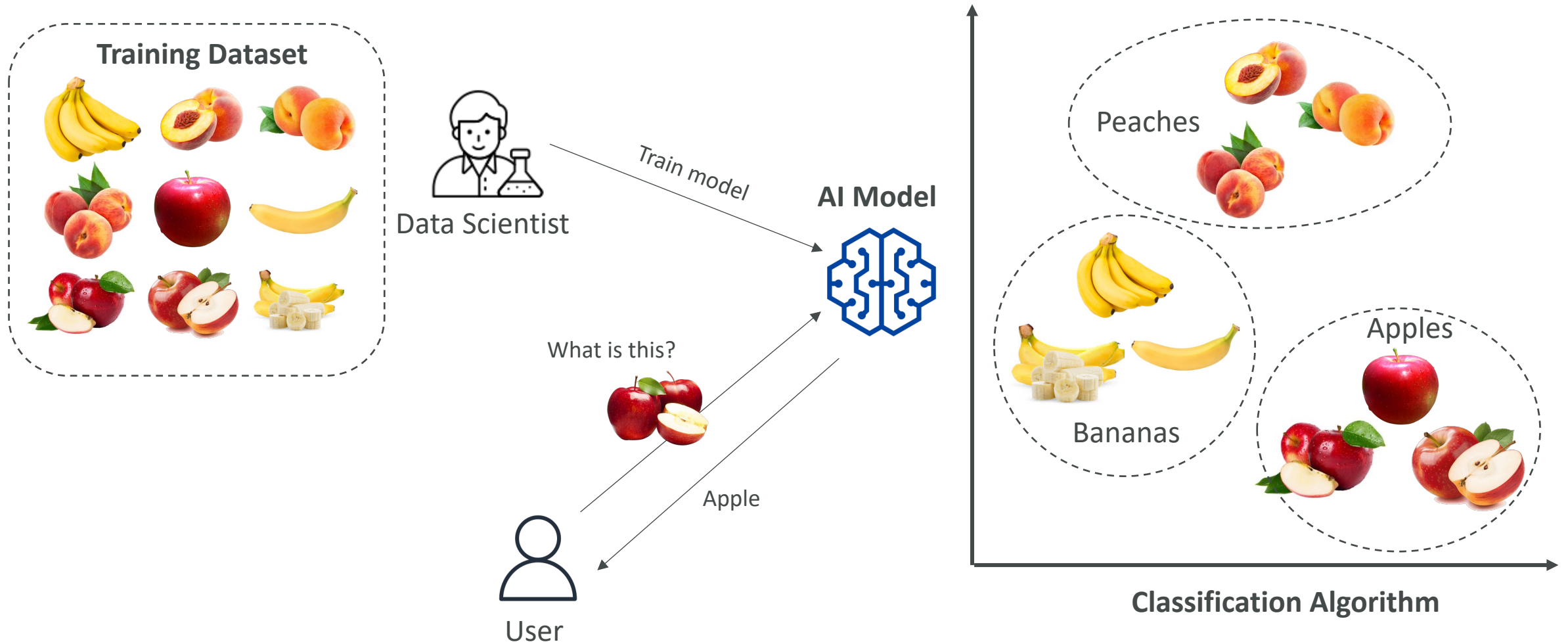


AI, Machine Learning and Generative AI

- **AI:** systems that perform tasks associated with human intelligence
- **Machine Learning (ML):** subset of AI, learns patterns from data instead of explicit rules
- **Generative AI (GenAI):** subset of ML, creates new content (text, images, code...)
- Every GenAI system is an ML system, and every ML system is an AI system
- So what is true of ML is true of GenAI:
 - It needs training data
 - It runs inference on new inputs
 - It can drift as the world changes



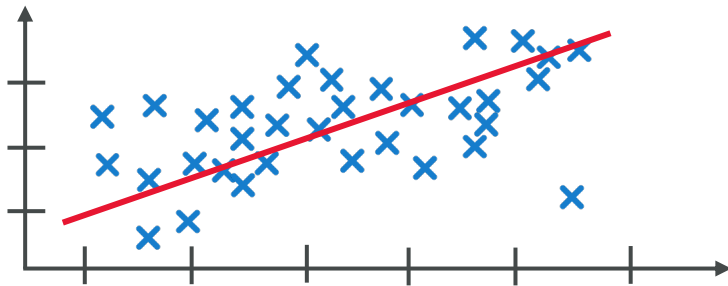
How does AI work?



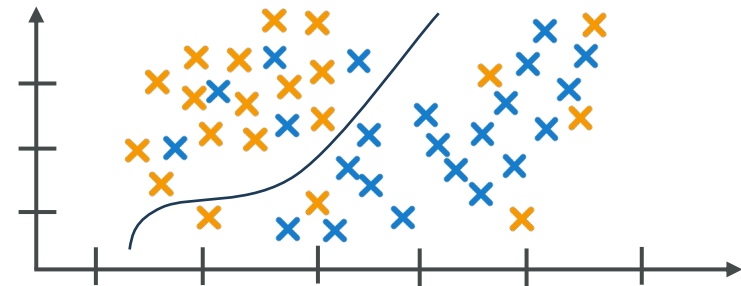
What is Machine Learning (ML)?

- ML is a type of AI for building methods that allow machines to learn
- **Data** is leveraged to improve computer performance on a set of tasks
- Make predictions based on data used to train the model
- No explicit programming of rules

Regression

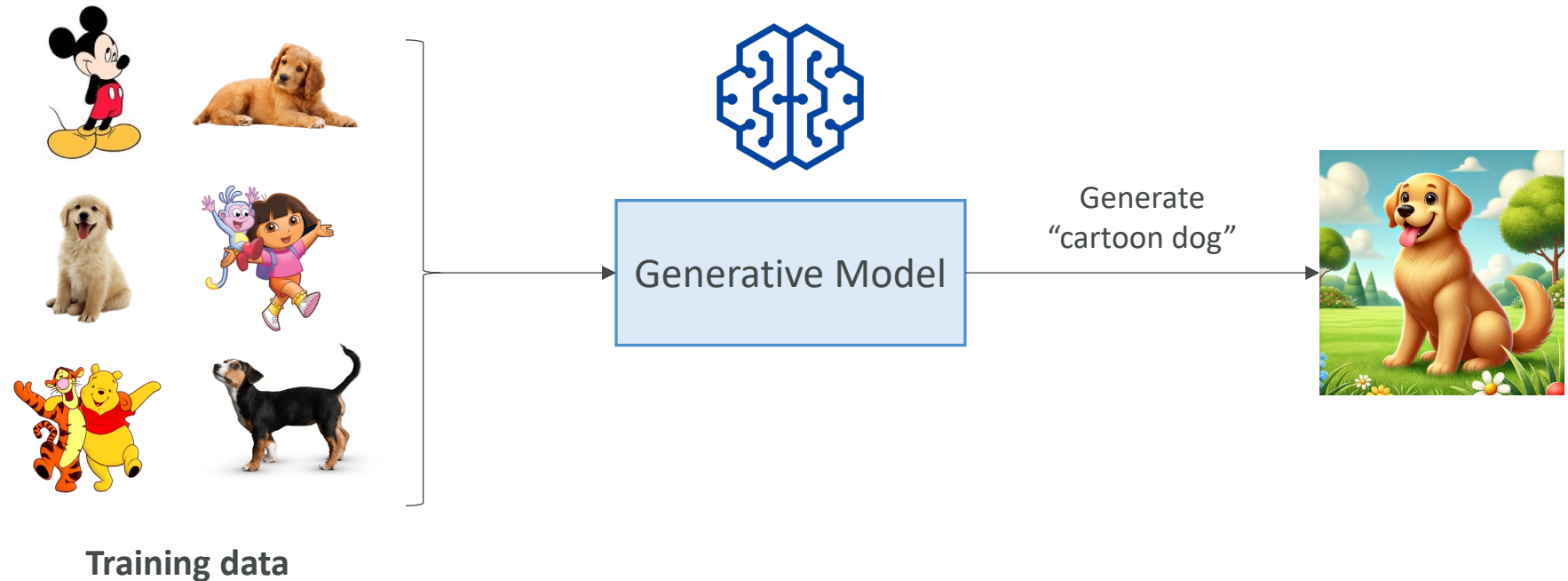


Classification

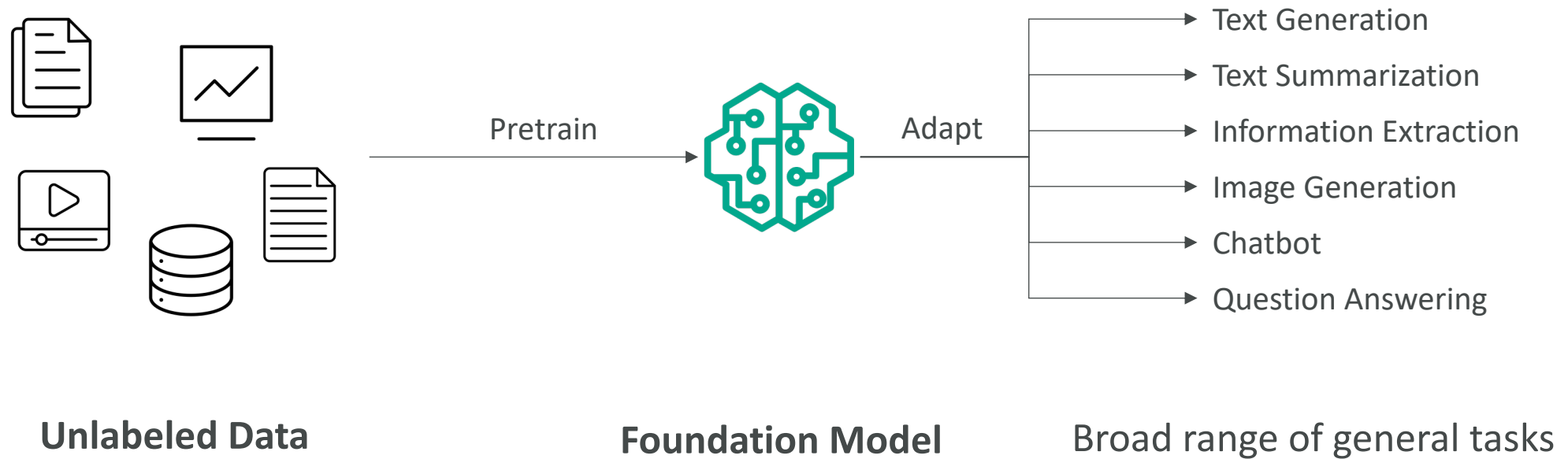


What is Generative AI ?

- Generative AI (Gen-AI) is a subset of Machine Learning
- Used to **generate new data** that is similar to the data it was trained on
 - Text
 - Image
 - Audio
 - Code
 - Video...



What is Generative AI ?



ML or GenAI? Read the Output, Not the Verb

- Look at what comes out of the system
 - A number, a label, a ranking → ML (scores, sorts, classifies)
 - Something new (text, summary, image, code) → GenAI (creates)
 - “Generates a ranked list” → still ML, not GenAI
- The verb in the scenario at the exam is often misleading

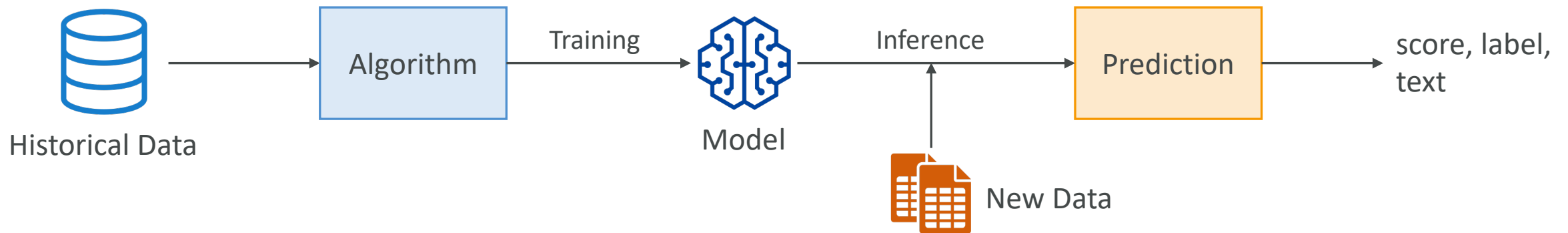
Output of the System	Layer
Probability that a customer churns	ML
Ranked list of the best leads to call	ML
Summary of a support call	GenAI
Draft of a product description	GenAI

Mini Quiz: ML or GenAI?

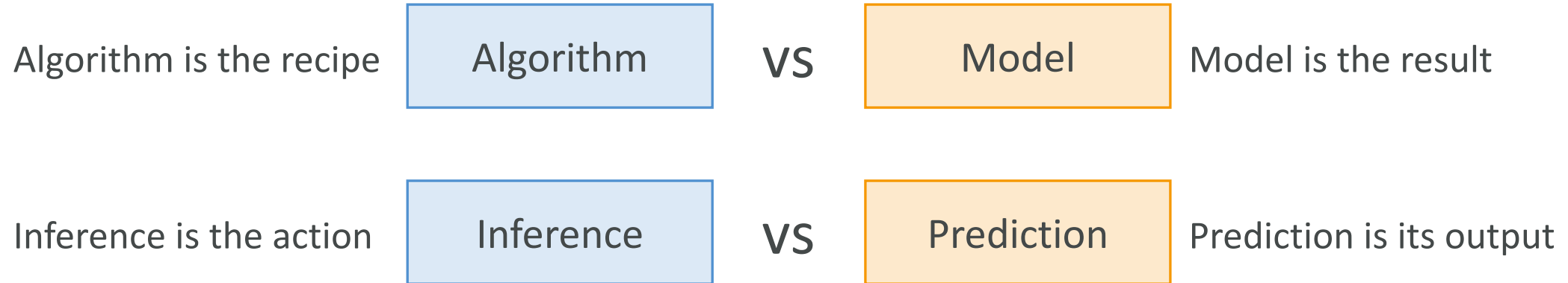
What the system does	Its output	ML or GenAI?
Predicts next month's demand per store	a number	ML
Drafts a reply to a customer email	new text	GenAI
Flags a transaction as suspicious	a label	ML
Summarizes a 40-page contract	new text	GenAI
Ranks leads by likelihood to buy	a ranking	ML
Creates a product image from a description	a new image	GenAI

How a Model Comes to Exist

- **Algorithm:** the method (recipe) used to learn from data
- **Training:** running the algorithm on historical data
- **Model:** the output of training, the learned patterns
- **Inference:** using the model on new data
- **Prediction:** the output of inference (a score, a label, a text...)



Algorithm, Model, Inference, Prediction: Traps



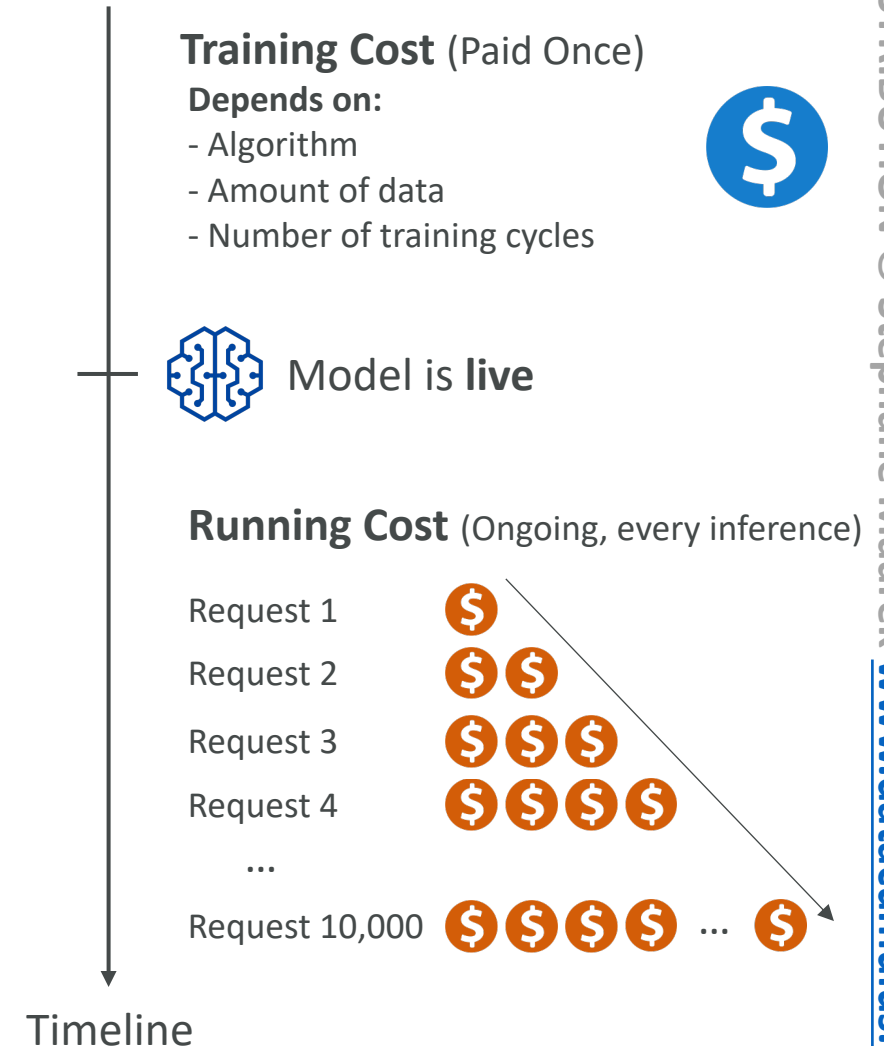
- **No model without training**, you can **NOT** hand-write a model
- No inference before training
- Data does not become a model by itself, **training is required**

Exam tip

The sequence data => algorithm => training => model => inference => prediction is tested at the exam

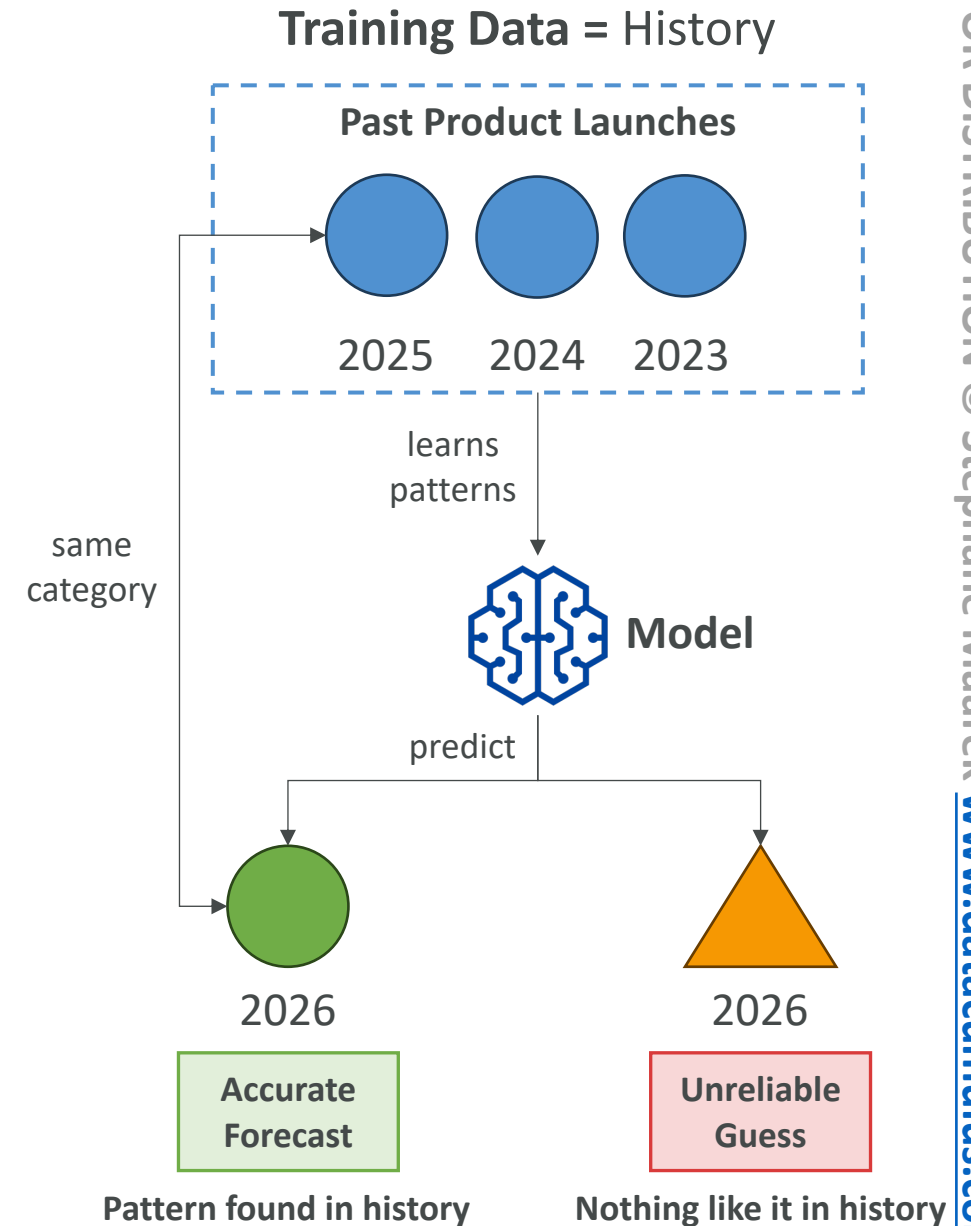
Training Cost and Running Cost

- **Training:** periodic, at launch and each time we retrain
 - Cost depends on the algorithm, the amount of data, the number of cycles
- **Inference:** every time the system is used, all day, every day
 - Cost is linear with the number of inferences: cost follows usage
- A model can be right at launch and wrong 6 months later (**Drift**, later in this section)



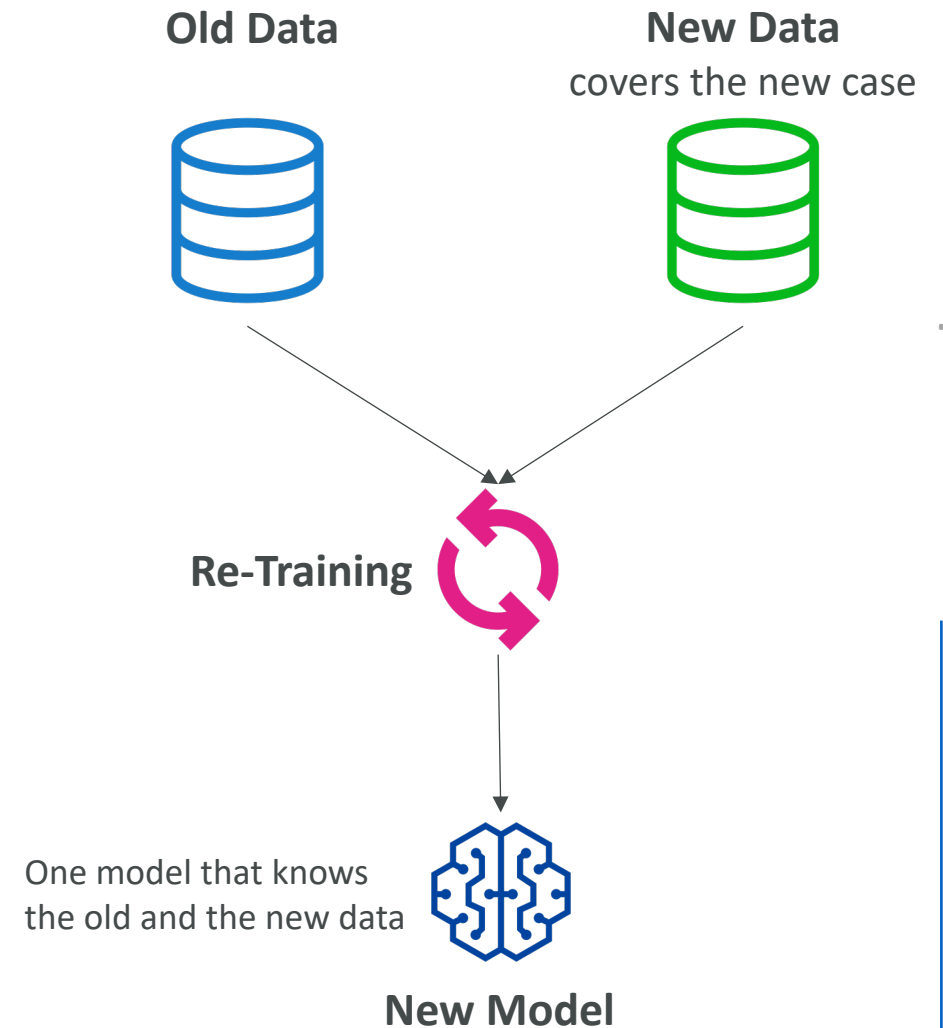
History Sets the Ceiling

- A model can only generalize from patterns it has seen in its training data
- **Seen** → past product launches in the data, so a new launch can be forecast
- **Never seen** → a new product category, segment or channel with no history at all
- Two questions to ask about any training data:
 - Which period does it cover?
 - Which population does it cover?



History Sets the Ceiling – Wrong Diagnoses

- “Running more inference”: does not add knowledge
- “Switch to another algorithm”: same data, same blind spot
- “Predictions are unreliable for new products, accept it”: the problem is fixable
- **The remedy** is to add data that covers the case, retrain, and the new model covers it



Structured Data

- Data is organized in a structured format, often in rows and columns (like Excel)

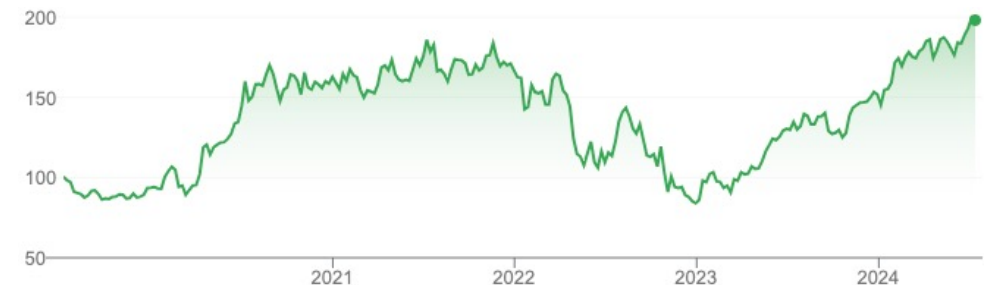
- **Tabular Data**

- Data is arranged in a table with rows representing records and columns representing features
- Example: customers database with fields such as name, age, and total purchase amount

Customer_ID	Name	Age	Purchase_Amount
1	Alice	30	\$200
2	Bob	45	\$300

- **Time Series Data**

- Data points collected or recorded at successive points in time
- Example: Stock prices recorded daily over a year



Date	Stock Price
01-07-2024	\$197.20
02-07-2024	\$200

Unstructured Data

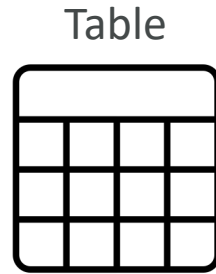
- Data that doesn't follow a specific structure and is often text-heavy or multimedia content
- **Text Data**
 - Unstructured text such as articles, social media posts, or customer reviews
 - Example: a collection of product reviews from an e-commerce site
- **Image Data**
 - Data in the form of images, which can vary widely in format and content
 - Example: images used for object recognition tasks



Review: Attended a yoga class at the new studio. The instructor was excellent, and the facility was well-maintained. Loved the variety of classes offered. Only downside was the parking situation.



Structured vs. Unstructured Data



Document



Audio



Video



Image



	Structured Data	Unstructured Data
Definition	Organized in rows and columns, fixed fields	No predefined format
Examples	Transactions, CRM records, sensor readings	Emails, contracts, call recordings, images
Suits	Score, forecast, classify (classic ML)	Assistants over documents and calls (unblocked by GenAI)
Typical Question	“what will happen?”	“What does this say?”

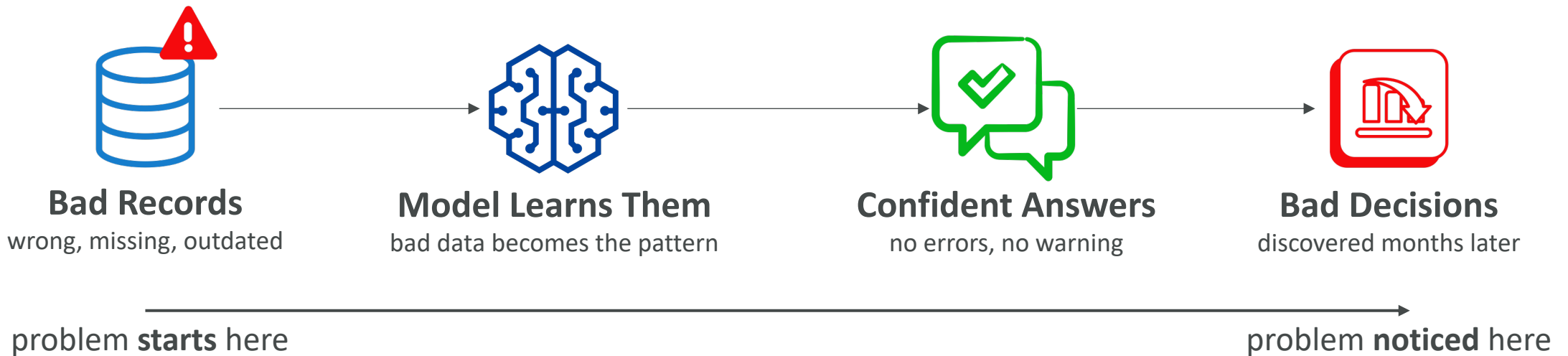
Match the Use Case to its Data

- Before committing money:
 - what data does this use case need?
 - do we have it, in what form?
- Volume never compensates for the wrong type or state of data
- Wrong type = wrong solution, no matter how good the model

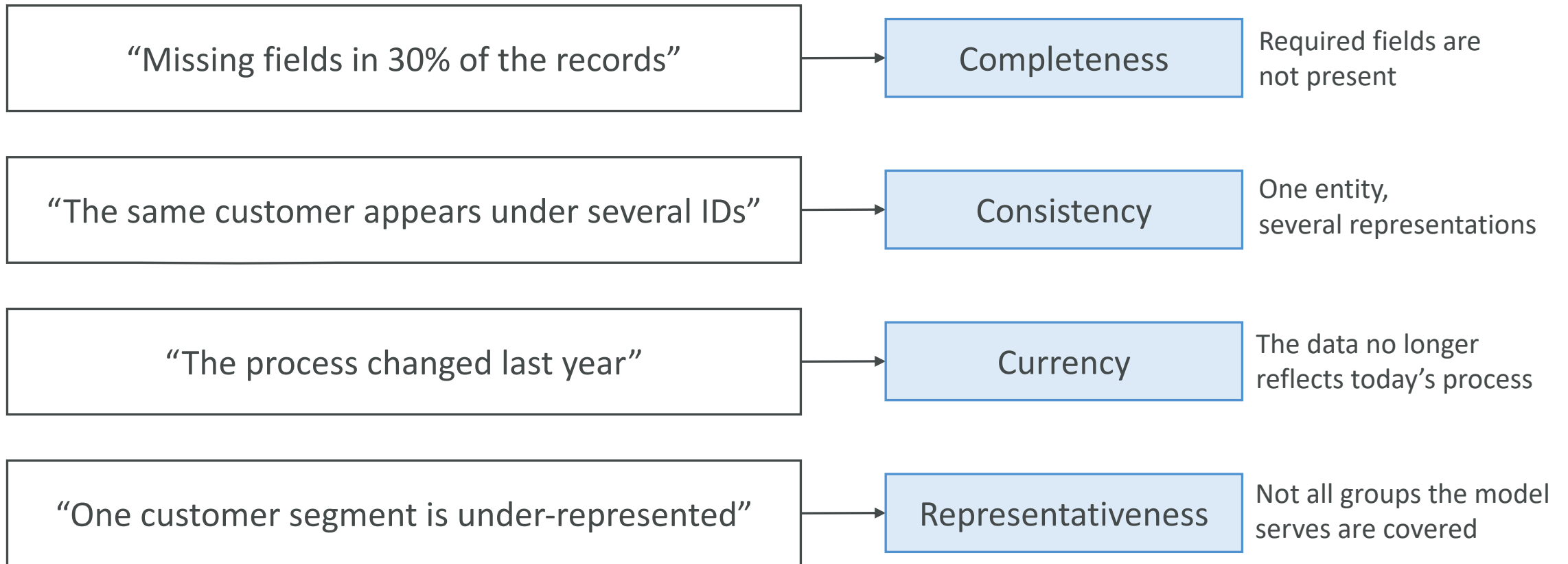
	Churn Score (ML) needs: structured data	Document Assistant (GenAI) needs: text documents to read & search
10 years of transactions type: structured customer, date, amount... 	 Fit	 Wrong Type
1 Million PDFs type: unstructured contracts, manuals, policies... 	 Wrong Type	 Fit

Data Quality – The Half that Fails Quietly

- Bad records don't get rejected, they become the pattern the Model learns
- The model stays producing confident outputs: no error, no warning
- The only symptom: bad business decisions, discovered later
- **Data Quality** is the dimension most often assumed rather than checked

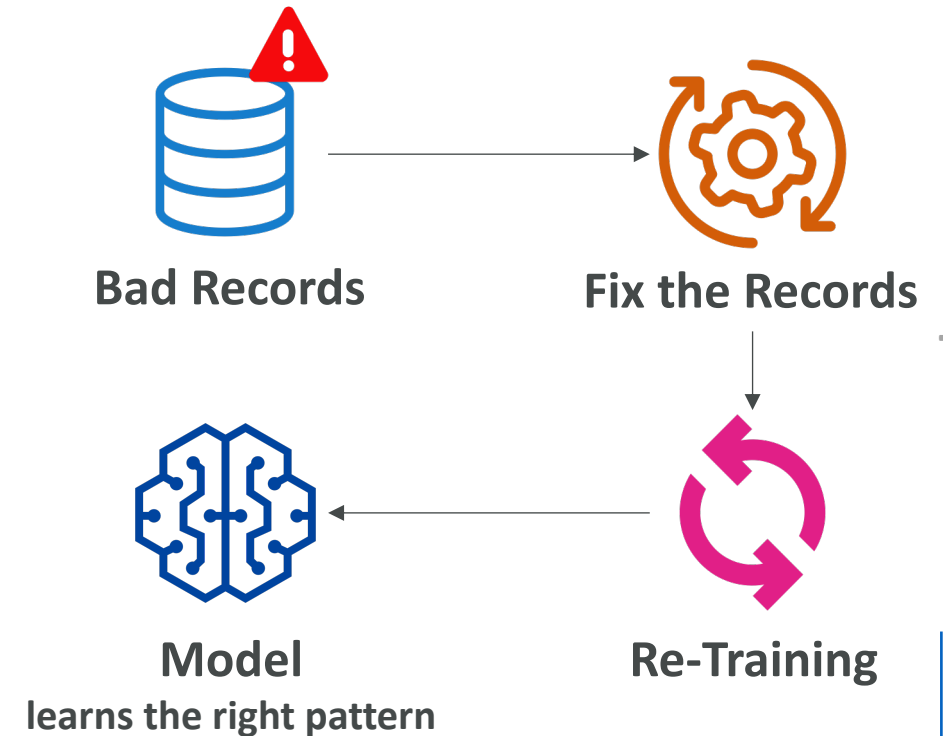


The Four Dimensions of Data Quality



Data Problems Need Data Solutions

- Fix the data at the source, then retrain
- What does NOT fix bad data
 - A better or more sophisticated model
 - More data of the same quality
 - Alerts, dashboards, more staff
- Retraining on bad data = learning the same mistakes again



Exam tip

Distractor: the exam will offer attractive model-side answers for a data-side problem, which is a wrong answer

The One Data Quality Question to Ask

- Before approving: *“Is the data this use case needs complete, consistent, current and representative?”*
- Make it specific to the data type
 - Documents: are these the current versions?
 - Transactions: one ID per customer?
 - Historical Outcomes: do they cover every customer group?
- If nobody can answer, **Data Readiness** is the first work item (Domain 4)



AI Business
Strategist



Before we approve,
is the data this use case needs:

- ☐ **Complete**
- ☐ **Consistent**
- ☐ **Current**
- ☐ **Representative**

Two International AI Standards: ISO/IEC 23053 and ISO/IEC 42001

AI Standard ISO/IEC 23053

- A framework for AI systems that use Machine Learning
- **Used for** terminology, what the parts are called (data, model, training, inference...)
- **Cue words:** “common vocabulary”, “describe the components”

AI Standard ISO/IEC 42001

- The AI Management System standard (AIMS)
- **Used for** governing AI across the organization: establish, implement, maintain, improve
- **Cue words:** “manage AI responsibly”, “certify our AI practices”

AI Standards: What Counts & What Does Not

- **What Counts:** an independent certification against an AI standard (ISO/IEC 42001)
- **What Does Not Count:** an information security certification (ISO 27001, SOC 2), that is security, not AI governance
- **What Does Not Count:** a vendor's own "Responsible AI" page, that is marketing



ISO/IEC 42001 Certification



ISO 27001, SOC 2 (security)



Vendor's "Responsible AI" page

Exam tip

Judge a provider's AI practices on independent certification, never on its own claims.

Rules or AI? The Deciding Question



Can you write the correct answer down in advance,
and will it stay true?

Yes

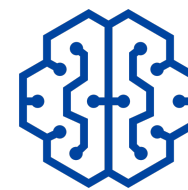
No

Use Rules

if / then logic



- The logic is known and fixed
- Only a few variables matter
- Inputs are structured



Use AI

learn pattern from data

- The relationships keep shifting
- Too many interacting variables
- Inputs are unstructured

Why Rules Are Underrated

- Cheaper to build and to run
- Explainable by reading them
- Identical result every time
- No Drift: no monitoring, no retraining
- Both wrong choices cost money, and fail differently
 - AI where a rule would do: paying for complexity you don't need
 - Rules where the pattern is complex: brittle logic, endless exceptions

	Rule	AI
Cost	Low	build + run + monitor
Explain It	read the rule	needs tooling
Result	identical	varies, probabilistic
Upkeep	none	monitor, retrain

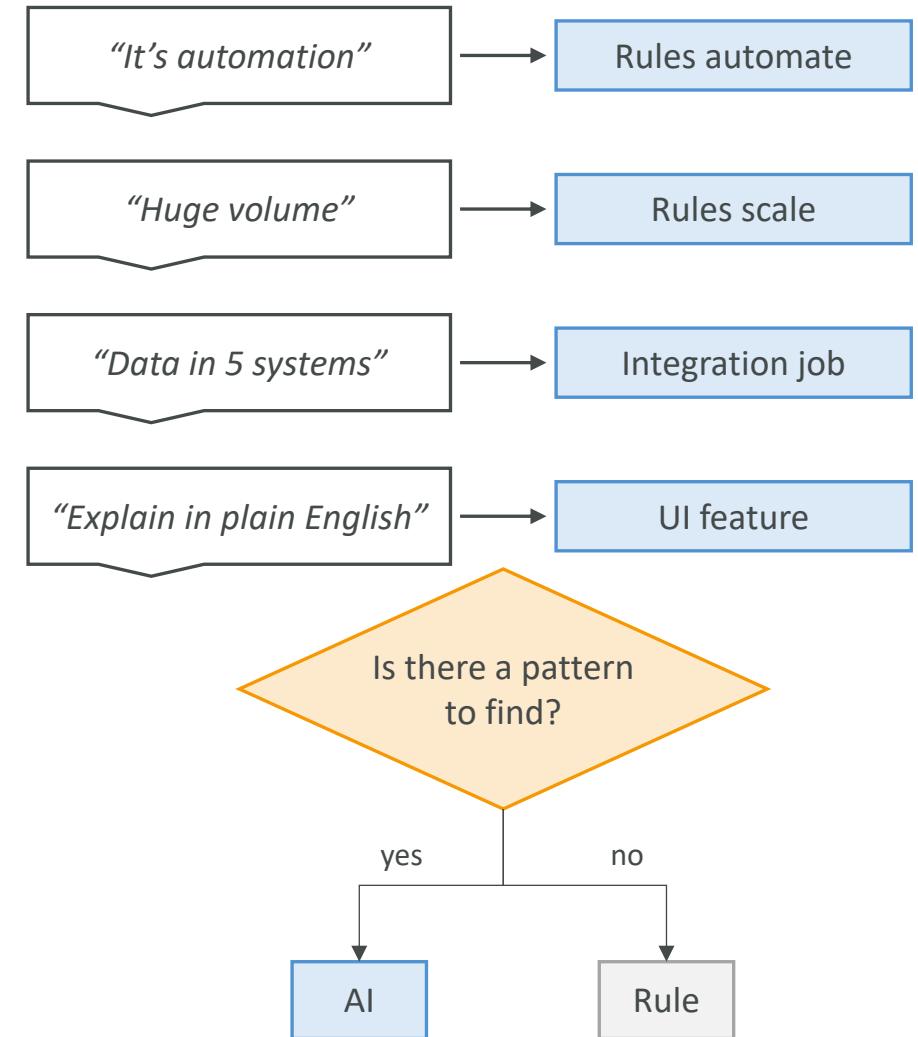
Wrong either way: AI for a stable rule burns money, rules for a complex pattern break on every exception

Sounds Like AI, Only Needs Rules

- Four arguments that sound like a reason for AI, and are not
 - “It’s automation”: rules automate too
 - “The volume is huge”: rules scale too
 - “The data sits in five systems”: that is an integration job
 - “Users want an explanation in plain English”: that is a UI feature

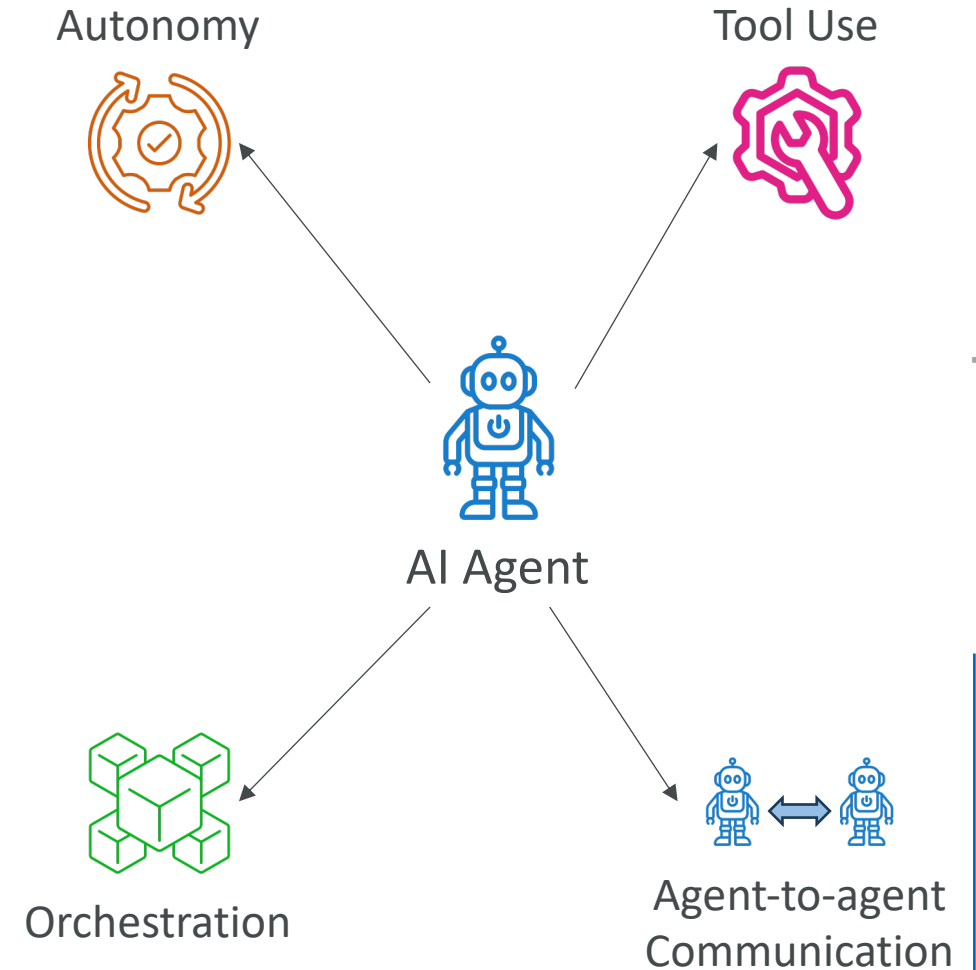
Exam tip

Name the pattern the system must find.
No pattern to find: it is a rule.



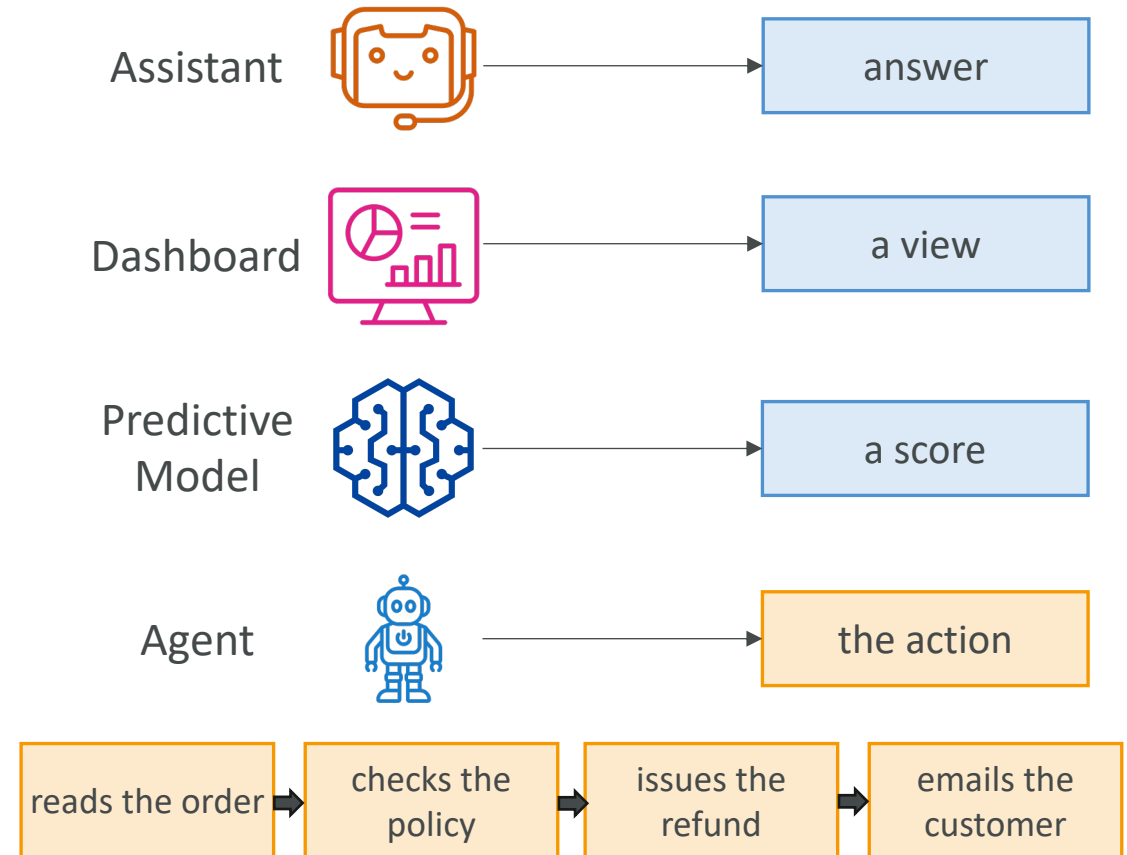
AI Agents

- An AI system that interprets a goal, plans the steps and acts, without a person at each step
- Four capabilities
 - **Autonomy:** decides the next step itself
 - **Tool Use:** calls APIs and systems
 - **Orchestration:** runs multi-step workflows
 - **Agent-to-agent Communication:** hands work to other agents



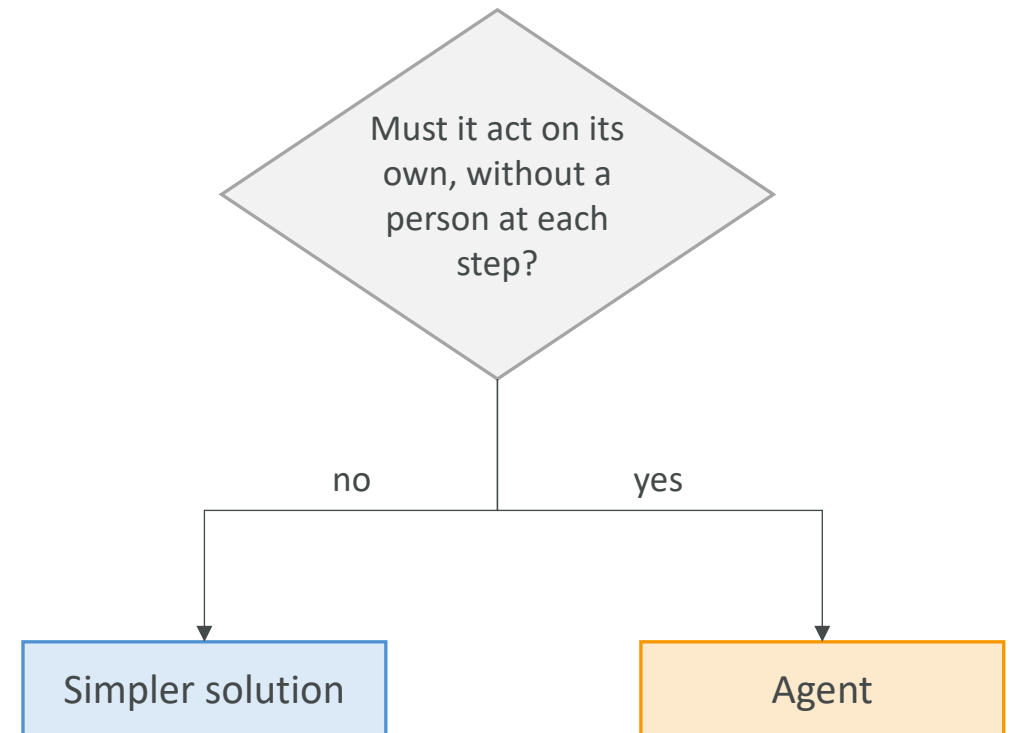
Four kinds of AI solution

- An **assistant** gives an answer (“what is our refund policy?”)
- A **dashboard** gives a view (“how many refunds this week?”)
- A **predictive model** gives a score (“is this refund request fraudulent?”)
- An **agent** takes the action (“process this refund”)
- Only the agent acts, on its own, without a person approving each step



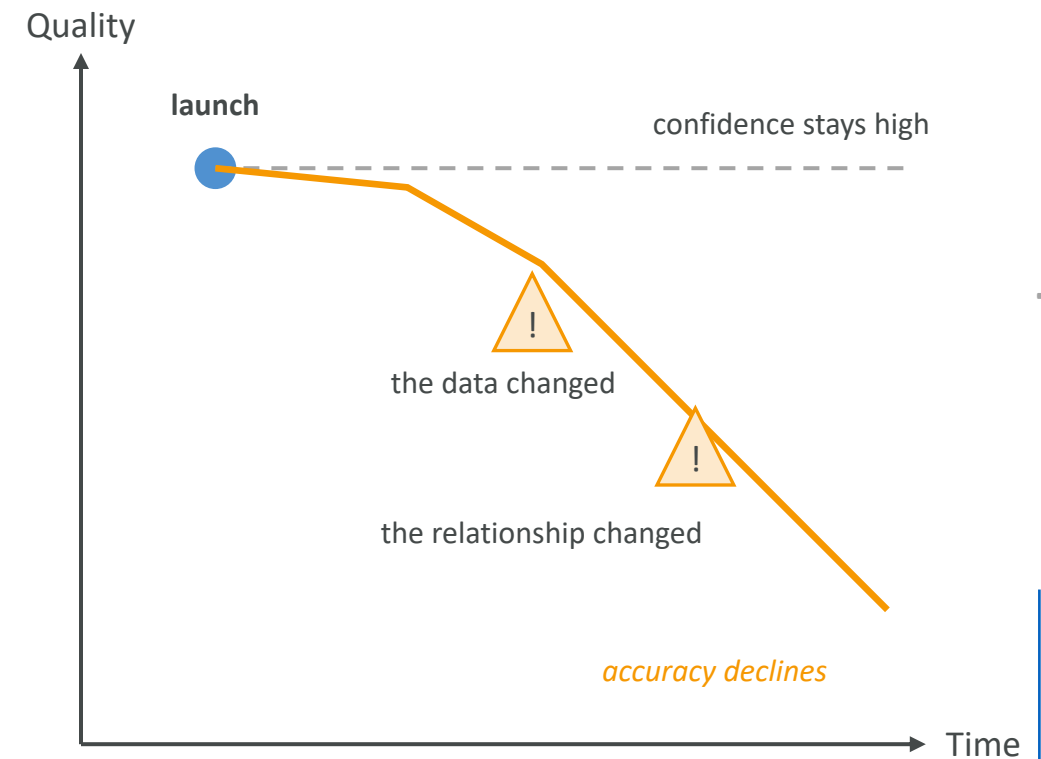
Agent or Not? The Deciding Question

- Does the system need to act on its own, without a person at each step?
 - **No**: an assistant, a dashboard or a model is enough (cheaper, easier to govern and explain)
 - **Yes**: an agent, with the governance an autonomous system needs (Domain 3)
- Three things that do NOT make something an agent
 - Speaking natural language
 - Being grounded in your company content (RAG)
 - Being built on a foundation model



Why Live AI Needs Monitoring: Drift

- The model is trained on a snapshot of the world, and the world keeps moving
- Nothing fails: outputs stay confident and get less correct
- Two causes
 - The input data changed (new customers, new products, new channels)
 - The relationship between inputs and outcome changed (new competitor, new regulation)



The Four Kinds of Drift

Data Drift

- Live inputs look different from the training data
- Example: a churn model trained on retail customers now scores mostly enterprise accounts

Quality-Output Drift

- Predictions become less accurate
- Example: fraud alerts that were 80% right are now 55% right after a product change

Bias Drift

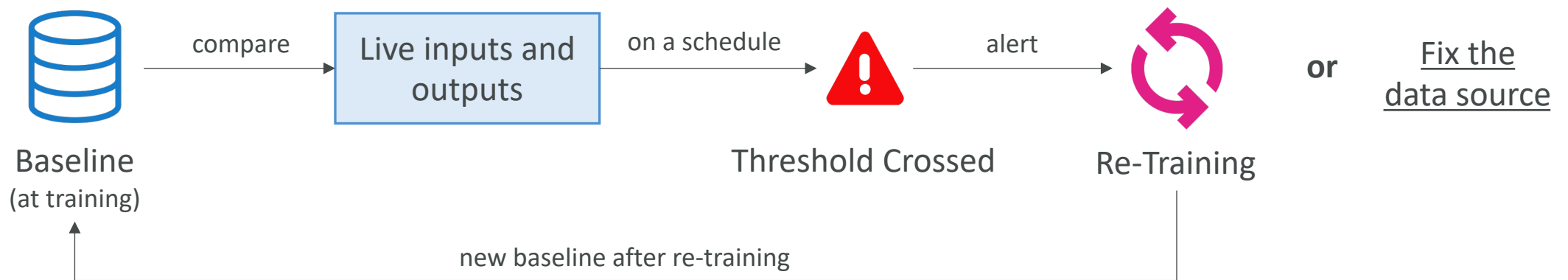
- Performance diverges across groups
- Example: a hiring screener stays accurate overall but misses more candidates from one region

Factor-Influence Drift

- Which inputs matter has changed
- Example: a pricing model that leaned on delivery time now leans on a competitor's price

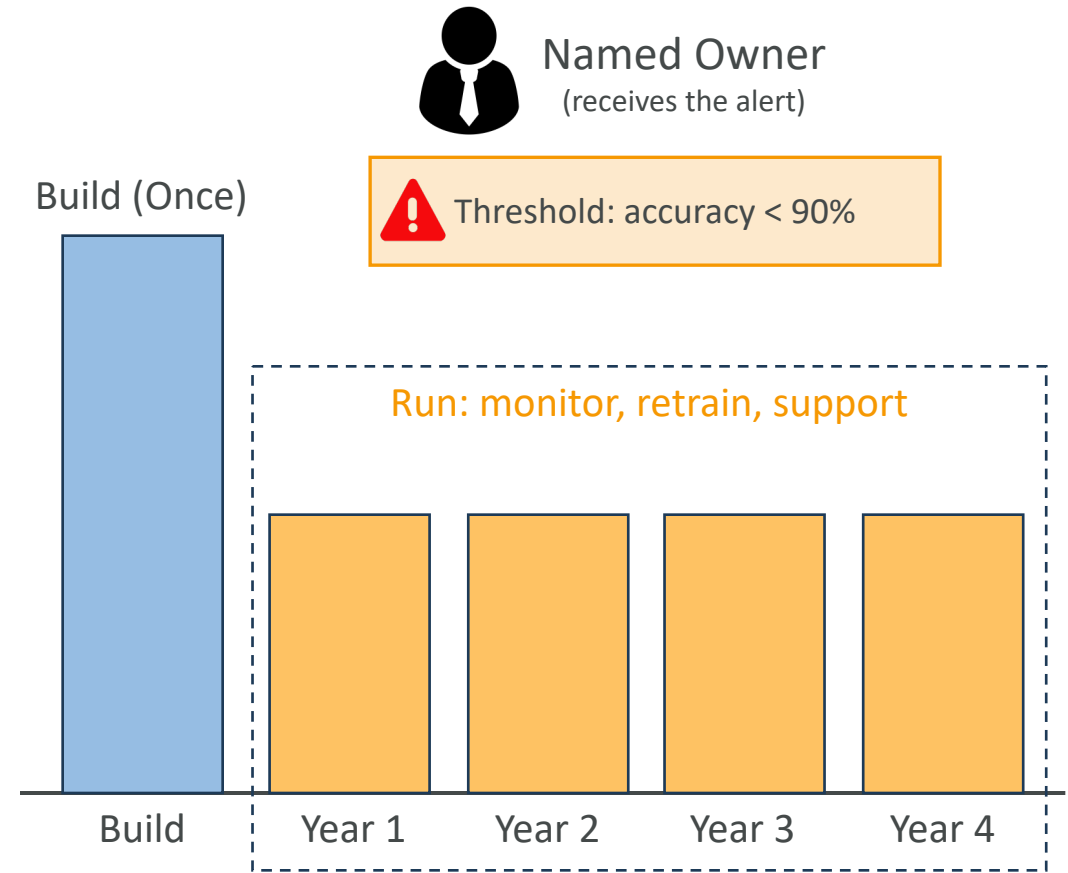
How Monitoring Works

- Baseline captured at training time
- Live inputs and outputs compared to the baseline on a schedule
- Threshold Crossed: alert, then retrain or fix the data source
- Things that look like drift but are not: a capacity limit, a bad prompt, normal day-to-day variation
- “Wait for an error” never works: drift raises no error



AI Is an Operating Commitment

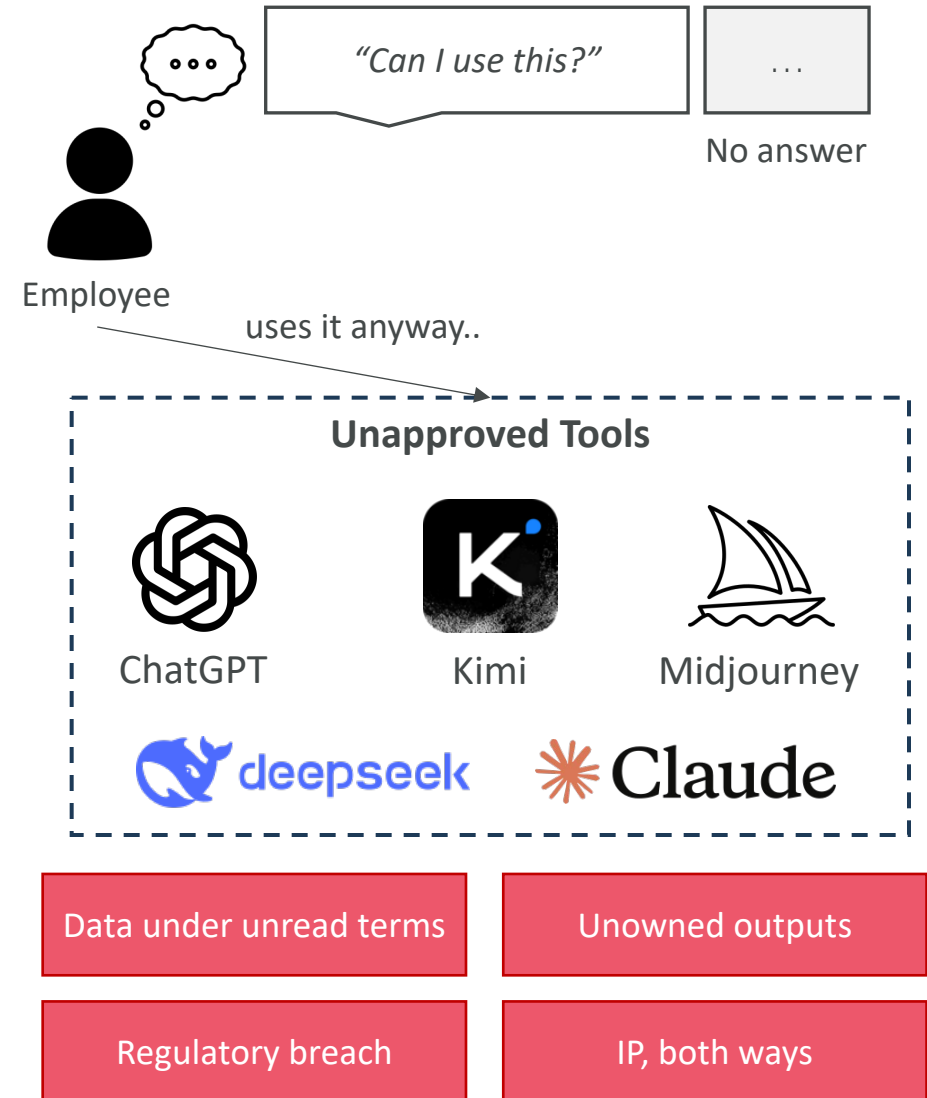
- Monitoring, retraining, support... for as long as the system runs
- A business case that funds only the build is incomplete
- A monitoring control needs a named owner and an alert threshold, or nobody reacts
- Same idea in Domains 2, 3 and 4 (cost planning, governance, production-grade)



the business case must fund the whole chart, not the first bar

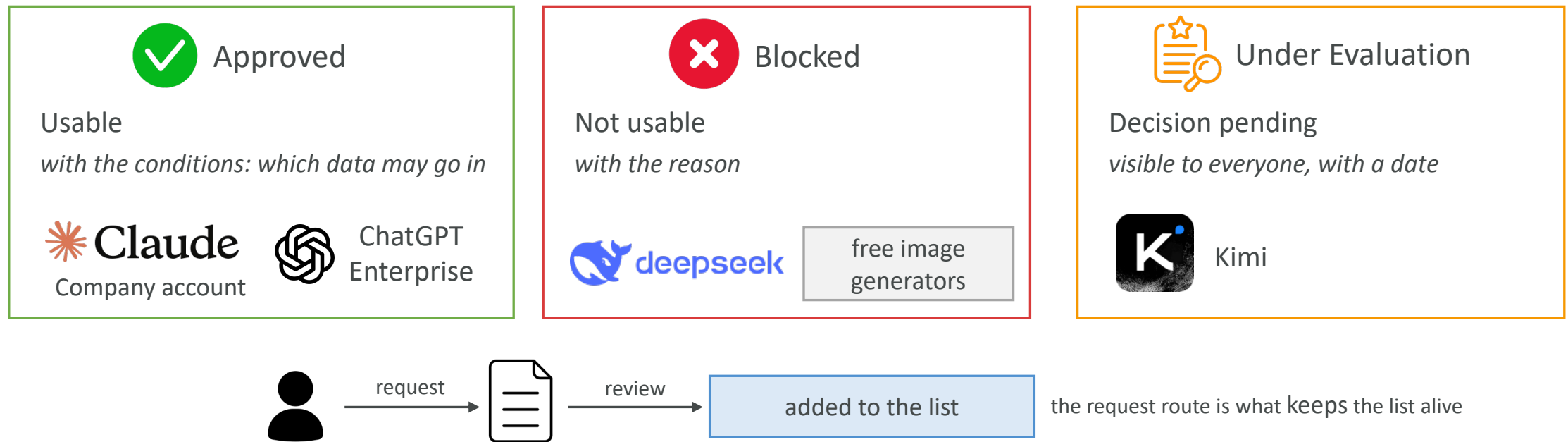
Shadow AI

- AI tools used in the business without review or approval
- Rarely malicious: usually born from an unanswered question (“can I use this?”)
- The risks
 - Company data sent under terms nobody read
 - Outputs nobody owns or checks
 - Regulatory breach (personal data, regulated decisions)
 - Intellectual property, in both directions



The Transparent Tool Classification

- One published list of tools, in three states
- It holds only if it is current, easy to find, and paired with a request route



First Response When You Discover Shadow AI

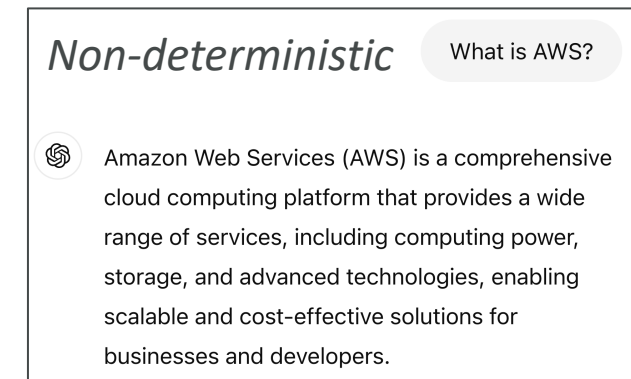
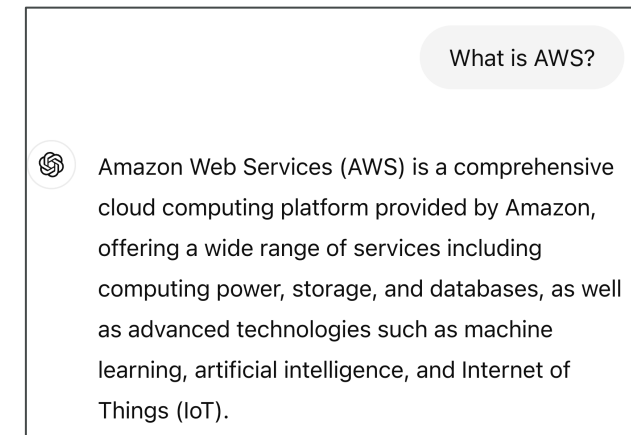
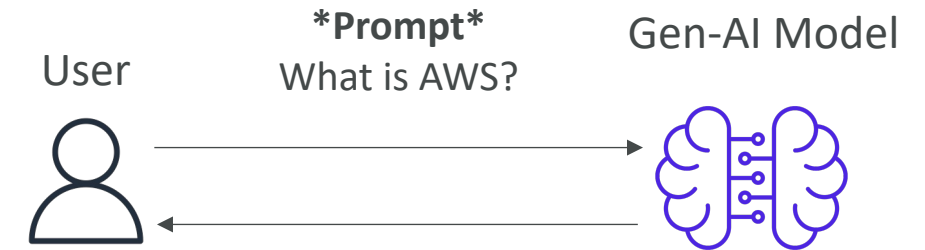
- Publish the classification, with the request route
- Why the alternatives fail
 - Blanket block: drives usage further underground
 - Mandatory training: does not answer “can I use this?”
 - Manager-by-manager approval: slow and inconsistent
- Shadow AI is a symptom of missing governance, not of bad employees

The Published List

Tool	Status	Condition or Reason
Claude (company account)	Approved	no customer personal data
ChatGPT Enterprise	Approved	internal documents allowed
DeepSeek	Blocked	data leaves the region
Kimi	Evaluating	decision by 30 Oct.

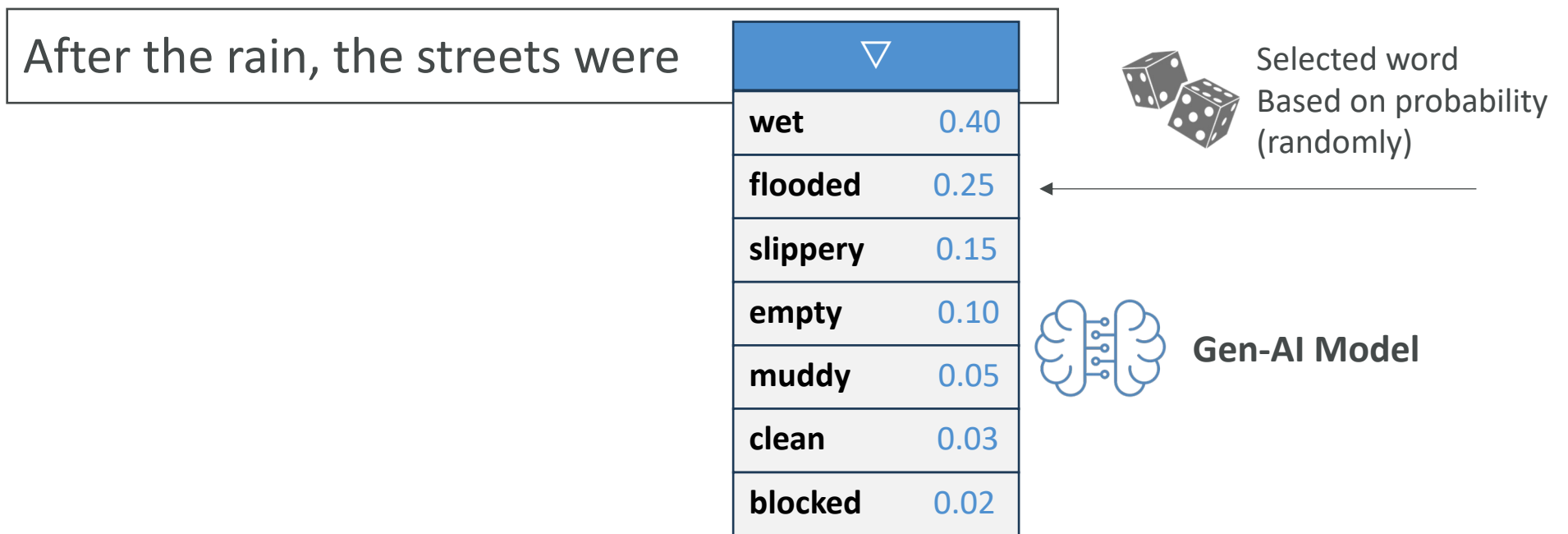
Generative Language Models

- We usually interact with the LLM by giving a **prompt**
- Then, the model will leverage all the existing content it has learned from to generate new content
- **Non-deterministic**: the generated text may be different for every user that uses the same prompt



Generative Language Models

- The LLM generates a list of potential words alongside probabilities
- An algorithm selects a word from that list



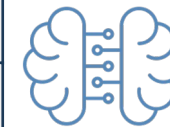
Generative Language Models

After the rain, the streets were flooded

▽
and 0.30
with 0.20
but 0.15
from 0.12
until 0.10
because 0.08
. 0.05



Selected word
Based on probability
(randomly)



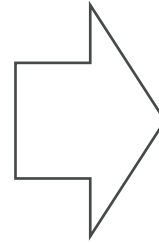
Gen-AI Model

Generative AI for images

Generate images from text prompts

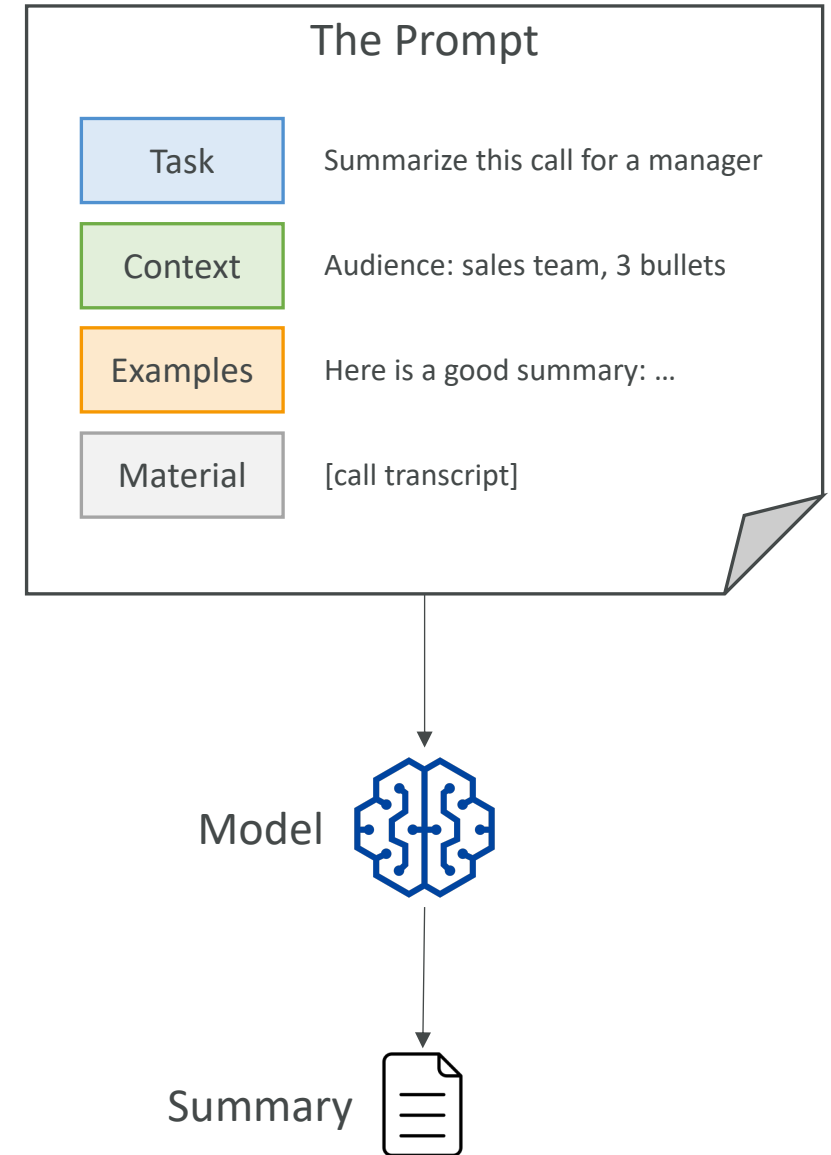
Prompt

Generate a blue sky with white clouds and the word “Hello” written in the sky



Prompt Engineering

- **A prompt:** the input we give the model
- **Prompt Engineering:** shaping the prompt to get the tone, style and focus we want
 - The task, the context, examples, the material to work on
- No change to the model: the cheapest lever



Prompt Engineering – Four Principles

- Be specific about the task and the output form
- Supply the context the model lacks (audience, constraints, definitions)
- Show examples when format matters: 3 to 5 examples fix voice and format
- Change one thing at a time: a reusable prompt, not a lucky one



Simple Prompt vs Full Prompt

Simple Prompt

"Summarize this call."

[call transcript]

Result: a long, generic summary, in a different shape every time, addressed to nobody

Full Prompt

Task

"Summarize this support call in 3 bullets: problem, what was done, next step."

Context

"The reader is the sales manager, who has 30 seconds. Use plain words, no ticket IDs."

Examples

"Here are two summaries in the format I want: ..."

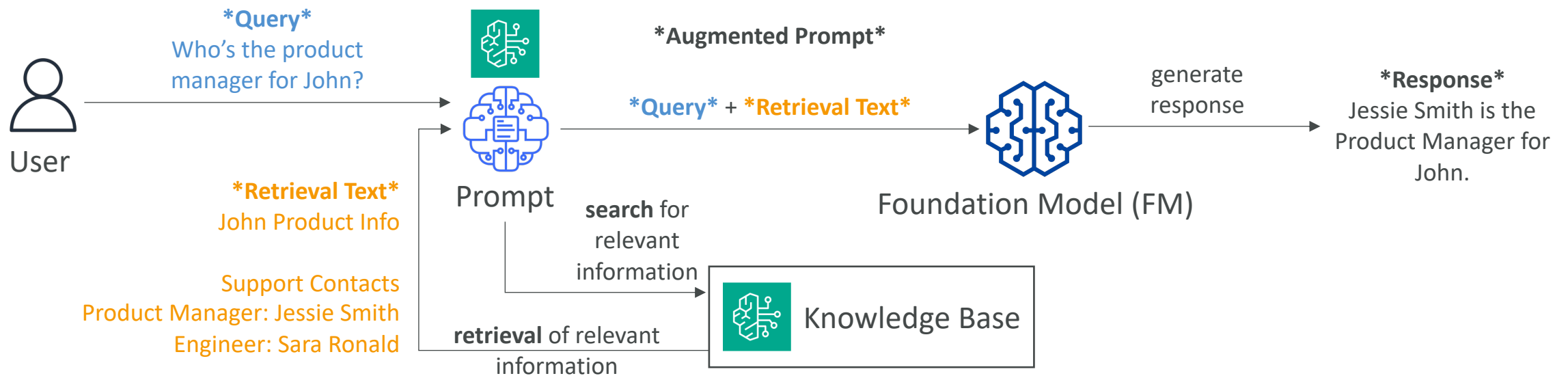
Material

[call transcript]

Result: 3 bullets, same shape every time, written for the manager

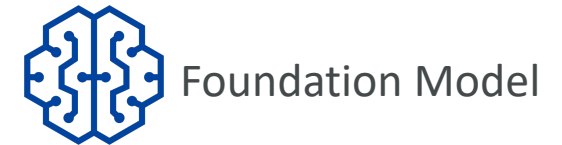
RAG - Retrieval-Augmented Generation

- Allows a Foundation Model to reference a data source outside of its training data
- Use where fresh data is needed to be fed into the Foundation Model

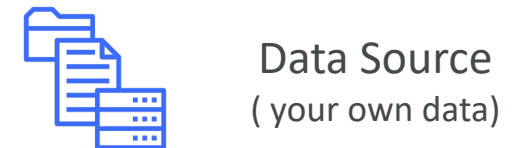


Fine-Tuning a Model

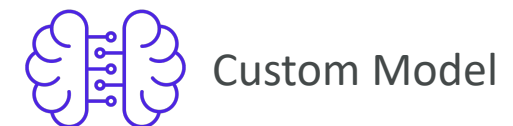
- Adapt a **copy** of a foundation model with your **own data, and own behavior directives**
- Fine-tuning will change the weights of the base foundation model (costly and compute-intensive one-time operation)
- Use cases:
 - A chatbot designed with a particular persona or tone, or geared towards a specific purpose (e.g., assisting customers, crafting advertisements)
 - Training using more up-to-date information than what the language model previously accessed
 - Training with exclusive data (e.g., your historical emails or messages, records from customer service interactions)
 - Targeted use cases (categorization, assessing accuracy)



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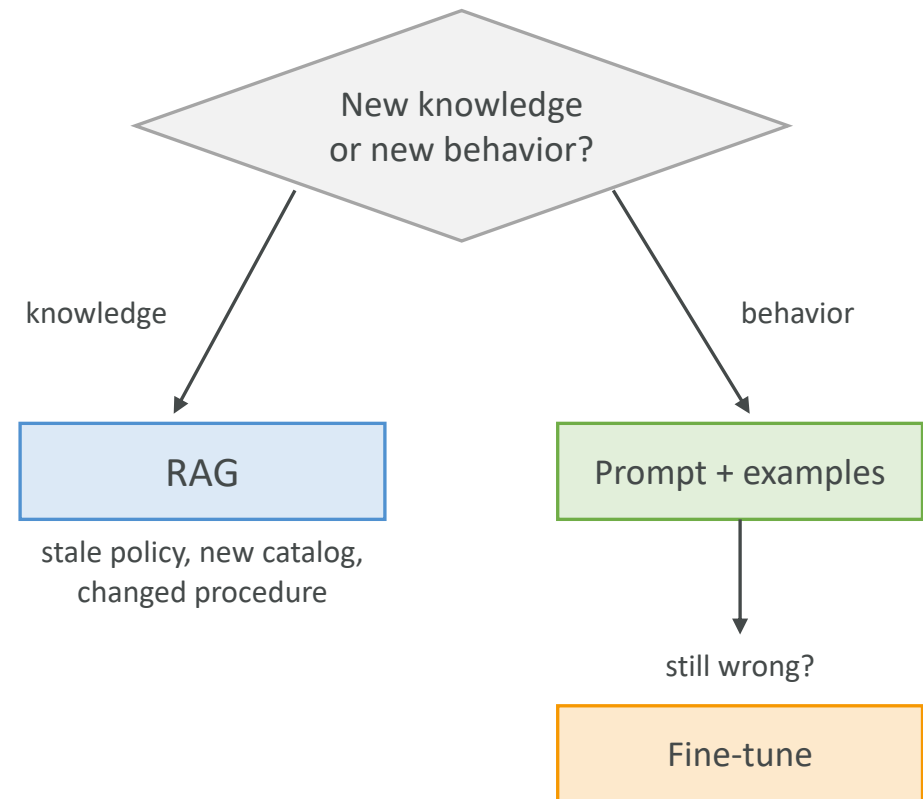


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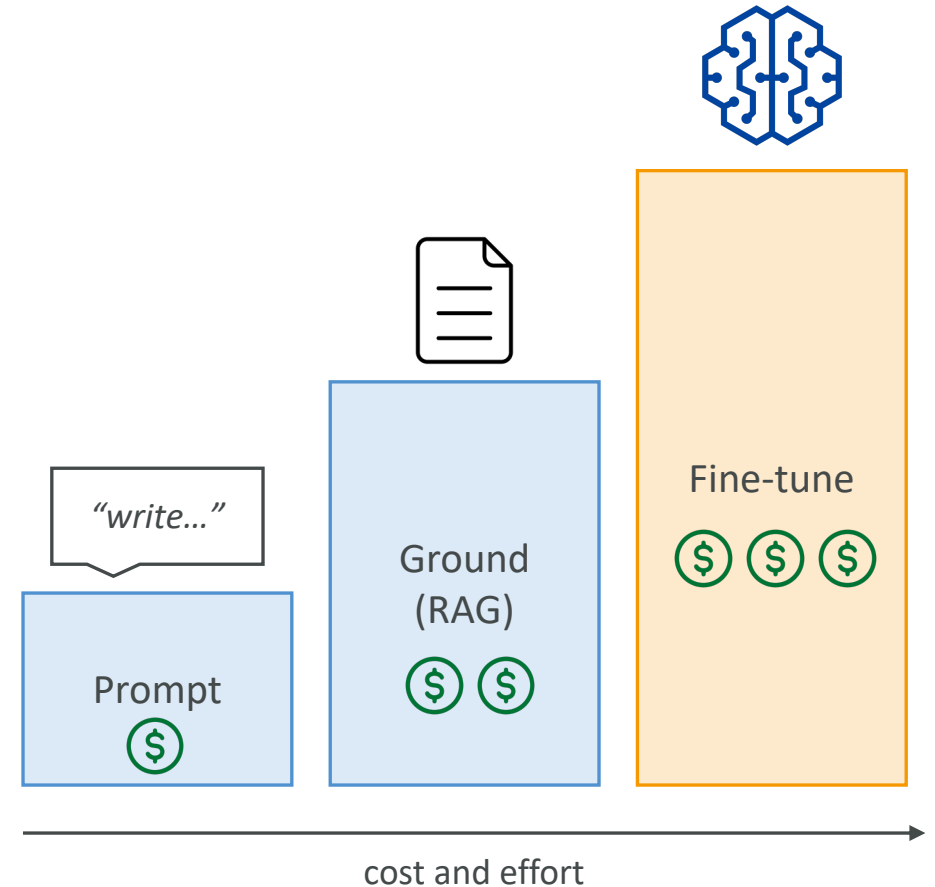
RAG vs Fine-Tuning – The One-question Test

- Does the model need new knowledge, or a new behavior?
- Stale answers: RAG, not more training
 - Facts change: retrieve them, don't retrain the model on them
- Right facts, wrong format: fine-tuning
 - Best for: tone, format, specialized tasks



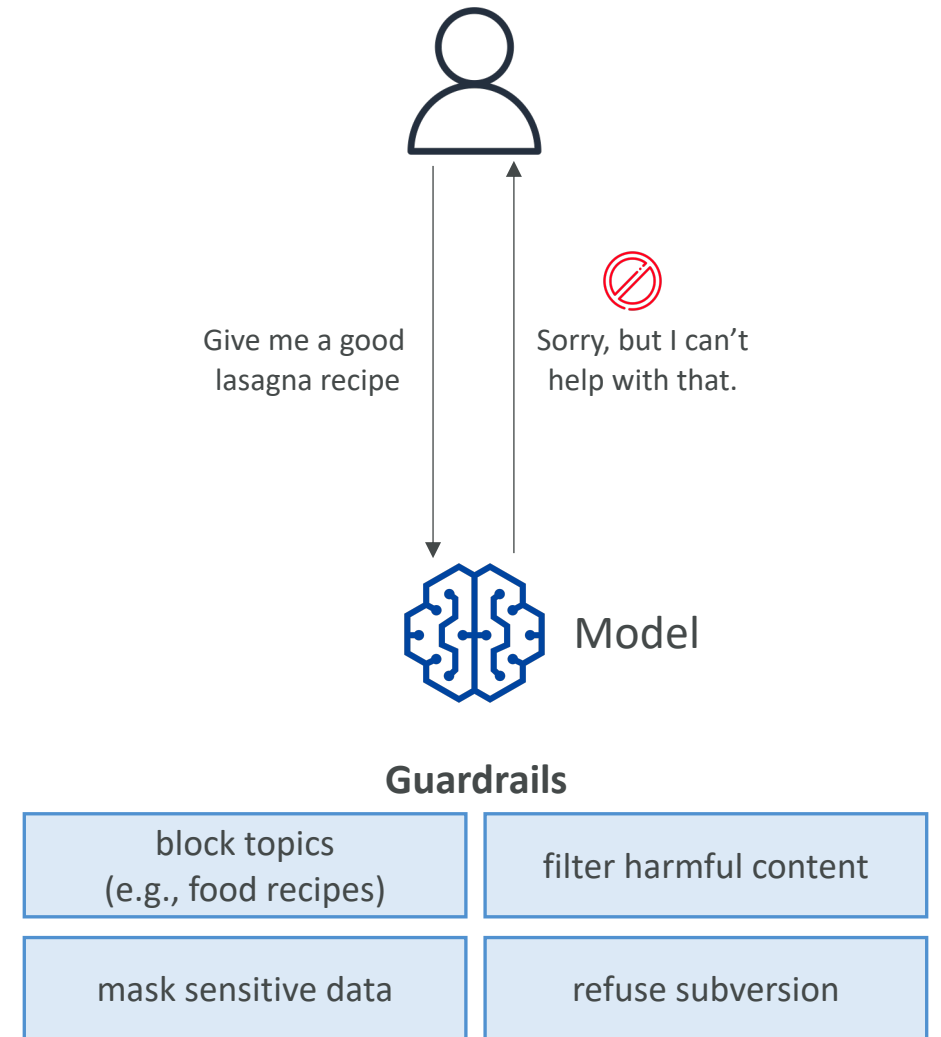
Improving a GenAI Answer The Adaptation Ladder

- Three levers, in increasing cost
 - **Prompt Engineering:** change the instructions (cheap, immediate)
 - **Grounding (RAG):** give the model your documents at inference time
 - **Fine-tuning:** change the model with your examples (most expensive)
- Always start with the prompt



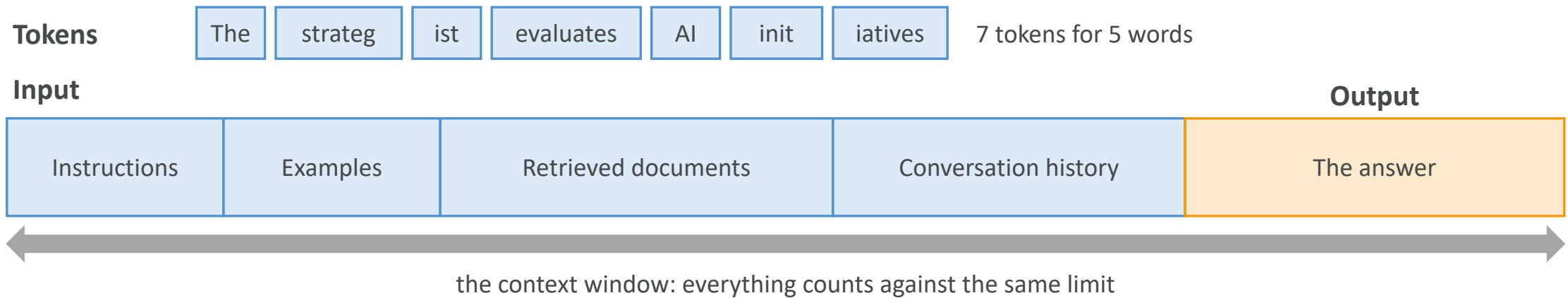
Guardrails – a Safety Layer Around the Model

- A **Guardrail** is a filter applied to what goes into the model and what comes out of it
- **What it does:** blocks topics, filters harmful content, masks sensitive data, refuses attempts to subvert the instructions
- **What it never does:** add facts, or fix a format
- Independent of the model: the same Guardrail can sit in front of any model



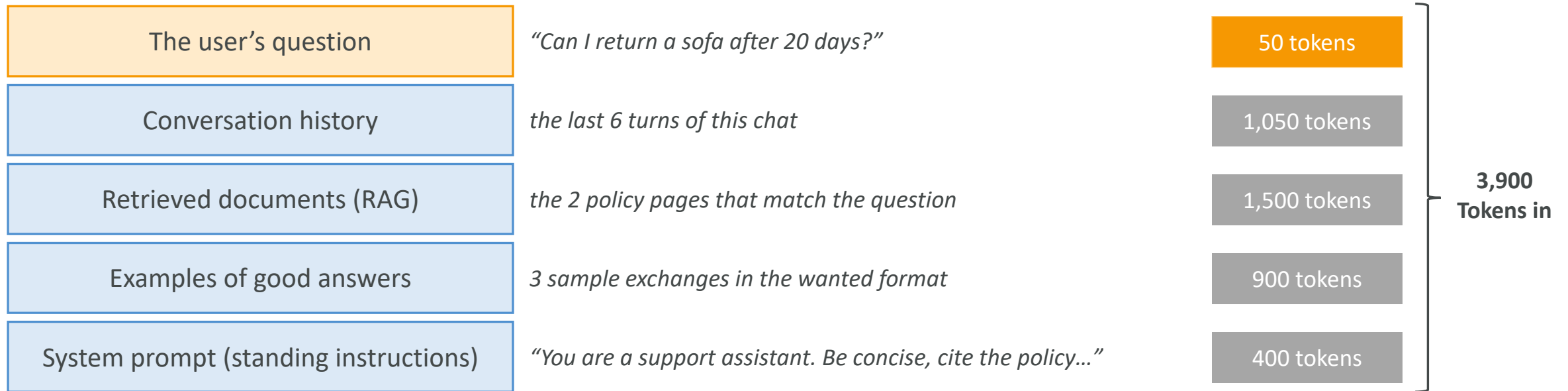
Tokens and the Context Window

- Token = the unit the model reads and counts (a piece of a word)
- Tokens measure capacity (how much fits) and cost (pay per token)
- Context window = the maximum tokens in one exchange
- One budget shared by input AND output



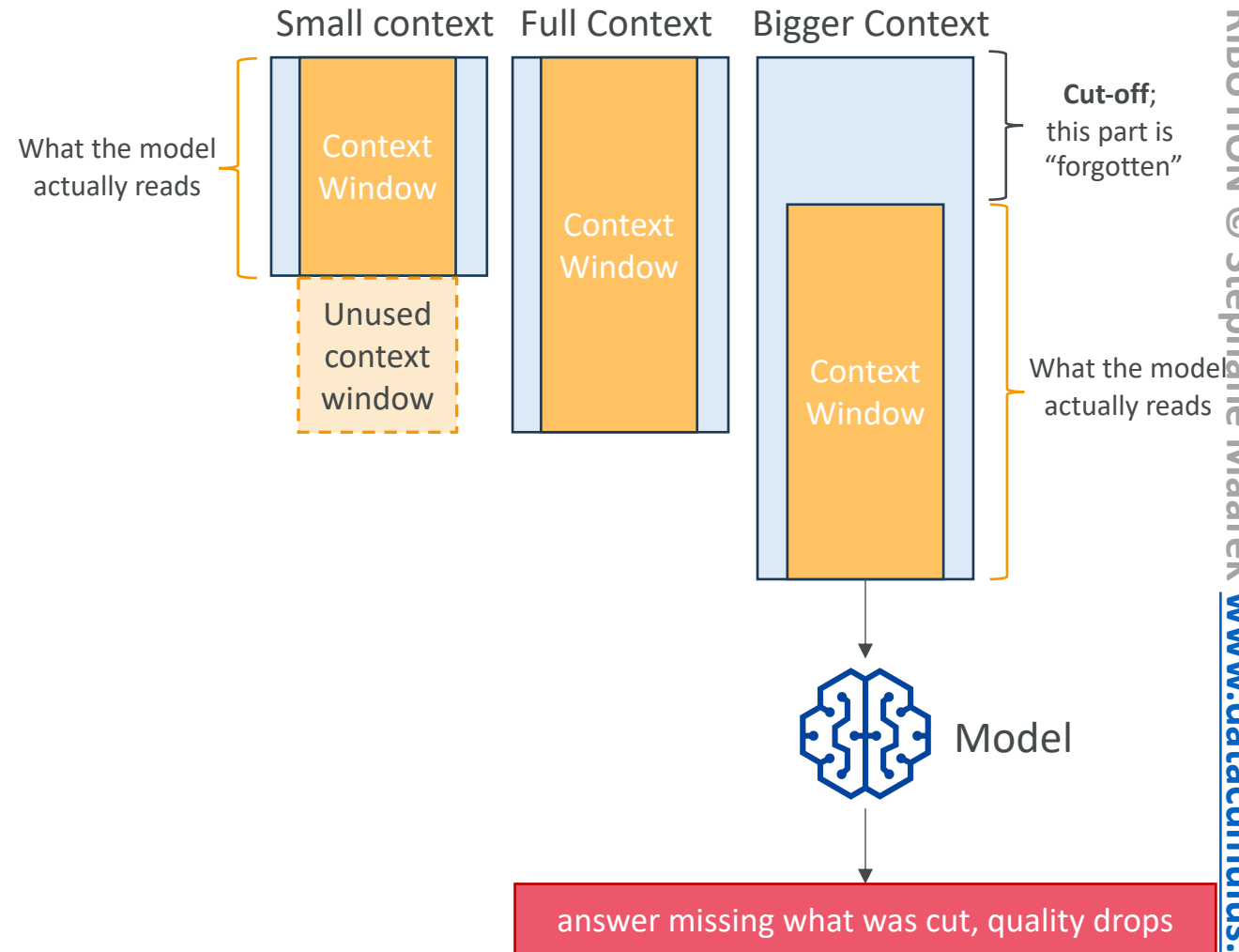
What Fills The Context Window (an Example)

- The user typed 12 words. The model received 3,900 tokens
- Everything the application adds around the question counts against the same window



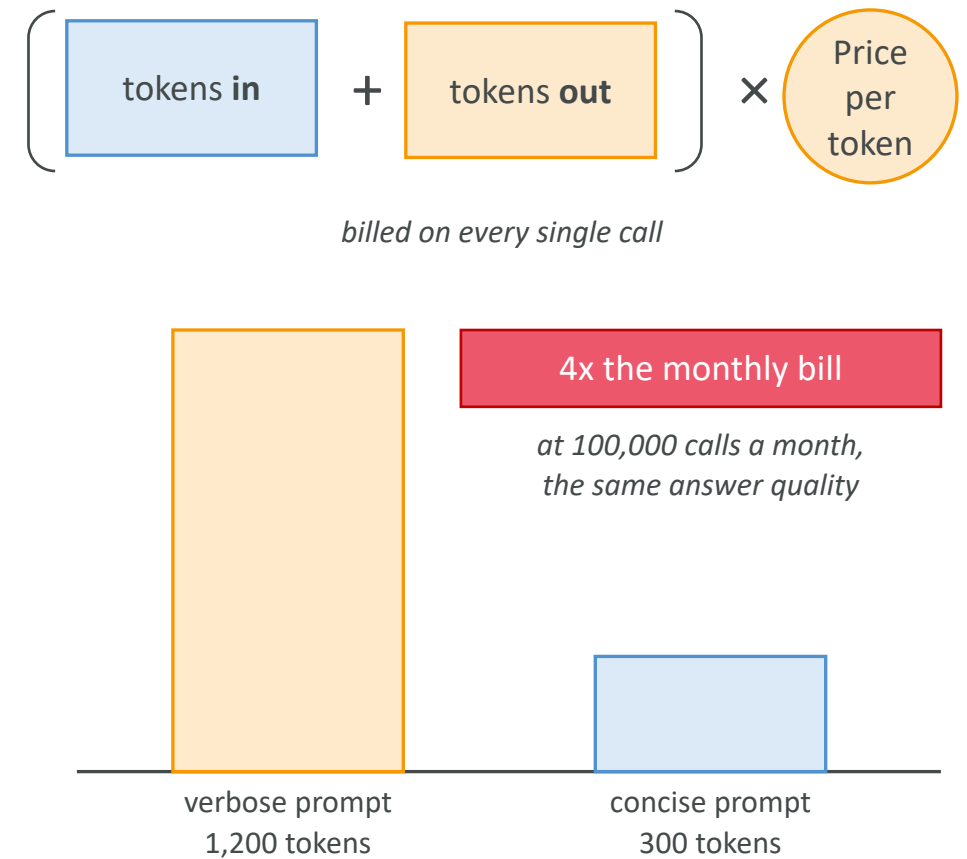
Symptoms of a Capacity Problem

- Earlier turns in a long conversation are forgotten
- Poor answers only on long documents
- Answers cut off mid-sentence
- The length test: if the problem tracks with length, it is capacity, not content
- The fix: manage what fills the window (summarize history, retrieve less)



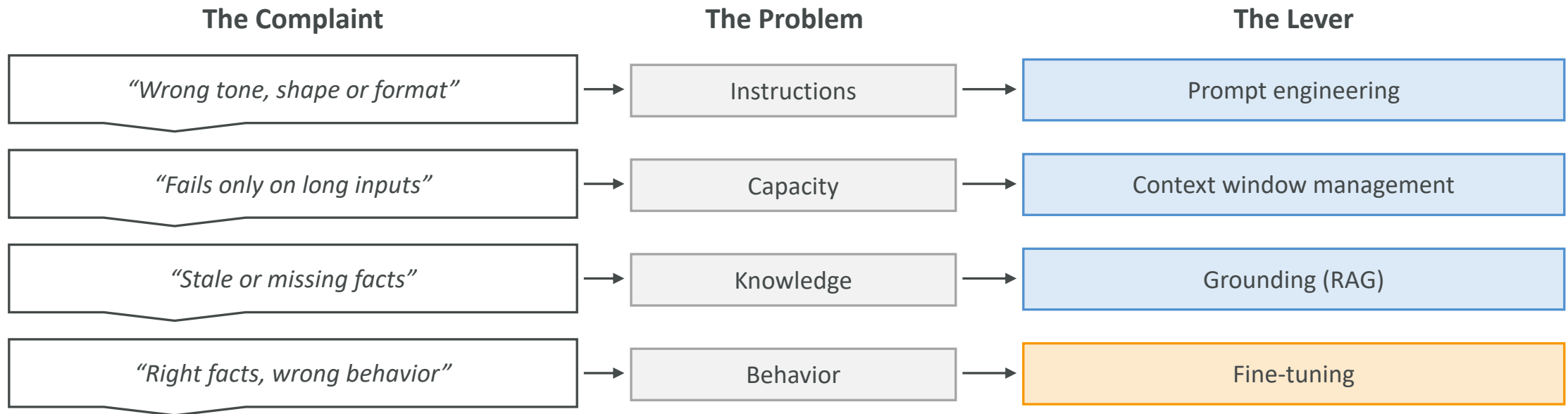
Tokens and Cost

- **Consumption Pricing:** every token in and out is billed
- A verbose prompt = a recurring charge on every call
- Retrieved documents and long histories cost tokens too
- Pricing affects cost, never capability: a cheaper plan does not make the model worse at the task



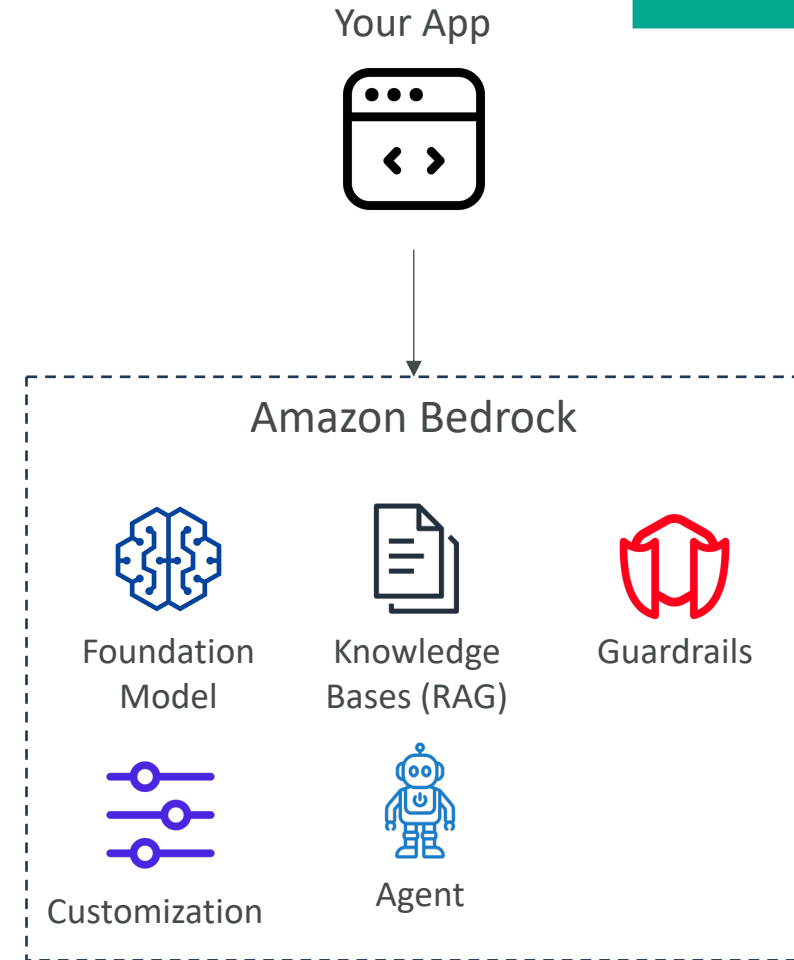
Levers to Use When You Have Complaints

- The same complaint (“the answers are bad”) hides four different problems
- “Give it more data” is never the fix by itself



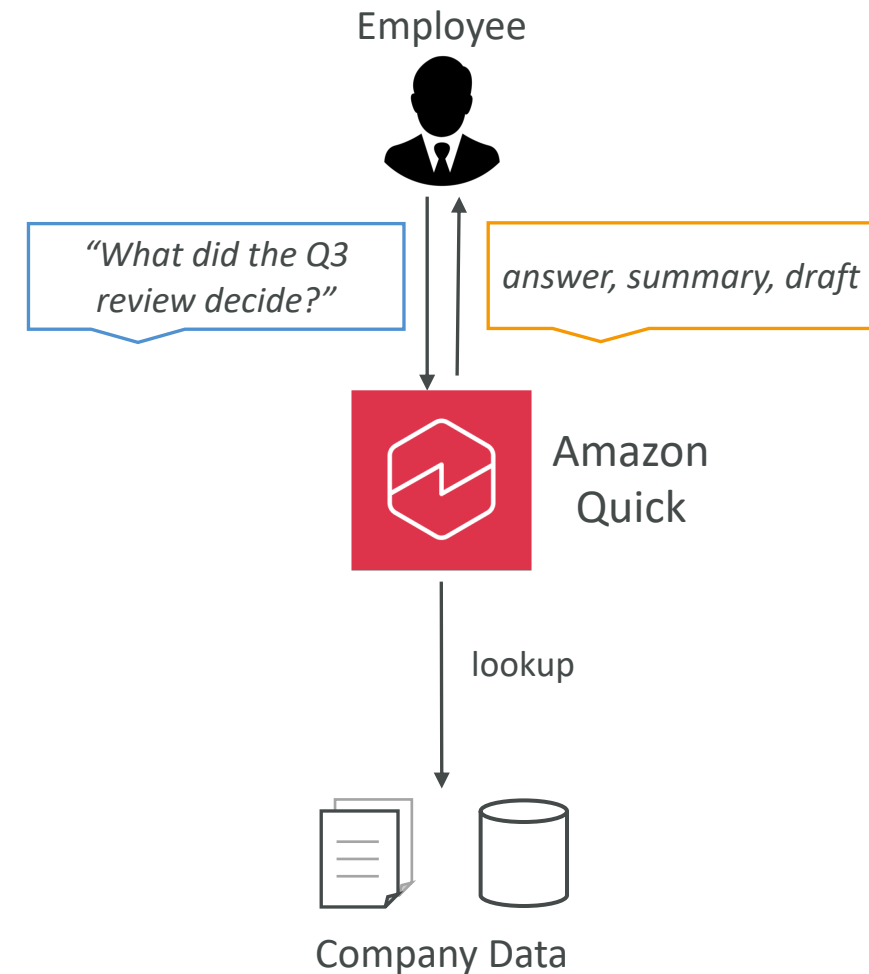
Amazon Bedrock

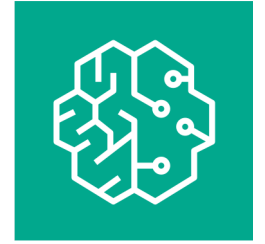
- Build GenAI apps on foundation models from several providers (Claude, DeepSeek, Llama...)
- You choose a model, you never train one from scratch
 - Knowledge Bases: grounding on your documents (RAG)
 - Guardrails: filter inputs and outputs
 - Customization: fine-tuning with your examples
 - Agents: multi-step actions with tools



Amazon Quick

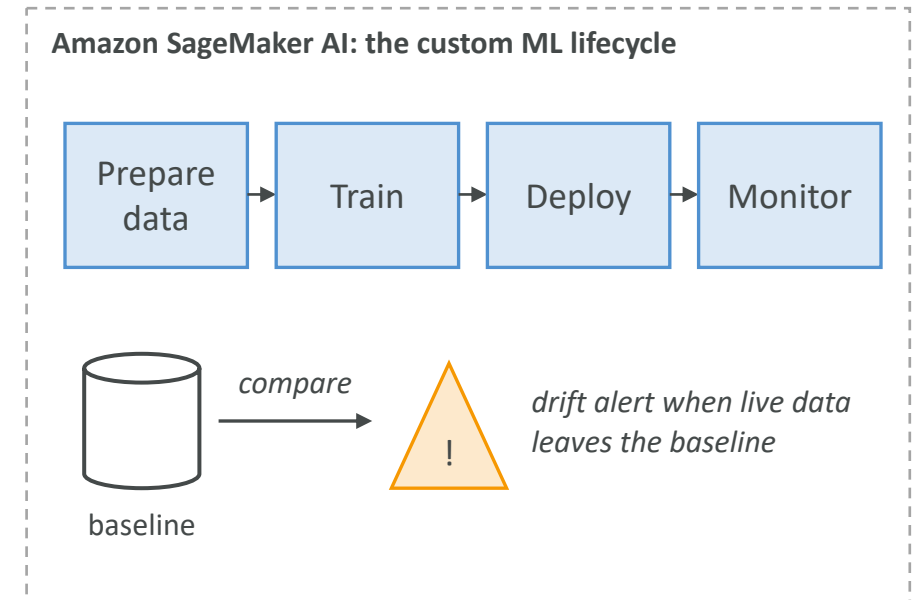
- Ready-made AI-powered business assistant connected to the company's own data
- It works through chat, dashboards and analytics, research, and automated workflows
- No model work: connect the data sources, set the permissions
- Respects each user's access rights to the documents
- Exam: an internal assistant needed quickly, based on your own data





Amazon SageMaker AI

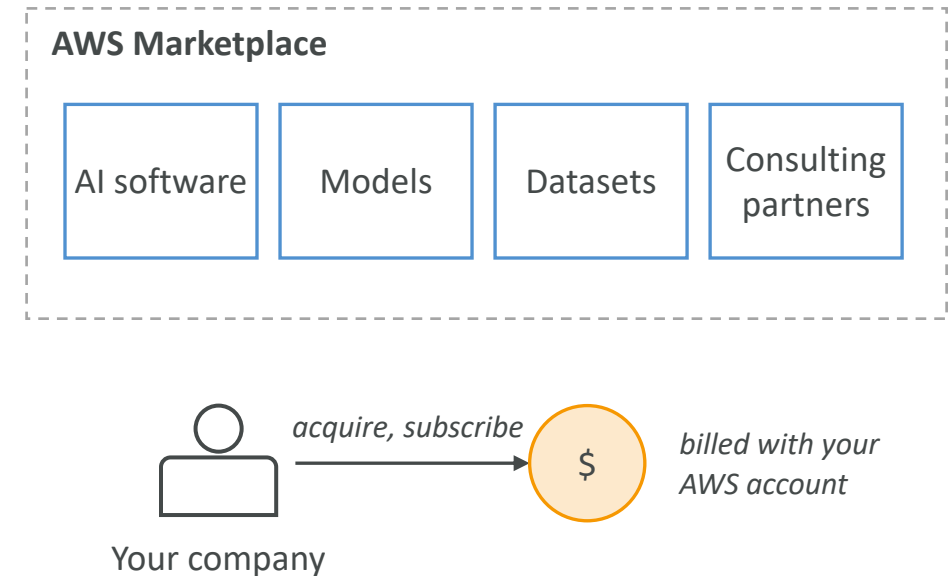
- The full custom ML lifecycle: prepare data, train, deploy, monitor
- For models on your own data, when the model is the differentiator
- Model monitoring: a baseline at training, live data compared to it
 - Answers “is the live model still good?”
- It is where deeper customization (fine-tuning) happens



AWS Marketplace



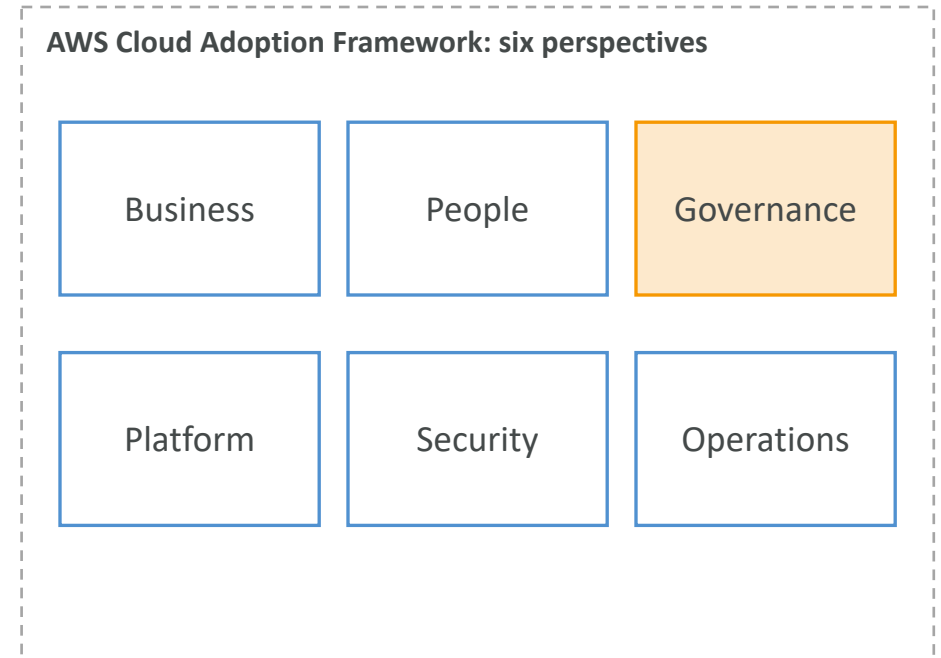
- Catalog of third-party software, models, datasets and partner services
- Bought and billed through your AWS account
- One way to have governance around approved AI tools to avoid Shadow AI





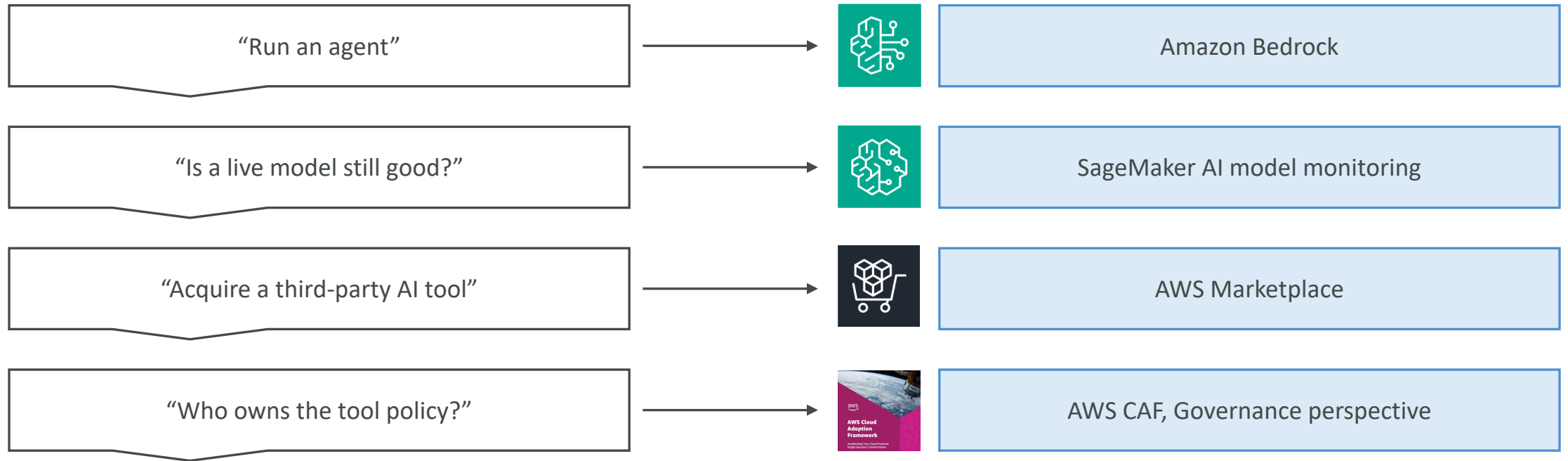
AWS Cloud Adoption Framework (CAF)

- AWS's framework for organizing a cloud and AI adoption across six perspectives
- Business, People, Governance, Platform, Security, Operations
- The Governance perspective is where accountability for AI policy, risk and tool decisions lives



Governance perspective: who owns policy, risk and accountability for AI

Map The Decision To The Service



Putting Domain I Together: The Method

1. Name the solution type: rule, ML, GenAI, agent
2. Name the data it needs, and ask the quality question
3. Is AI warranted? Is an agent needed? Which part can a rule handle?
4. What will be monitored once it is live?
5. If GenAI: improve the answer (prompt, then RAG, then fine-tuning)
6. Then, and only then, the budget



Meet CozyCo: The Company We Advise

- Online furniture retailer, 1,200 employees
- 40,000 support tickets a month, handled by 3 support teams
- Leadership wants “AI” in the plan, without saying what for
- Our role in this course: the AI Business Strategist at CozyCo
- Each domain adds a chapter: the vocabulary, the business case, the governance, the scaling

CozyCo

online furniture retailer

Employees	1,200
Support tickets	40,000 / month
Support teams	3
Leadership	“we need AI”

Wrap-up: Two AI Proposals at CozyCo

	A: Support Assistant for Policy Questions	B: Ticket Routing
Solution type	GenAI, grounded on the policy documents (RAG)	ML: reads a ticket, outputs a label (the owning team)
Data	Policy documents: are they the current versions?	Past tickets and who solved them: complete? consistent?
AI? Agent?	AI: questions come as free text. Agent needed if it must act on a system after answering requests	AI: too many patterns for rules. No agent needed: it only sorts
Monitor	Stale answers when a policy changes	Drift when new products change what customers write
Improve	The prompt first, then the documents it retrieves	Fix the ticket data, then retrain

Domain 2: AI Strategy and Business Value Creation

Section Overview

- The question of this domain: is this AI initiative worth funding, and how to prove it?
- The domain with the highest weight at the exam
- The thread: from a real use case to a business case that survives the CFO
- CozyCo in this domain: the ticket-routing initiative, from idea to business case

Task 2.1

AI Strategy Aligned with Objectives

use cases, build buy partner, portfolio, migration

Task 2.2

Measure and Demonstrate Value

KPIs, baseline, attribution, ROI, cost

Task 2.3

Position AI for Advantage

durable advantage, transformation, investment level

CozyCo in Domain 2: The Support Numbers

- We met CozyCo in Domain 1: online furniture, 1,200 employees
- 40,000 support tickets a month, 3 support teams
- A ticket takes 3.1 days to resolve; 14% bounce between teams
- Returns cost \$18M a year
- Data on hand: 6 years of orders, all tickets since 2019, product photos
- Leadership wants “AI” in the plan. Our job: use cases, numbers, decisions

CozyCo

online furniture retailer

Tickets 40,000 / month

Resolution 3.1 days

Bounced tickets 14%

Returns cost \$18M / year

Order history 6 years

A Use Case Has Three Parts

- The AI capability: what the system can do
- The mechanism: how it changes the work
- The business outcome: what improves, and by how much
- Missing one of the three: not a use case yet, a wish or a science project

	CozyCo: ticket routing
AI capability	Reads each incoming ticket and predicts which team should own it
Mechanism	Tickets land in the right queue on arrival, nobody re-routes them by hand
Business outcome	Bounced tickets from 14% to 3%, resolution from 3.1 days to 1.5 days

Start From The Outcome, Not The Technology

- “Let’s put a chatbot on the website”: a technology looking for a problem
- “Cut the time to resolve a ticket from 3.1 days to 1.5”: a use case, it names the work that changes
- Where to look for outcomes
 - Repetitive, high-volume work: ticket routing, invoice matching
 - Judgment-heavy work with a lot of input: damage assessment on returns
 - Work where the delay itself costs money: quotes, refunds, restocking

Three Decisions, In This Order: Outcome, Feasibility, Budget

1. Outcome: what improves, and by how much
 2. Feasibility: do we have the data, the skills and the time to do it now
 3. Budget: how much this outcome is worth spending
- Reversing the order is the classic mistake
 - Budget first: a fixed AI envelope spent on whatever is easiest to start
 - Feasibility first: the team builds what the data allows, not what the business needs
 - A “safe” pilot in an internal function, while the customer-facing problem everyone can see stays unsolved

Before AI: Could a Rule Do It?

(Recap from Domain 1)

- Domain 1 gave the test: if the same input always gives the same correct output, write a rule
- What Domain 2 adds: write that condition into the proposal, as a sentence
 - “We will not use AI for refunds under \$20: the policy is a table”
 - “We use AI for routing: 40,000 free-text tickets, no rule set captures them”
- Why it matters for the business case: a rule has no run cost, no monitoring, no drift
- CozyCo: refunds under \$20 stay a rule, ticket routing and damage assessment go to AI

Build vs Buy vs Partner

- A use case says what to get. How to get it is a second decision: ask where the advantage lives

Build	Buy	Partner
What you do: train or assemble your own model on your own data	What you do: subscribe to a service that already does it	What you do: bring in a firm with the skills or the domain
You own: the model, the pipeline, the know-how. It takes data, skilled people, months	You own: a subscription and your configuration. It takes days to weeks	You own: the outcome, delivered with someone else's expertise. It takes a contract and a scope
Fits when: the advantage is in your own data or logic	Fits when: you need speed and the capability does not set you apart	Fits when: you lack expertise you cannot acquire in time
CozyCo: demand forecast on 6 years of its own sales	CozyCo: support assistant for policy questions	CozyCo: damage assessment on returns, nobody in-house, 4 months

Exam Tip

The answer follows the use case, not a company-wide policy: the same company builds, buys and partners on different initiatives.

Five Constraints That Can Change The Answer

- The advantage question gives a default
- Five practical constraints can override it
 - **Budget:** building costs more upfront than buying
 - **Timeline:** a fixed date rules out a build that needs months
 - **Internal Capabilities:** no data scientists means no build, whatever the data
 - **Vendor Proposals:** a strong offer on the market makes buying the rational default
 - **Regulatory Compliance:** a regulated domain can push toward a partner who knows the rules
- Missing skills change the path, never the decision to proceed

Exam Tip

“We lack the skills, so we delay until we have hired” and “we lack the skills, so we terminate” are both distractors: missing skills mean buy or partner.

Rank The Portfolio on Four Criteria

- A portfolio is a set of bets, not a to-do list: rank them, then decide per initiative
- **Business value:** what it earns or saves
- **Feasibility:** data, skills and timeline, available now
- **Sustainability:** can it be operated and funded for its whole life
- **Strategic alignment:** does it serve the strategy

Initiative	Value	Feasib.	Sustain.	Align.	Priority
Ticket Routing	Orange	Green	Green	Green	1
Damage assessment	Green	Orange	Orange	Green	2
Demand forecast	Green	Red	Orange	Green	3
Website chatbot	Red	Green	Orange	Red	4

Three Verdicts For Every Initiative: Scale, Pause, Terminate

- A portfolio review ends with a decision per initiative, never with “let’s see”

Scale

When: the pilot met its KPIs against the baseline, and the foundation can carry more

Do: fund the rollout, with the run costs

CozyCo: ticket routing, 85% correct after 6 weeks, extend to all 3 teams

Pause

When: a gap blocks it and can be fixed in a known time. A pause has a condition and a date

Do: name the gap, the fix, the date to reassess

CozyCo: demand forecast, product data not unified, fix by Q2

Terminate

When: a review shows it will not pay off, whatever is added

Do: stop it, record the reason, free the budget

CozyCo: the website chatbot from the wish list, 2% of visitors use it, no path to value

Four Portfolio Traps

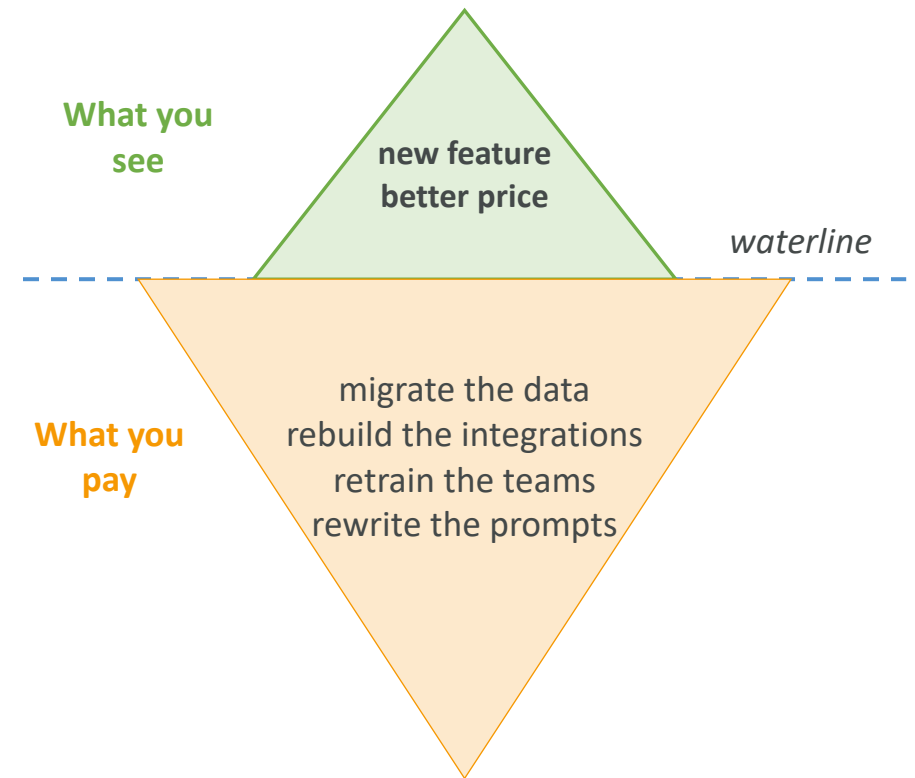
- **Sunk Cost:** money already spent. It is gone whatever you decide, so it is never a reason to continue
 - “We already spent \$2M on the chatbot”: irrelevant to whether the next \$500k will pay off
- **The Quiet Drain:** an initiative nobody stops keeps consuming budget and attention without a review
- **The Platform Switch:** moving an unsound initiative to another platform changes the bill, not the outcome
- **Data Not Ready is a Feasibility Gap:** it calls for a pause and a data fix, not for “use rules instead”

When a Live Process Moves to AI: Four Checks Before the Switch

- CozyCo is about to route tickets with AI instead of the manual desk.
- Four checks:
- **Business Continuity:** the old way keeps working during the switch, with a way back
 - CozyCo keeps the manual desk for 2 months in parallel, with one switch to turn AI routing off
- **Cost:** the one-time migration and the new monthly cost are both known
 - CozyCo: \$60k of integration, then a cost per ticket instead of a fixed team
- **Data Readiness:** the data the new process needs exists and is reachable, checked before the switch
 - CozyCo: ticket text and team labels since 2019 exist;
- **Performance:** parity on the measures that matter, proven before the full switch
 - CozyCo: 85% routing accuracy on the two pilot teams before the third team switches

Switching AI Platforms: The Iceberg

- The same four checks apply when you move from one AI platform to another
- The visible part is what pulls you: a new capability, a better price
- The hidden part is what you pay: data to migrate, integrations to rebuild, people to retrain
- Switch only when the platform is the problem, not the model or the data



KPIs: Chosen Before Launch, of Two Kinds

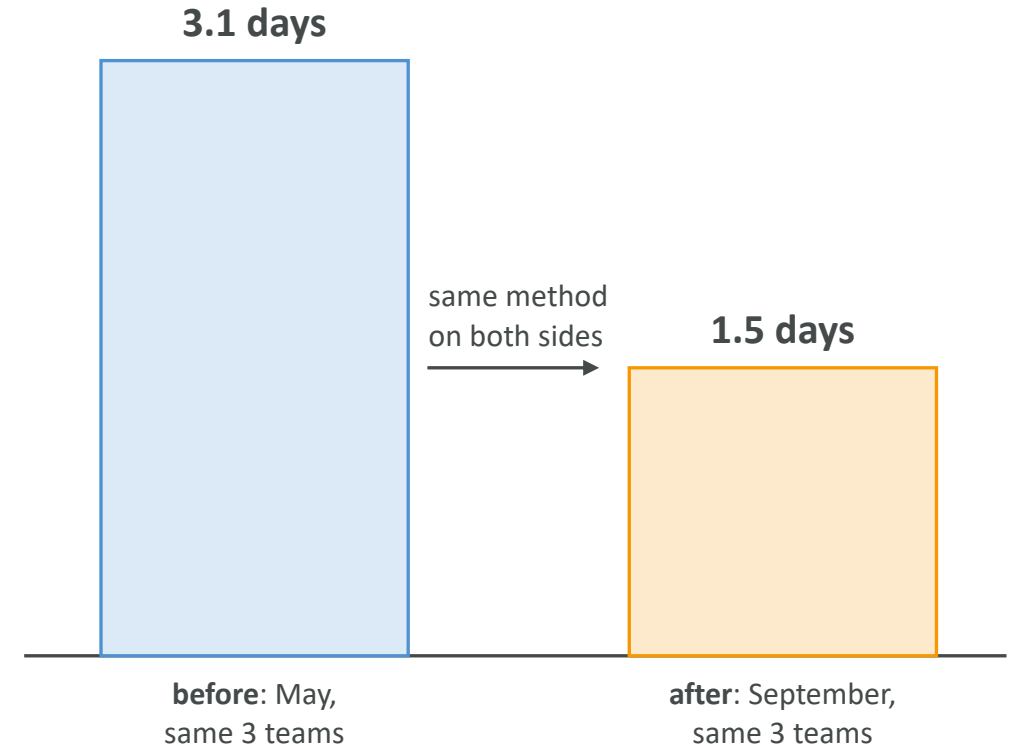
- A KPI (**Key Performance Indicator**) is the measure that says whether the initiative did what it was funded to do
- Chosen before launch, so that the before-state can be recorded
- **Tangible KPI:** converts directly into money
 - CozyCo: resolution time (3.1 days), cost per ticket (\$11), returns cost (\$18M)
- **Intangible KPI:** measurable, but not directly convertible into money
 - CozyCo: customer satisfaction score (3.9 / 5), agent productivity, decision speed
- Leaving the intangibles out understates the value of the initiative

What a Good KPI Measures, and What It Does Not

- It measures the outcome and the quality of the process being automated
 - CozyCo routing: bounced tickets, resolution time, first-contact resolution
- It does not measure organizational inputs: headcount, budget, number of agents
- It does not measure infrastructure health: uptime, latency of the model
- It does not measure company-wide results too broad to attribute: total revenue, share price
- One question per candidate KPI: if the initiative works, does this number move because of it?

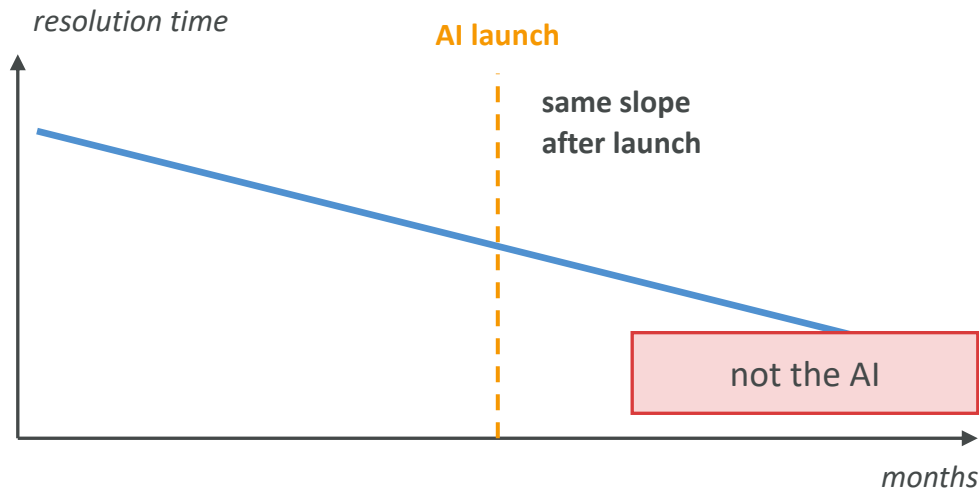
The Baseline: The Before-state, Measured The Same Way

- Recorded before launch, with the method that will measure the after-state
- Without it an improvement is a claim, not evidence
- Three questions make it credible
 - Recorded before launch, not remembered afterwards?
CozyCo: May export
 - Same measurement on both sides?
Same 3 teams, same helpdesk timer
 - Comparable period?
Volume flat, 40,000 tickets both months
- “Things got faster since we launched” fails the first question: nobody recorded where it started

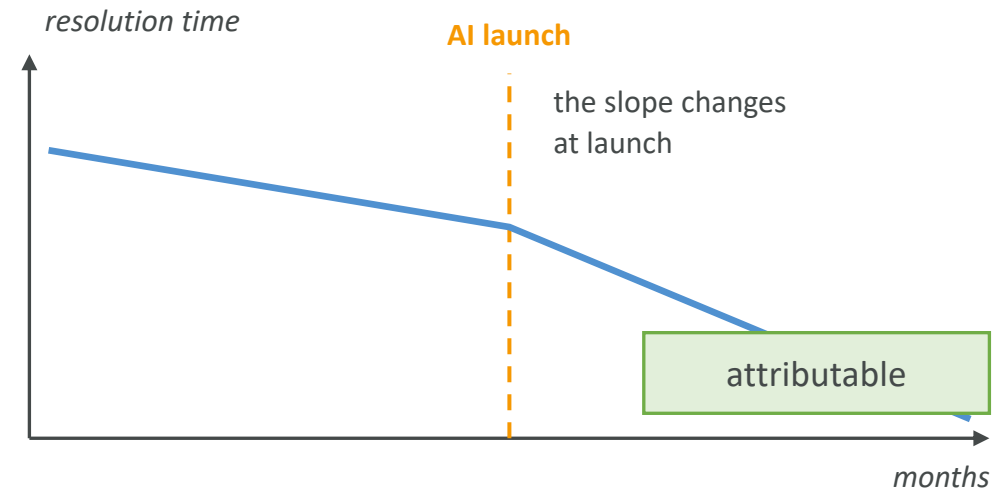


Attribution

- **Baseline:** the number moved
- **Attribution:** the AI moved it



- Already improving before launch, same pace after. The AI changed nothing



- The pace changes at launch. The change is attributable

The Value Statement The CFO Accepts

- The CFO's question is always the same: "How do you know the AI caused it?"
- A credible value statement answers it in three parts
 1. The measured change against the baseline
 2. The mechanism: what the AI did that produced the change
 3. The causal logic: what else moved, and why it does not explain the change

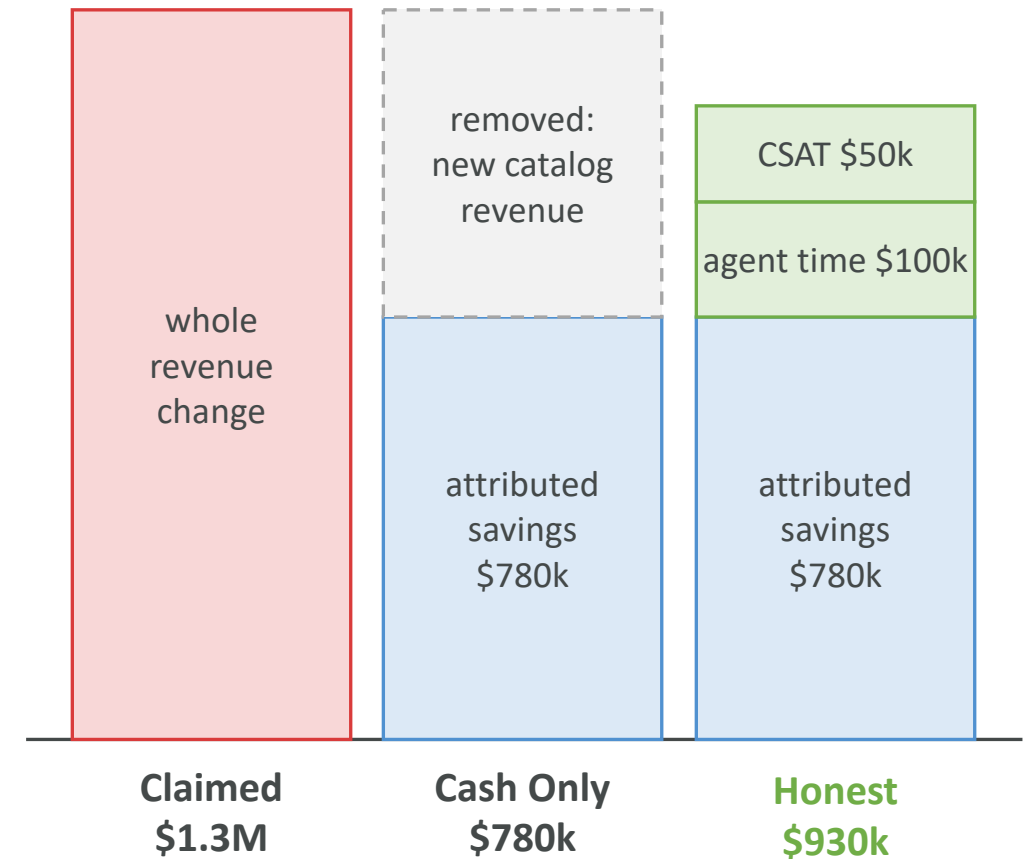
CozyCo: "Resolution time fell from 3.1 to 1.5 days (same teams, same clock). The AI routed 91% of tickets to the right team on arrival, so 14% of tickets stopped bouncing. Volume and staffing were flat over the period, and the seasonal dip comes in December, not September."

ROI: The Inputs Are The Problem

- $ROI = (\text{benefits} - \text{costs}) / \text{costs}$
- CozyCo routing, year 1: benefits \$930k, costs \$310k, $ROI = 200\%$
- The formula is rarely wrong
- The two inputs (benefits & costs) can be mistaken
 - Costs Undercounted: the build is counted, the years of running it are not
 - Benefits Misstated: too high when the whole revenue change is claimed, too low when intangibles are dropped
- Payback period (how long until benefits cover costs) shares the same inputs, so the same errors

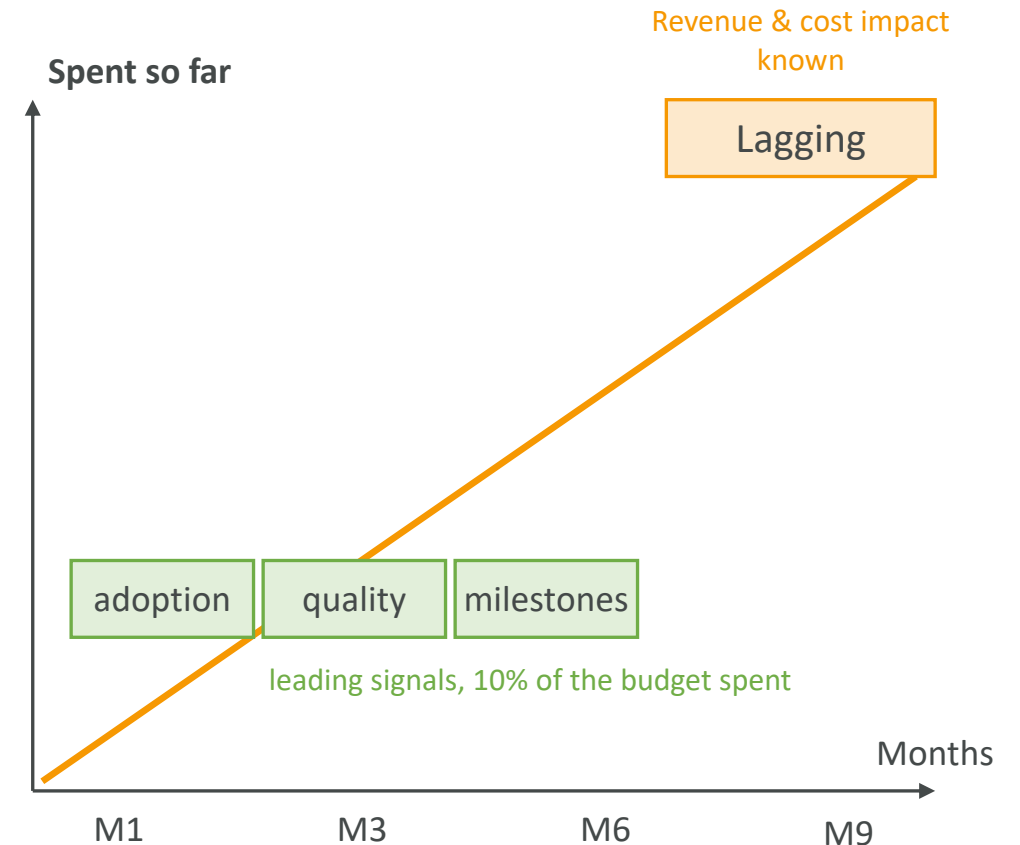
Benefits: What Makes The Bar Grow or Shrink

- Start from the whole change
- Remove what the AI did not cause: the new catalog's revenue
- Add the intangibles you can measure: agent time freed, customer satisfaction
 - **Customer Satisfaction Score (CSAT)**, 1 to 5, from the post-ticket survey. 3.9 to 4.2, worth \$50k in avoided repeat contacts
- Claim everything: overstated
- Cash only: understated
- Honest: in between



Leading Indicators

- **Lagging Indicator:** the cost or revenue impact, known months after launch
- **Leading Indicator:** an earlier signal that predicts it
- Three to read in months 1 to 3
 - **Adoption:** are people using it for real work?
CozyCo: 62% of tickets routed
 - **Quality Trend:** is accuracy holding as usage grows?
CozyCo: 85% in month 1, 86% in month 3
 - **Delivery Milestones:** are the dates holding?
CozyCo: Third team on time
- By month 9 the budget is spent either way.
- The leading signals arrive at 10% of it



Exam Tip

Activity measures (tickets processed, budget spent to date) are not leading indicators: a project can be on budget, busy, and failing

Month 3: The CFO Asks “Are we on track?”

- Do report the leading indicators: 62% adoption, 86% correct routing, milestones on time
- Do not guess the final number (“we will save about \$1M”): a guess is not a signal
- Do not answer with the spend (“we have used 33% of the budget”): on budget says nothing about on track



62% Adoption, 86% Correct, on time



“We will save about \$1M”



“We have used 33% of the budget”

Improvement and Competitive Advantage

- **Operational Improvement**
 - Better at what you already do
 - CozyCo's ticket routing: any rival can buy the same routing service next quarter
- **Competitive Advantage**
 - An improvement rivals cannot easily match
 - **Durable:** it rests on something hard to copy
- CozyCo's demand forecast is a competitive advantage
 - Proprietary Data: 6 years of CozyCo's own sales and returns, nobody can buy that dataset
 - A Data Flywheel: every season's actual sales feed the forecast, it improves, purchasing relies on it more, more data flows back
 - Depth of Integration: the forecast drives purchasing end to end, rivals must copy the whole process, not a tool

What Does NOT Make an Advantage Durable

- Being First: only if the lead compounds. CozyCo launched routing first; a rival with the same vendor caught up in a quarter
- Spending More: a bigger budget can be matched by a bigger budget
- Building Rather than Buying: a sourcing choice, not a source of advantage.
 - Buying a foundation-model service gets you started fast
 - Only building on your own data can make it a differentiator.
 - Use the fast path to learn, the owned path to stand apart
- A gain everyone can copy is still worth having: keep it

Business Model Transformation Through AI

- **Context:** leadership asks whether an AI initiative is “transformational”?
- **Improvement:**
 - the existing business runs better
 - Ex: faster routing, fewer returns
- **Transformation:**
 - a new value proposition
 - a new revenue stream
 - a new market position
- The reading test: follow where the value travels
 - It stays inside, the company does what it did, only better: improvement
 - It reaches new customers or earns money in a new way: transformation
- CozyCo example: selling its demand forecast to furniture makers as a paid service is a transformation. Routing tickets faster is not

Four Kinds of AI Initiatives

- Knowing the kind tells you what to expect from it. Only transformation sends value outside

Scaling	Optimization	Process improvement	Transformation
• The same capability applied to more of the business • CozyCo: extend ticket routing to the returns team	• A better decision inside an existing process • CozyCo: simulate the warehouse to plan delivery slots	• An internal process redesigned so it runs better • CozyCo: use the routing model to triage supplier emails	• New customers, a new revenue stream or a new market position • CozyCo: sell the demand forecast to furniture makers

Controlling Cost and Its Ownership

- GenAI is billed per token: every request costs money, for as long as it runs
- The right spend is the one the KPIs justify, not the lowest possible
 - Starve it (cap requests, cheapest model): you cap the value.
 - Let it run unchecked: you pay twice for the same answers
- **Experiments:** paid from a central AI budget, teams try things without fearing the bill
 - Chargeback from day one kills experiments.
- **Proven initiatives:** the cost moves to the team that benefits, billed on usage
 - A central pool forever kills accountability
 - CozyCo: \$0.02 per routed ticket, on the support team's cost center
 - Ownership moves with the cost: the team watches its spend against its own KPIs

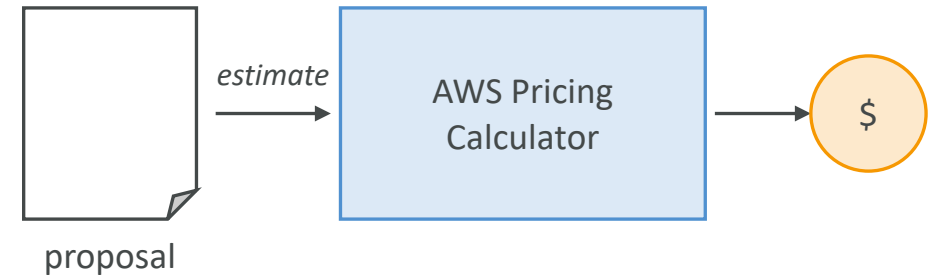
AWS Services in an AI Strategy

- Recognition only: what each service stands for when the strategy is discussed

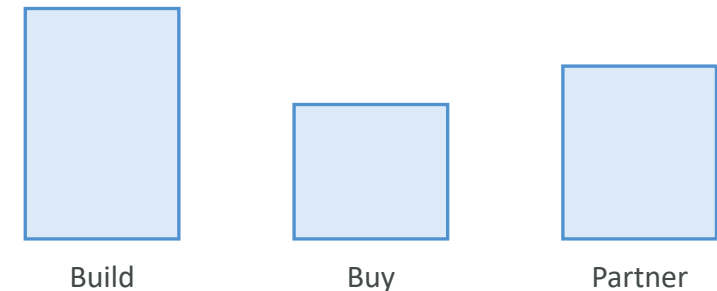
	What to remember	What it stands for
Amazon Bedrock	Managed foundation models: you start in days. Being early helps only for a while, rivals can buy it too	The buy path, and the fast path
Amazon SageMaker AI	Your own model on your own data: the choice when the model or the data is the durable differentiator	The build path, and the owned path
AWS Marketplace	Third-party tools, models and consulting partners. A differentiator acquired here can be acquired by rivals too	The buy and the partner paths
Amazon Quick	KPIs and leading indicators, tracked and shared in dashboards. It is not a cost-reporting tool	Tracking the result

AWS Pricing Calculator

- Free tool to estimate the AWS cost of a solution before anything is built
- Turns a proposal into an estimate, not a guess: the planning half of cost control
- Use it to compare the cost of build, buy and partner before committing
- It has no role in the model lifecycle: it does not train, monitor or run anything



Before committing, compare the three paths



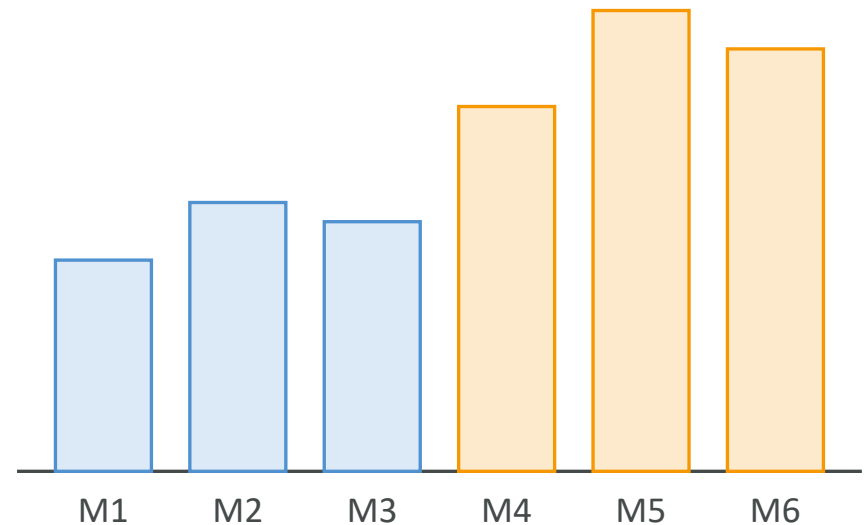
AWS Cost Explorer



- Shows the real AWS spend over time, filtered by service, account or tag
- Tag the resources of each initiative to see what it costs month by month
- The measuring half of cost control: it reports actual costs, it does not estimate

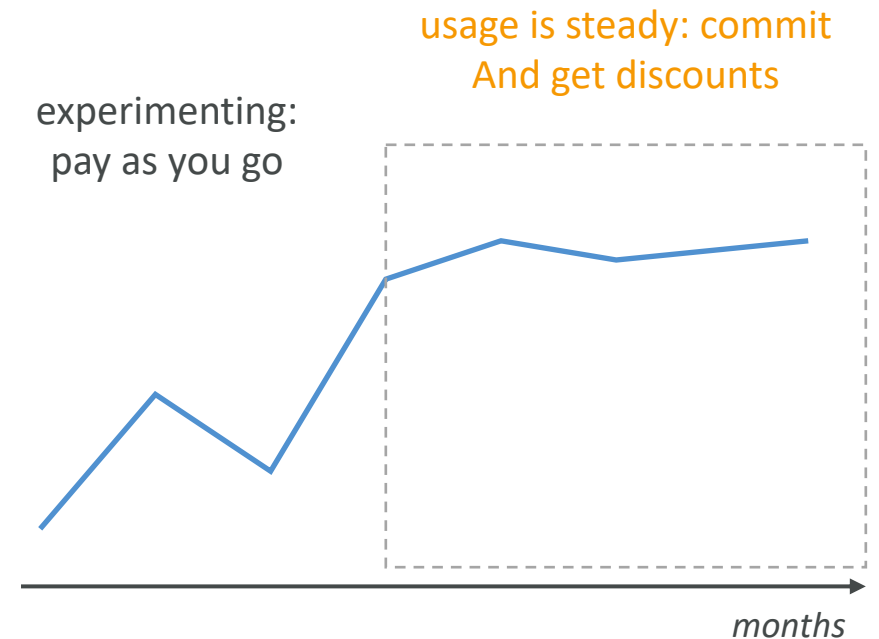
Real spend per month, tag = ticket routing

routing: +40% in M4



Savings Plans

- A commitment to a steady usage for 1 or 3 years, in exchange for a lower price
- Only once usage is steady: committing during experimentation locks in a guess
- Lowers the run cost of a proven initiative; it never decides whether it is worth running



Cost Control: Estimate Before, Measure During

- Before committing: an estimate, not a guess.
AWS Pricing Calculator turns the proposal into a number
- Once live: measure the real spend per initiative.
AWS Cost Explorer, with one tag per initiative
- Once usage is steady: commit to lower the price.
Savings Plans
- The business indicators live elsewhere:
Amazon Quick reads the KPIs, not the bill

Wrap-up: The Business Case, In Order

1. The outcome in one sentence, with no technology in it
2. Where the advantage lives, and therefore build, buy or partner
3. The condition under which you would not use AI
4. One tangible and one intangible KPI, with their baselines recorded before launch
5. The costs a first draft misses: the years of running it
6. The attribution question, answered before the CFO asks it
7. One leading indicator for the first month
8. Improvement or durable advantage, and what would have to be true

Wrap-up: CozyCo's Ticket-Routing Case

- The whole domain on one page, for ticket routing

Outcome	Resolve tickets in 1.5 days instead of 3.1; bounced tickets from 14% to 3%
Path	Buy: routing on a managed model, our advantage is not the model but the ticket history
Not AI if	The routing logic could be written as a table (it cannot: 40,000 free-text tickets)
KPIs and baselines	Resolution time 3.1 days (May export); CSAT 3.9 / 5 (May survey)
Real cost	\$310k build + \$130k a year to run: \$700k over 3 years
Attribution	Volume and staffing flat; seasonal dip in December; mechanism: 91% routed on arrival
Leading indicator	Adoption: share of tickets routed by the model, month 1 target 50%
Advantage?	An improvement, not a durable advantage: rivals can buy the same routing. Worth doing anyway

Domain 3: AI Governance and Responsible AI Leadership

Section Overview

- The question of this domain: who decides, who is accountable, who watches?
- Responsible AI is a leadership job: principles first, then structures, then the risks in production
- CozyCo in this domain: PayLater, a financing model to govern before it goes live

Task 3.1

Responsible AI In Decisions

dimensions, tradeoffs, governance by design, human oversight

Task 3.2

Governance and Compliance

governance body, risk classes, compliance, access control

Task 3.3

Enterprise AI Risks

monitoring, bias, harmful content, IP, reliability

CozyCo PayLater: The System This Section Governs

- Installment financing at checkout: “pay in 4” on any order
- A model reads the application and the order history, 9,000 applications a month
- 94% accurate in testing. It decides 70% of applications on its own, 30% go to a credit specialist
- It cannot give the reason for an individual decision
- Leadership wants it live next quarter
- For the generative topics (hallucination, guardrails, IP) we reuse the support assistant from Domain 2

What Responsible AI Is?

- **Responsible AI:** designing, building and running AI so that its benefits do not come at the expense of people, customers or the company
- Why it exists: an AI system decides at scale. One flaw in the design repeats on every decision, silently
- It covers the whole lifecycle: the idea, the data, the build, the launch, the years in production
- It applies to every AI system, not only generative AI: PayLater is a classic model and needs it most
- In practice: a set of questions asked while decisions are still cheap to change, and owners who answer them

The Eight Dimensions of Responsible AI (1/2)

Dimension	The question it asks	CozyCo
Fairness	Does it perform comparably across groups of people?	PayLater: same approval chance for equal repayment histories
Explainability	Can someone say why this particular output happened?	PayLater: why was this application declined?
Privacy and security	Is data obtained, used and disclosed as consented and as required?	PayLater trained on browsing history nobody consented to: a privacy failure
Safety	Can it produce, or be pushed into producing, something harmful?	The support assistant giving a customer dangerous assembly advice

The Eight Dimensions of Responsible AI (2/2)

Dimension	The question it asks	CozyCo
Transparency	Do people know AI is involved, and on what basis it decides?	Customers told that a model decides their financing
Veracity and robustness	Does it hold up on unusual, incomplete or adversarial input?	PayLater on an application with a foreign address format
Controllability	Can a person steer it, override it or stop it?	A specialist can overturn any PayLater decision; the CFO can switch it off
Governance	Does someone own it, with the authority to act?	PayLater has an owner and a council, not “the data team”

Two Pairs That Are Easy to Confuse

- Transparency is not explainability
 - **Transparency:** people know that AI is involved (a notice at PayLater checkout)
 - **Explainability:** someone can give the reason for this outcome (why this application was declined)
 - A system can be fully transparent and unable to explain a single decision
- Accuracy is not fairness
 - PayLater is 94% accurate overall
 - It still approves 61% of one group and 48% of another with equal repayment histories
 - Accuracy is measured on everyone at once; fairness is measured group by group

Why Tradeoffs Exist?

- You cannot maximize every dimension at once. Some pull against each other by nature
 - **Accuracy vs Explainability:** the models that predict best are usually the hardest to explain; a simple, explainable model gives up a few points of accuracy
 - **Speed vs Safety:** a human review, a grounding check or a filter adds time and cost to every decision
 - **Personalization vs Privacy:** the more personal data a model sees, the better it tailors, and the more it exposes
- A business objective usually sits on one side: launch faster, predict better, personalize more
- The tradeoff is not a sign of bad design. It is the normal situation, and it has to be decided, not discovered

How to Resolve a Tradeoff?

1. Name the two principles in tension, explicitly
2. Let the context decide which one is non-negotiable
 - A regulatory obligation, physical or financial safety, or a legal right makes a principle non-negotiable
 - The same tension resolves the other way in a low-stakes internal tool: nothing is non-negotiable there
3. Add something that keeps both, rather than abandoning one
 - An explanation layer, a review step, data minimization
- The decision is made by the accountable owners and written down.

Exam Tip

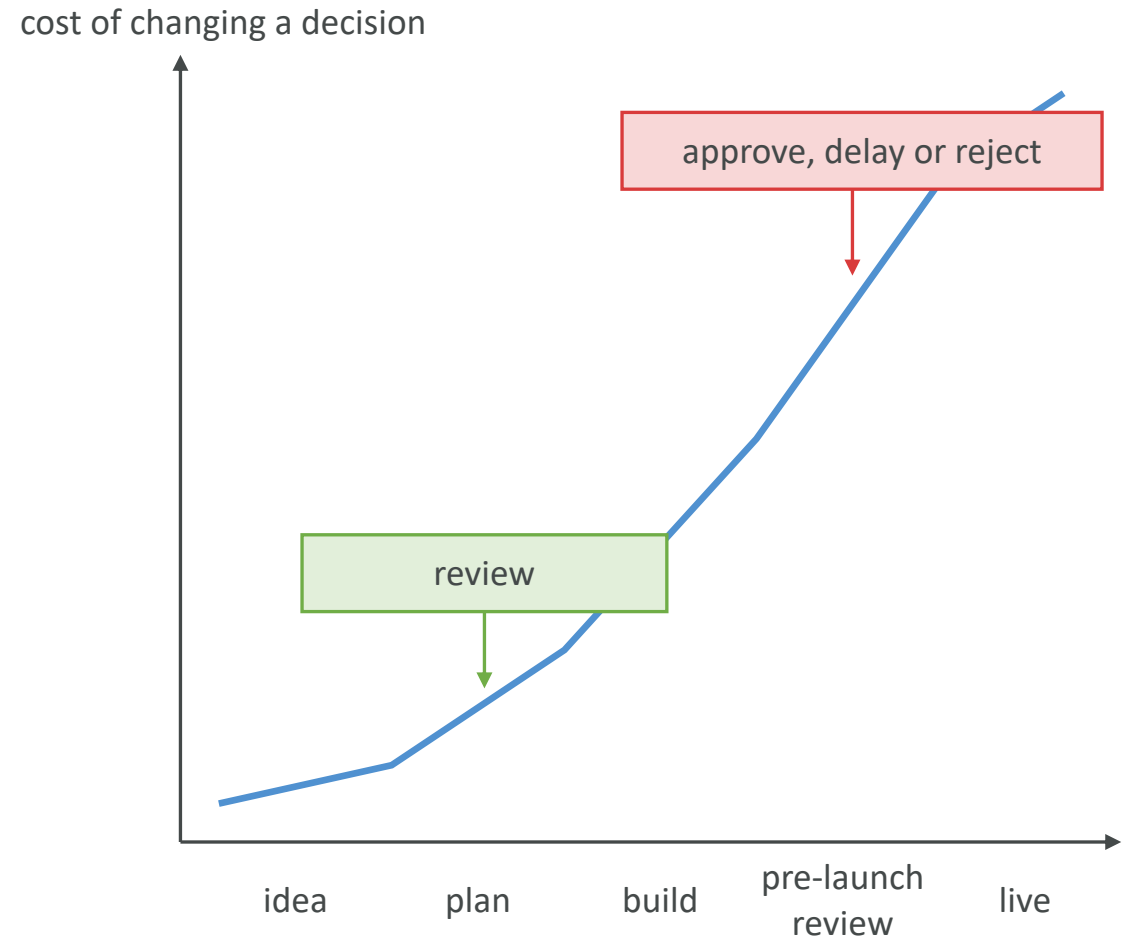
The exam never asks which principle matters most in general. Every question gives a context, and the context decides.

Applied to CozyCo PayLater

Step	CozyCo PayLater
1. The tension	Accuracy vs explainability: the 94% model cannot give the reason for a decline
2. What the context makes non-negotiable	Consumer lending rules require a reason for every automated decline. Explainability is not optional here
3. What keeps both	An explainable model, or an explanation layer on the current one, giving a reason per decision. Accuracy drops to 92%
The decision, written down	The council accepts 92% with reasons over 94% without. Owner: the CFO. Launch moves by six weeks

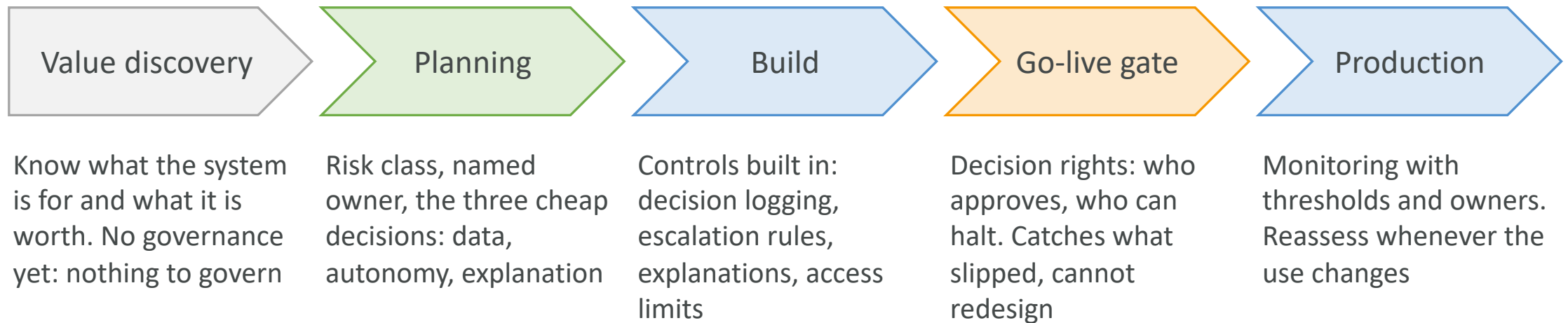
Governance Starts at Planning, When Changes Are Cheap

- Governance by design: the responsible AI questions are answered while the system is being planned, not checked at the end
- Three decisions are cheap to change at planning and expensive later
 - What data the system uses
 - What it is allowed to decide on its own
 - How an outcome will be explained
- A review at the end can only approve, delay or reject.
- A review during planning can still shape the system



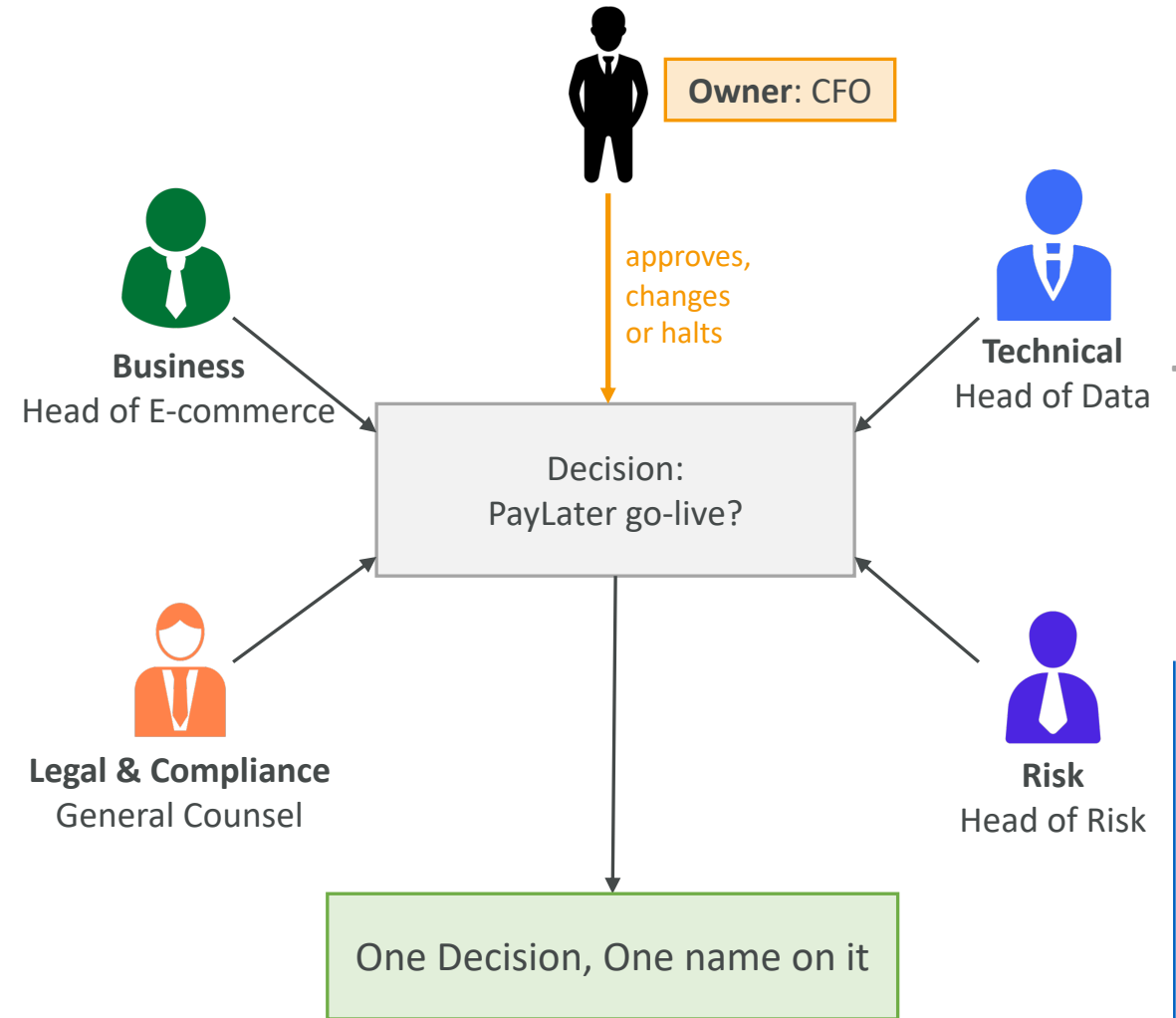
The Governance Touchpoints Along a Project

- Governance is a series of touchpoints, not one review.
- Each stage has its own job



A Governance Body: Four Functions, One Owner

- Four functions, each brings what the others cannot
 - **Business:** the outcome.
 - **Technical:** the limits
 - **Legal and Compliance:** the obligations.
 - **Risk:** the tolerance
- One owner per system, with the authority to approve or halt
- Same people without the owner: a committee that discusses
- With the **owner**: a body that decides
- Depth of review follows the risk class
- To model the structure: ISO/IEC 42001, the AWS CAF governance perspective

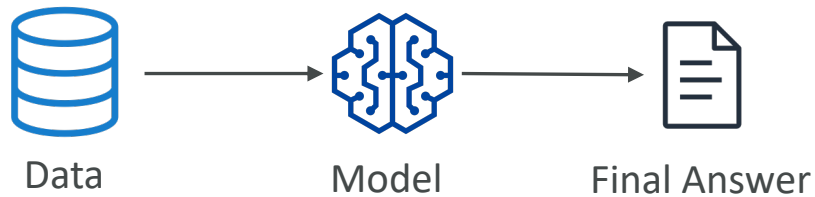


How Governance Structures Fail

- Symptoms you can read
 - Decisions stall with everyone in the room: nobody is accountable
 - Legal objects once the design is fixed: a function was missing from the start
 - CozyCo: PayLater was built for five months before counsel saw it, then rebuilt to implement the feature to log a reason per decline
- Three structures that look like governance and are not
 1. A center of excellence that alone approves or rejects: every team must ask it for permission and wait its turn, so teams start using AI tools without asking (shadow AI)
 2. Each business unit governs its own AI: three rulebooks, no enterprise owner
 3. Deploy first, govern later: that is remediation, and compliance cannot be retroactive

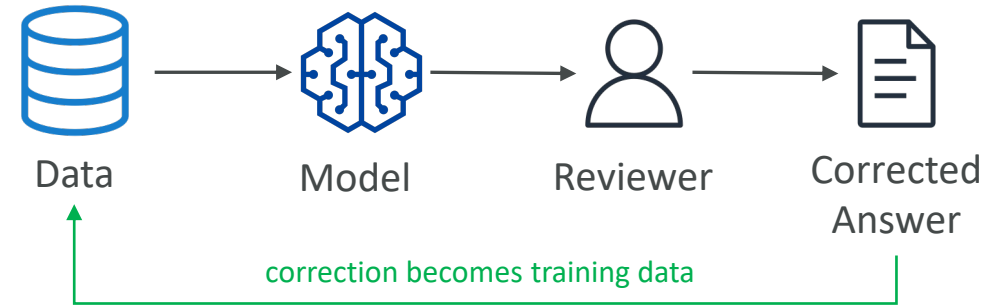
Human Oversight: What It Is and When It Is Required

AI Decides Alone



- The stakes are low
- The action is easy to undo
- The model is confident
- A person reviews the draft anyway
- No regulation applies

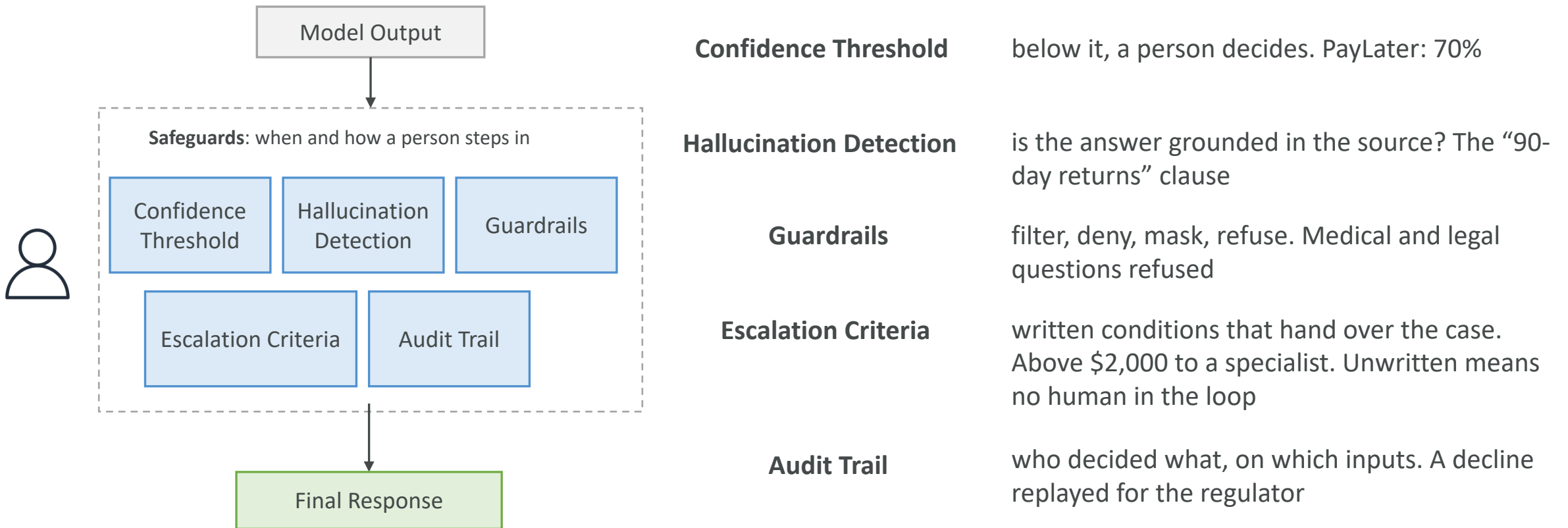
Human in The Loop



- The decision affects rights, health, safety, money or access to a service
- The outcome is consequential and hard to reverse
- The context is regulated
- The model's confidence is unreliable

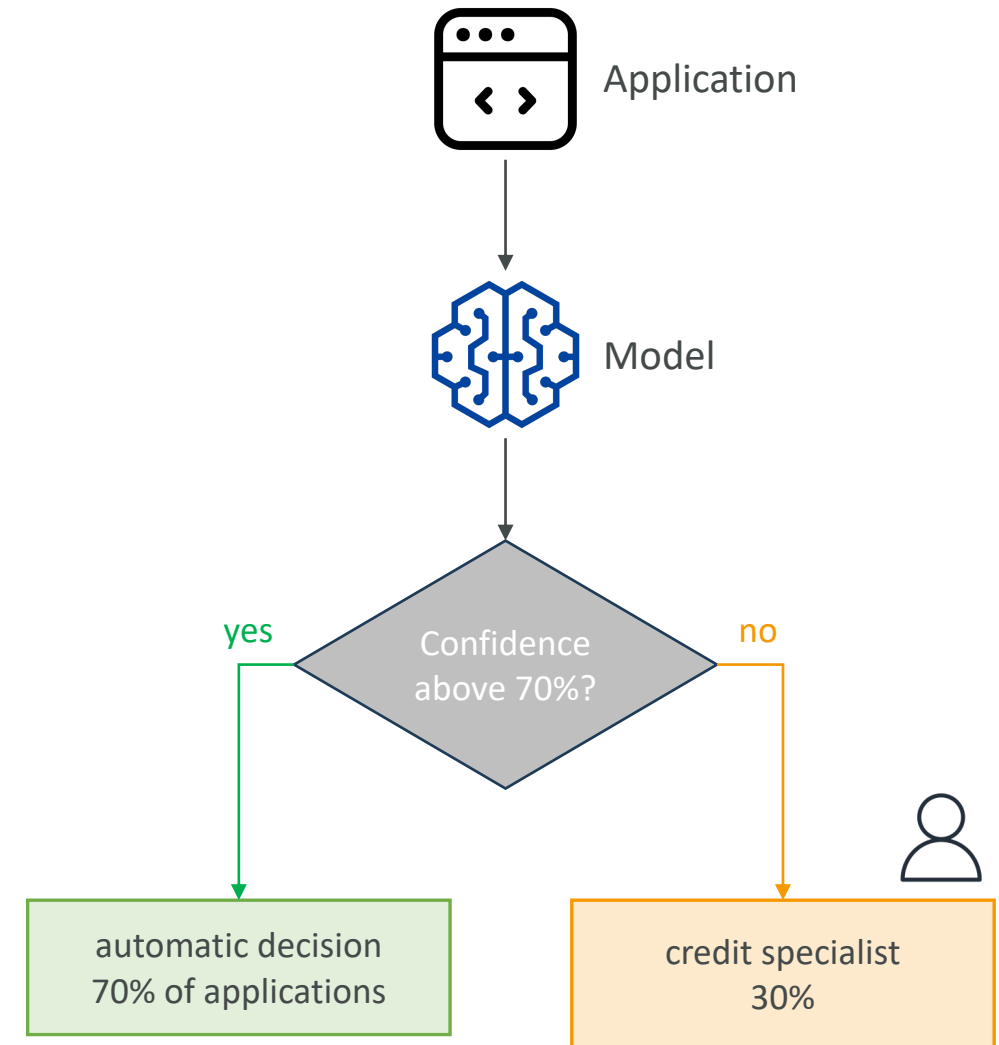
Five Safeguards Decide When a Person Steps In

- They sit between the model's output and the final answer



Confidence-based Routing: The Design That Keeps Both

- The pattern: the system decides when it is confident, a person decides the rest
- Errors are prevented where they are likely, and the efficiency is kept where they are not
- The alternatives each give one up: review everything (no efficiency), sample afterwards (no prevention), a disclaimer (neither)
- PayLater: 70% of applications decided in seconds, 30% reach a specialist the same day



The Risk Classification Matrix

- **Class** = severity of wrong outcome & likelihood
- **Severity**: consequence to people, reversibility, reach
- **Likelihood**: how often the system is wrong, and whether a person acts on the output or the system acts on its own (autonomy)
- Regulation sets a floor: a regulated use never sits below high
- The class sets the depth of oversight
- Reassess when the use changes: the assistant moves risk classification the day it faces customers, with no change to the model

		Severity of a Wrong Outcome			
		Minor	Moderate	Major	Extreme
Likelihood	Frequent	Medium ticket routing	High	Critical	Critical
	Likely	Low	Medium support assistant	High PayLater automatic	Critical
	Unlikely	Low	Low demand forecast	Medium PayLater referred	High
	Rare	Low assistant, internal pilot	Low	Medium	High

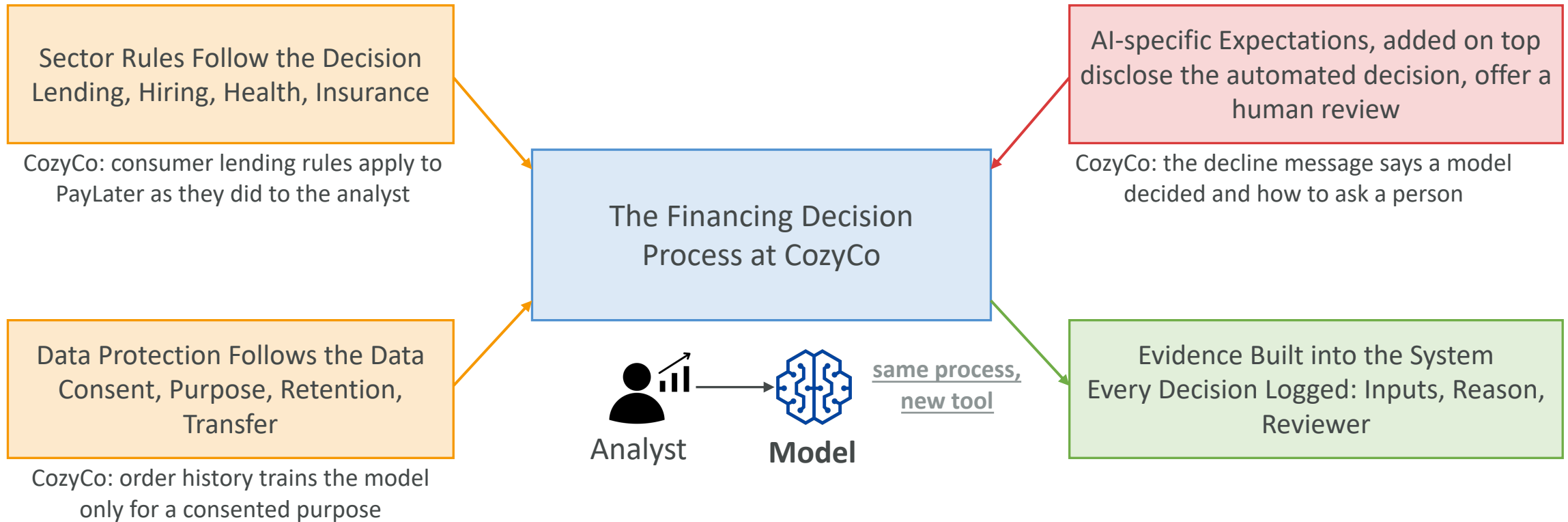
same model, now customer-facing

Risk Comes From The Use, Not From The Technology

- The same foundation model drafts marketing emails and decides financing: one model, two uses, two different risk classes
- The class does not change because the model changed. It changes when the use changes
- Reassess when the scope expands, when a pilot goes to production, when the purpose changes
 - CozyCo: reusing the PayLater model to pre-screen business customers is a new use, so a new assessment
- Triggers that are too late: a regulator's letter, an audit finding, a customer complaint

Obligations Attach to the Process, Not to the Technology

- Swap the analyst for a model: the obligations stay, a few are added

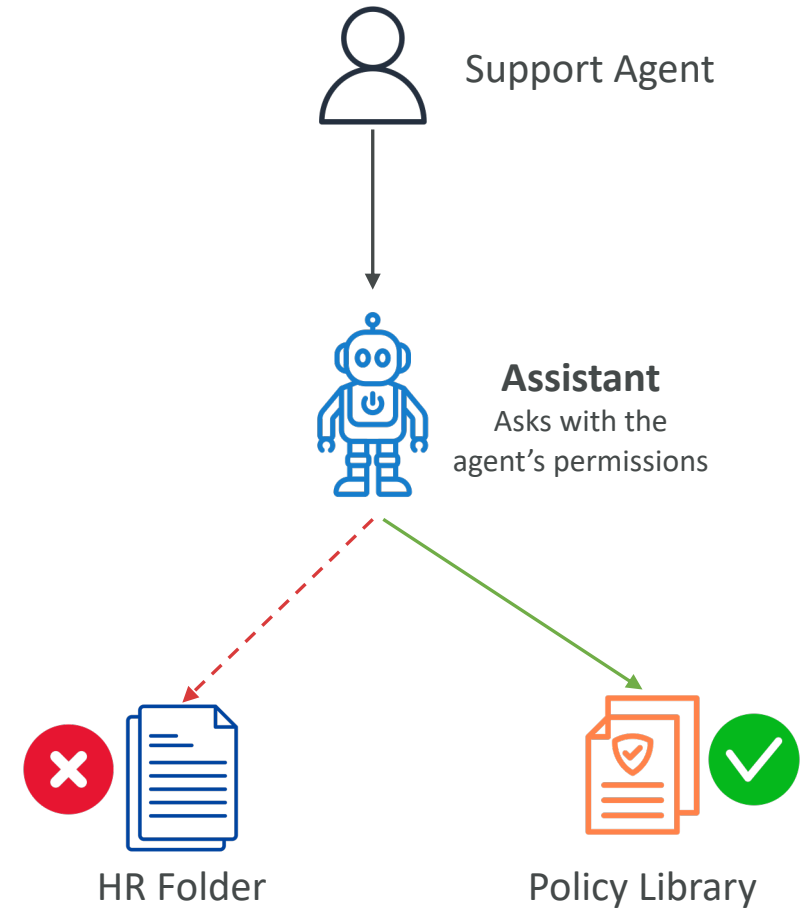


The Method: Identify, Map, Evidence

1. **Identify** which regulations apply to that process, not to “AI” in general
2. **Map** the controls that satisfy each obligation: a reason per decision, a review route, a retention rule
3. **Build** the evidence into the system: every PayLater decision logged with its inputs, its reason and who reviewed it
 - Evidence generated as you go is cheap; evidence reconstructed under audit is expensive and often impossible
 - Automation never removes an obligation, and no AI-specific rule replaces the sector rule that already applied

Least Privilege

- **Least Privilege:** each person and each system gets the minimum IT access for its task needs
- Otherwise it becomes a route around access control: an agent asking the assistant about salaries
- The system is a user too: an assistant that retrieves internal content must respect the asking person's own permissions

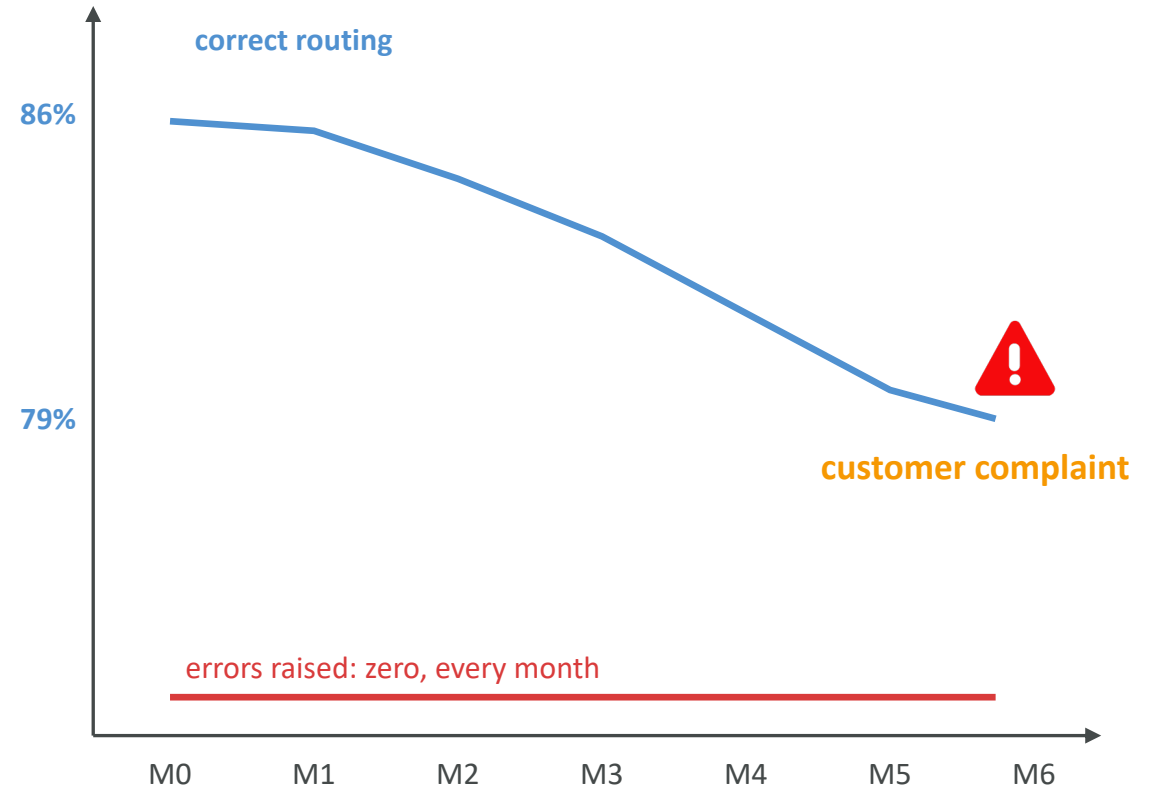


Three Families of Data Security Measures

- **Protect data at rest and in transit: encryption**, on by default in managed services
- **Mask or remove sensitive fields** where the task does not need them: the assistant never sees card numbers
 - Amazon Bedrock Guardrails can mask sensitive data in prompts and answers
- **Log access so that use can be audited:** who asked what, which documents were read, which decision was made
 - Logging is a governance control as much as a security one
 - No log, no proof for the auditor

Production Fail Silently

- CozyCo routing: 86% correct at launch, 79% by month 6.
- Nothing crashes, no error is raised, outputs stay confident and get less right
- The first signal was a customer complaint
- The risk profile does not settle at launch: data shifts, users improvise, the world moves
- A business case that funds only the build, an approval that names no monitoring owner: both unfinished



A Control Has Three Parts, and It Runs Continuously

- **A control:** something measured, a threshold that says what counts as a problem, a named owner who receives the alert and can act
 - PayLater: approval rate by group, a gap above 5 points for a month, alert to the credit risk lead
 - A dashboard nobody owns is not a control.
 - A policy with no measurement is not a control
- The threshold turns a trend into a decision: act on a number you've defined
- Continuous controls, not periodic

Four Signals To Watch, and The Remedy Each One Points To

Signal	What It Detects	Remedy When It Moves
Live Inputs vs The Training Baseline	The world changed: new customers, new products, new kinds of tickets	Retrain on recent data
Output Quality vs Known Answers	The system got worse on cases you can check	Investigate the cause: source, model or prompt
Behavior Across Groups	A group is treated differently	Fix the data or the target, tighten the threshold
Rate of Escalation to a Person	More cases the system cannot decide, or an upstream feed broke	Fix the upstream source, narrow the scope

Bias Enters at Four Stages (1/2)

- AI bias happens when an artificial intelligence system gives unfair, skewed, or discriminatory results because of flaws in its data or design

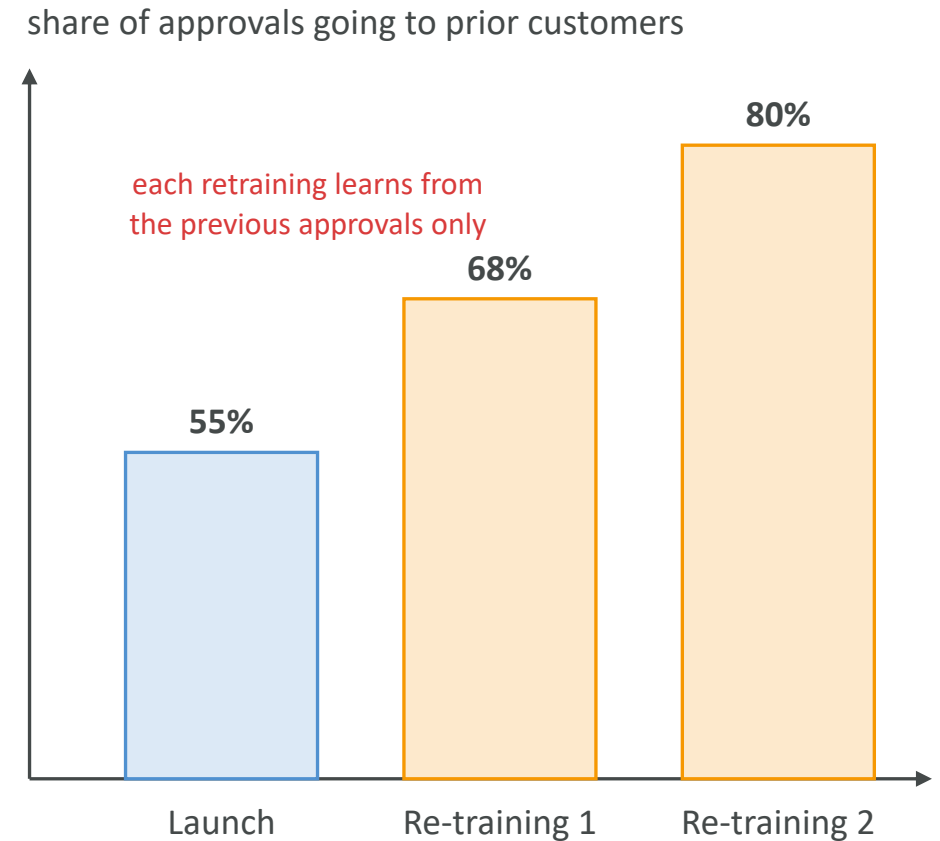
	1. Framing	2. Data
How bias gets in	The question itself carries the past: what the model is asked to predict	A group is rare in the history, or the process changed since the data was recorded
At CozyCo PayLater	Target chosen: “customers like the ones we approved before”. The old approval habits (biased) become the goal before a single row is loaded	80% of the order history is repeat customers, so new customers look risky. Last year’s returns policy changed, old repayments describe another process
The fix	Choose a target that measures the outcome you want: “paid in full within 4 installments”, for everyone	Represent every group. Refresh the data when the process changes

Bias Enters at Four Stages (2/2)

	3. Evaluation	4. Production Use
How bias gets in	One overall number hides a gap between groups	A fair model, and the people around it add bias back in how they use its output
At CozyCo PayLater	94% accurate overall. By group: 61% of one group approved against 48% of another, with equal repayment histories	Specialists review the referred 30% and overturn declines more often for customers who call back and argue
The fix	Measure by group, not only overall	Monitor how it is used (overturn rate by group, by specialist), not only what it outputs

Bias Drift, Seen Over Time

- Bias drift: the model's own decisions shape the data it learns from next, so a small tilt grows at every retraining
- Year 1: the model approves customers with a prior CozyCo order a little more often
- Only approved customers repay or default, so only they produce training data
- Year 2 retraining: the data is mostly prior customers, the model trusts them even more
- Nobody changed the model. The tilt grew because the model wrote its own training set
- The fix is not a retrain: watch the approval share by group, and learn about declined applicants too



Why Retraining Alone Does Not Fix Bias Drift

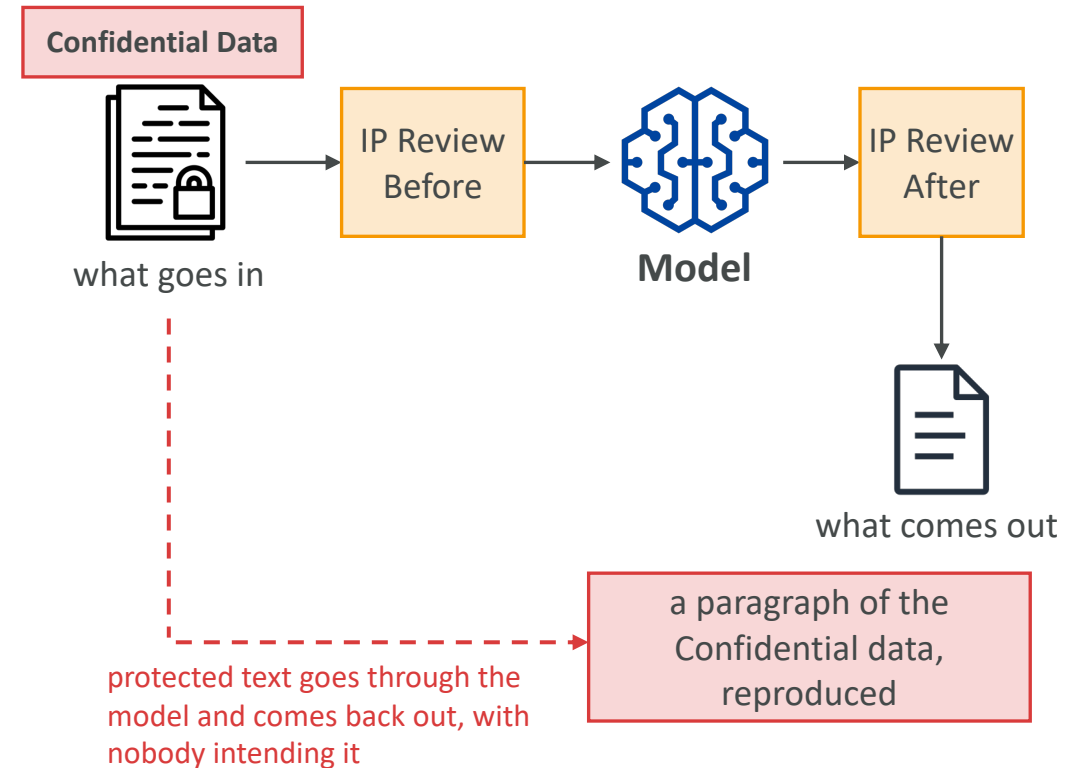
- Retraining on data the system skewed reproduces the skew, with fresher dates
- What works: ongoing fairness monitoring by group, a threshold, an owner
 - PayLater: approval rate by group every month, a gap above 5 points opens an investigation owned by the credit risk lead
 - Plus a way to learn about declined applicants: approve a small random sample and observe repayment

Harmful Content: Control The Boundary

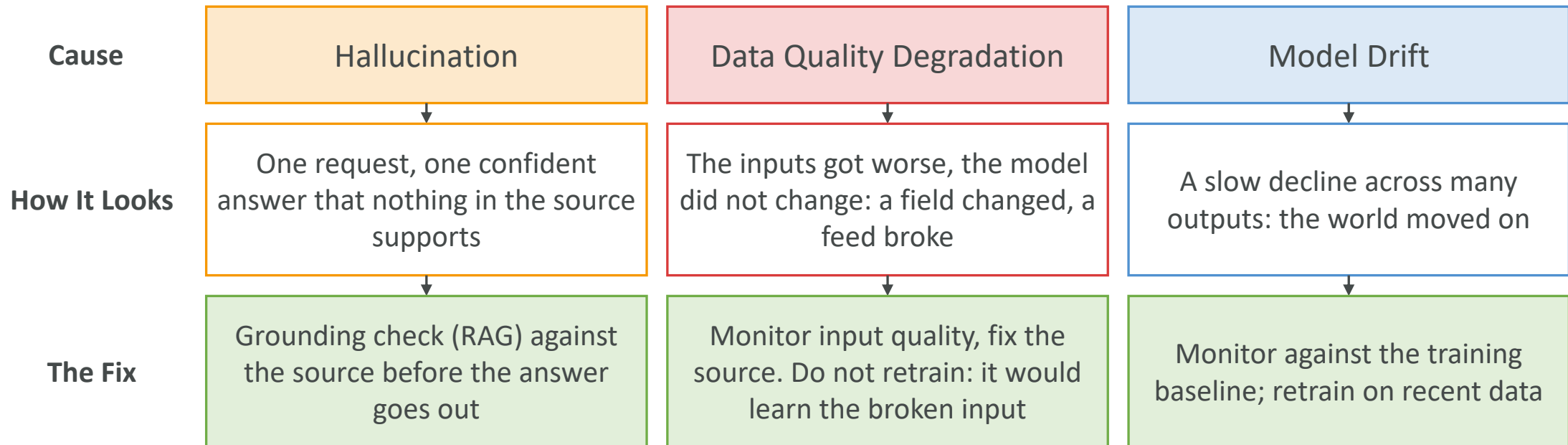
- Harmful content appears three ways: unprompted, through ordinary use, or deliberately elicited by a user
- Generative systems are more exposed because the output is open-ended
- The control sits at the boundary, on inputs and on outputs: filter categories, deny topics, mask data, refuse
 - The support assistant: denies medical and legal topics, refuses instructions that remove a safety part, masks card numbers
- Limiting the assistant to employees-only is not a control: employees can be harmed by a bad answer too, and anything it writes can be copied outside
- Output safety is your responsibility, not the model provider's

Intellectual Property

- IP runs both ways: what goes in, what comes out
- In: prompts, documents, code
 - Pasting a contract into a public tool: the leak is done, a summary does not undo it
- Out: may copy protected text from the model; rights of what is created are not automatically yours
- The response, sized by risk class
 - Rules on what goes into which tool
 - Tool terms reviewed before approval
 - A record of what each system was given
 - Legal review before output is published or sold

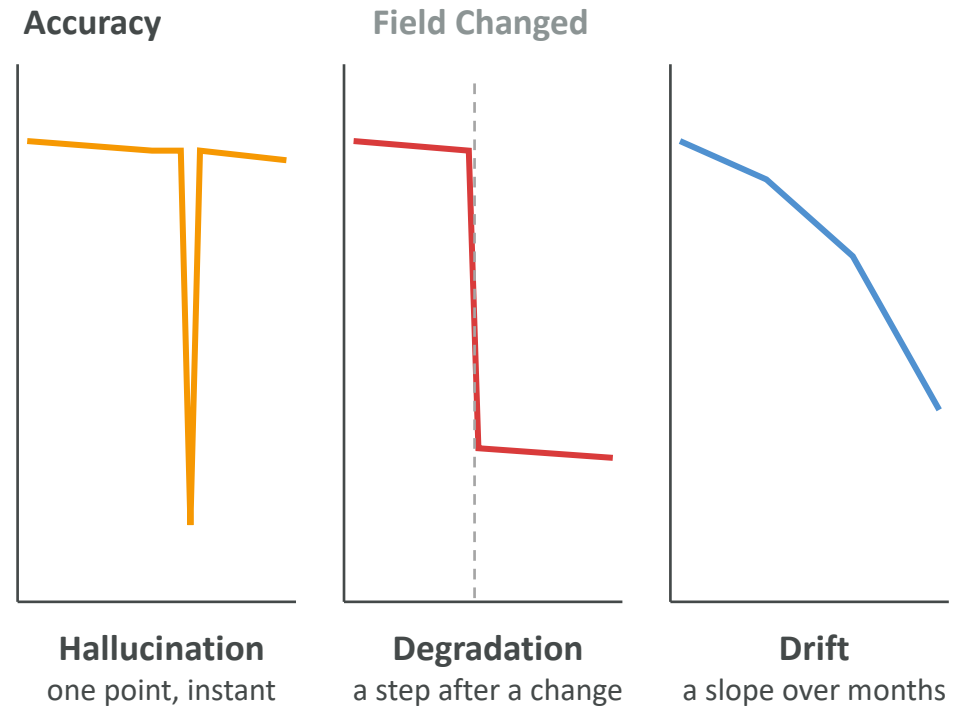


System Gives a Wrong Answer? Troubleshooting



Read The Shape Over Time

- One confidently wrong answer after months of accuracy: **Hallucination**
- A sudden drop right after an identifiable change upstream: **Degradation**
- A slow decline over months across many outputs: **Drift**
- A pattern by group: **Bias Drift**
- Retraining is the right remedy for drift only
- Retraining on degraded input teaches the degradation



AWS Well-Architected Framework: Responsible AI Lens



- A Lens: a set of questions added to a Well-Architected review for one topic
- Frames the responsible AI dimensions as design questions to answer before and after launch
- Tells you what to monitor for; it does not monitor anything itself
- Use it when the team asks “which questions should we ask at design time?”

Responsible AI
Lens



Your Design Review

Is it fair across groups?

Can we explain an output?

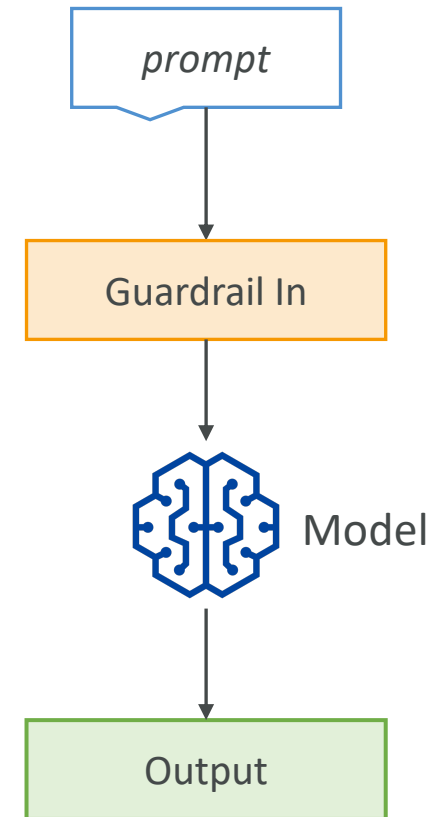
What do we monitor?

Who can stop it?



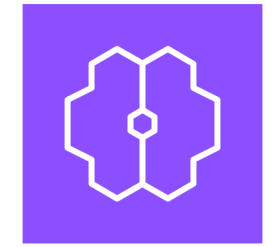
Amazon Bedrock Guardrails

- Safeguards applied to the prompts and the answers of a model on Amazon Bedrock
- Filter harmful categories, deny topics, mask sensitive data, block attempts to subvert the model
- Contextual grounding check (RAG)
 - flags an answer the source documents do not support
 - good against hallucinations

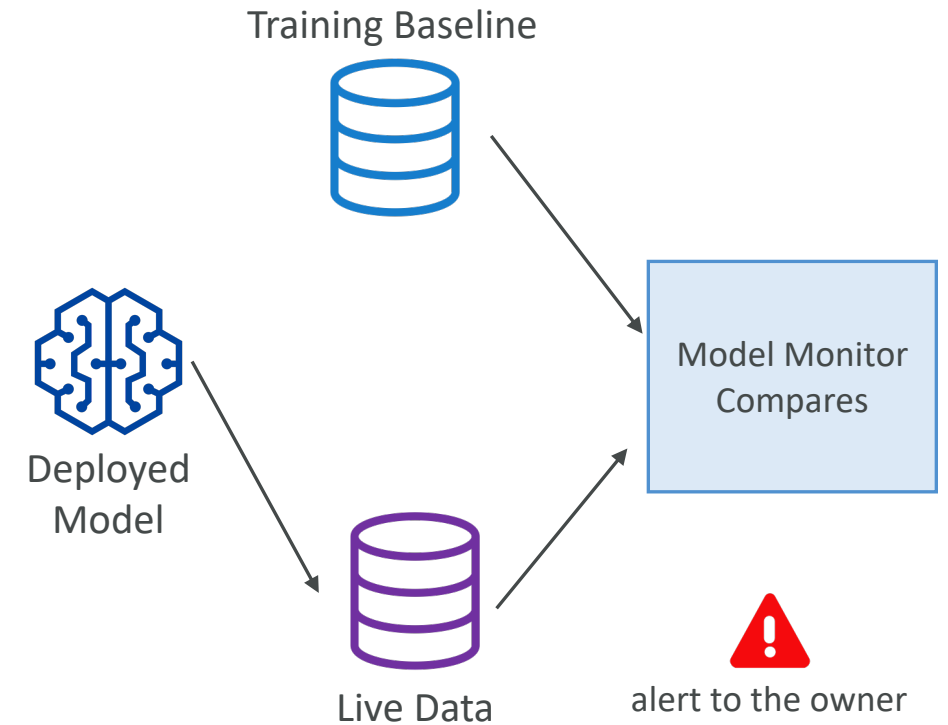


filter harmful categories, deny topics, mask sensitive data, block prompt attacks, flag answers not grounded in the source

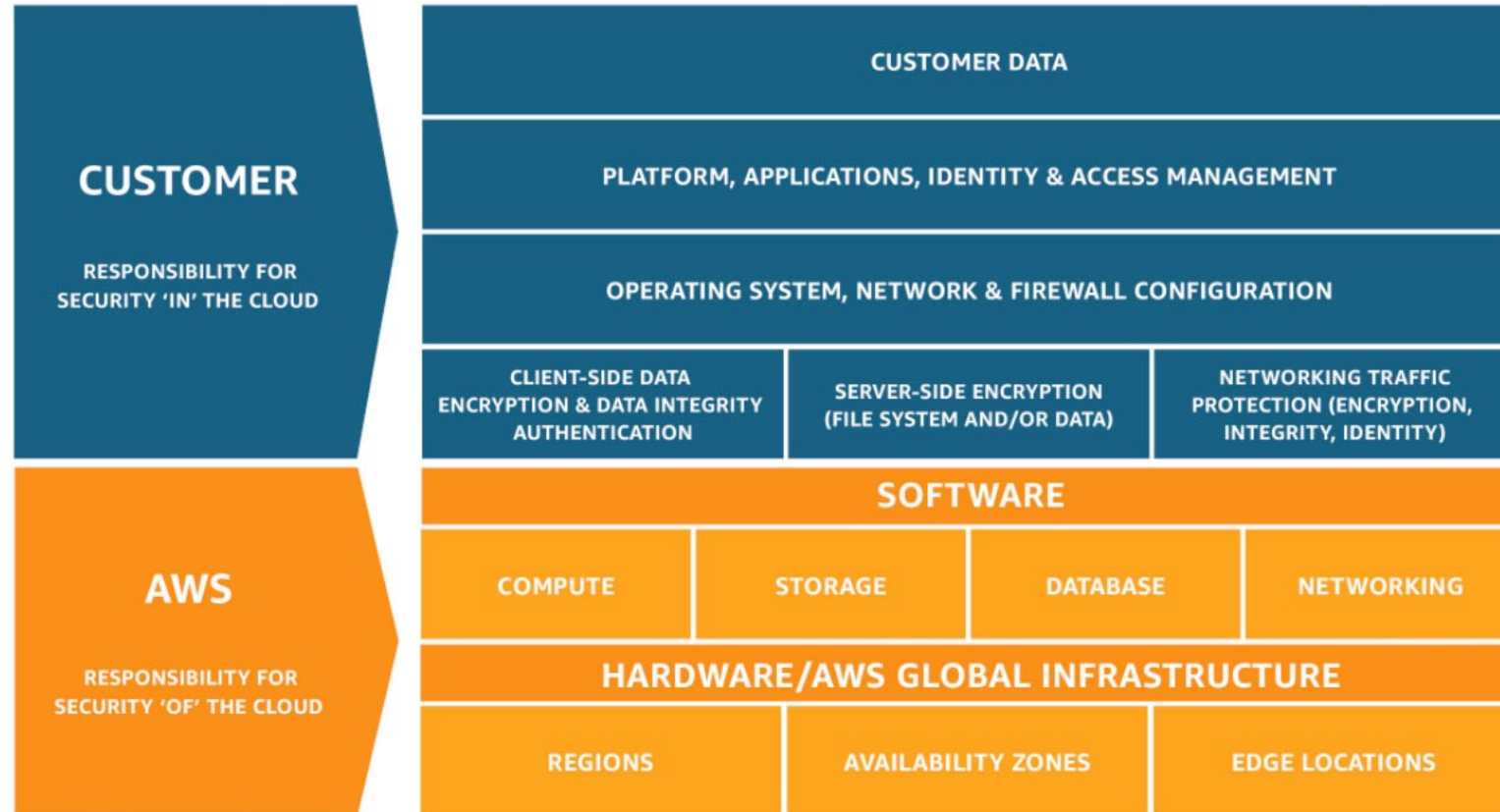
Amazon SageMaker Clarify and Model Monitor



- **Clarify:** reports on bias in data and in model outputs, and on which inputs drive a prediction (explainability)
- **Model Monitor:** watches a deployed model against its training baseline: data quality, drift, bias drift
- The evidence for the council and the regulator, and the alert for the named owner
- Use them to explain the model's decisions and to detect a gradual decline over months



Shared Responsibility Model Diagram



<https://aws.amazon.com/compliance/shared-responsibility-model/>

Shared Responsibility, Applied to CozyCo's AI Systems

- AWS: security of the cloud (data centers, hardware, hypervisor, the operation of managed services)
- You: security in the cloud (your data, its classification, who accesses it, how the service is configured)
- A managed service reduces configuration work, never responsibility
- CozyCo on a managed model service still owns what customer data goes into prompts

AWS: security **OF** the cloud
data centers, hardware, hypervisor,
operation of the managed services

You: security **IN** the cloud
your data and its classification,
who may access it, how the service is configured,
what you send to a model

Wrap-up: Governing An AI System, In Order

1. The responsible AI dimensions at stake, and where the gap is
2. The tradeoff, resolved by what the context makes non-negotiable
3. Decided at planning: the data it uses, what it decides alone, how it explains
4. A person where the stakes require one, with written escalation rules
5. One owner, one governance body, and a risk class that sets the depth of review
6. The obligations of the process, least privilege, and the evidence logged
7. In production: signals, thresholds, and an owner who receives the alerts

Wrap-up: PayLater, Governed End-to-End

- The whole domain on one page, for PayLater

Dimensions	Explainability is the gap: 94% accurate, no reason per decision. Fairness watched: 61% vs 48% approved
Tradeoff	Accuracy vs explainability. Lending rules decide: 92% with reasons over 94% without
Governance by Design	Decided at planning: order and payment history only, automatic decisions below \$2,000, a reason per decision
Human Oversight	A specialist decides below 70% confidence or above \$2,000; escalation rules written down
Structure and Class	Council of e-commerce, data, counsel, risk. Owner: the CFO. Highest class: money, regulated, hard to reverse
Compliance and Security	Every decision logged with inputs, reason and reviewer. The model reads order and payment data, nothing else
Monitoring	Approval rate by group (gap above 5 points), referral rate (above 40%), inputs vs baseline. Alerts to the credit risk lead

Domain 4: Business Readiness, Leadership, and AI Transformation

Section Overview

- The question of this domain: the pilot worked, can the company do it everywhere?
- CozyCo in this domain: the routing pilot goes from one team to the whole company

Task 4.1

Readiness and Maturity

maturity stages, readiness, capability gaps

Task 4.2

Data and Infrastructure

data readiness, silos, infrastructure

Task 4.3

Change and Skills

sponsors, champions, communication, culture, roles

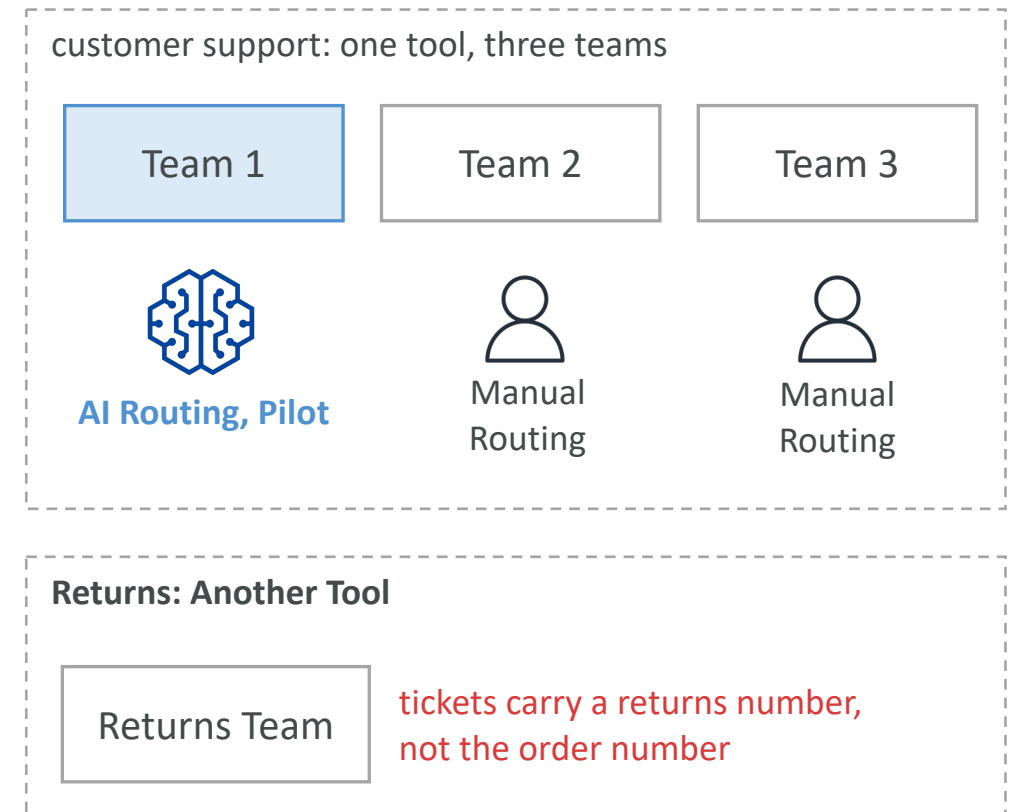
Task 4.4

From Pilot to Enterprise

phases, quick wins, CoE, metrics, rollout

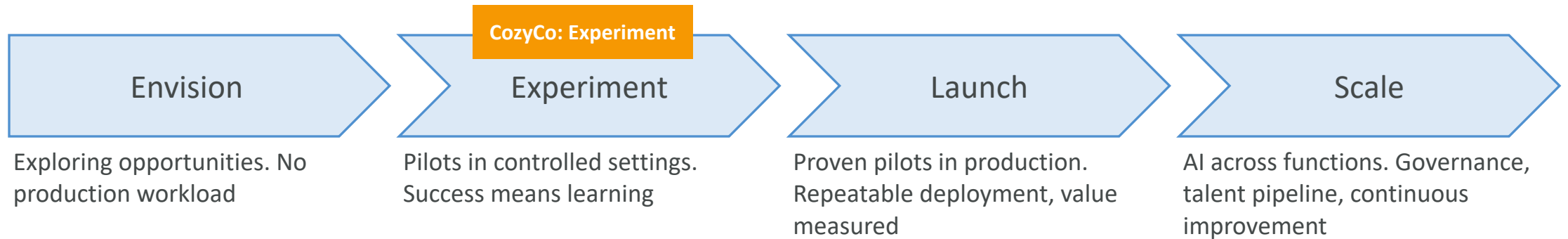
CozyCo in Domain 4

- Customer support has 3 teams, one ticketing tool.
- The routing model ran in team 1
- Result: 86% of tickets routed to the right team, resolution from 3.1 to 1.5 days
- Leadership wants the 3 support teams and the returns team by year end
- The routing model was built by two data scientists
- No handoff documentation, no deployment procedure
- The returns team uses a different tool (different order #)
- The head of support was not consulted
- Agents in teams 2 and 3 fear job cuts



Maturity & Readiness

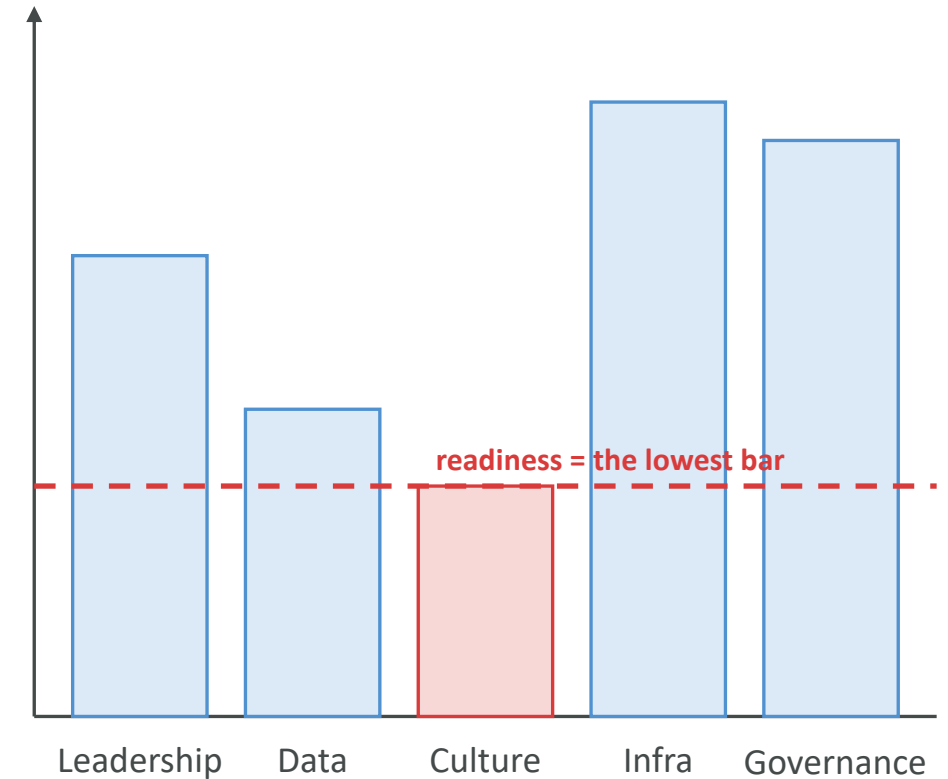
- **Maturity** is where you are today (not where you intend to be)



- **Readiness:** can you move to the next stage? Five dimensions:
 1. **Leadership alignment:** executives agree on what AI will do, and fund and defend it
 2. **Data quality:** accessible, accurate, governed. The dimension most often assumed, least often checked
 3. **Cultural preparedness:** the workforce accepts AI-assisted work. Culture decides adoption
 4. **Technical infrastructure:** it exists and can run the workload. Check it, no deep technical audit
 5. **Governance frameworks:** what is allowed, who is accountable, how risk is classified

Measuring Readiness

- Not an average, not a majority vote
- Strong on three, weak on two: blocked by the two
- One overall score (say 7 out of 10) hides which dimension blocks you
- The output of an assessment: the weakest dimension and what is missing in it
 - “No owner for the returns data” is better than “low readiness”



CozyCo Scored For Company-wide Routing

- **Leadership:** the CEO wants it, the head of support was not asked. **Gap**
- **Data:** ticket text since 2019 exists; returns tickets do not carry the order number. **Gap**
- **Culture:** teams 2 and 3 expect job cuts. **Gap**
- **Infrastructure:** a managed service that scales. **Fine**
- **Governance:** the AI council and an owner exist. **Fine**
- **Weakest:** culture and data. Company-wide routing is blocked there, whatever the CEO says

What To Invest In: The Four Capability Gaps

- The five dimensions of Readiness say whether you are ready
- The four gaps say what to invest in to become ready
- **People:** skills, headcount, champions
- **Process:** ownership, a repeatable deployment workflow, feedback loops
- **Technology:** data pipelines, compute, integration
- **Governance:** risk framework, policies, accountability

How To Prioritize The Investments

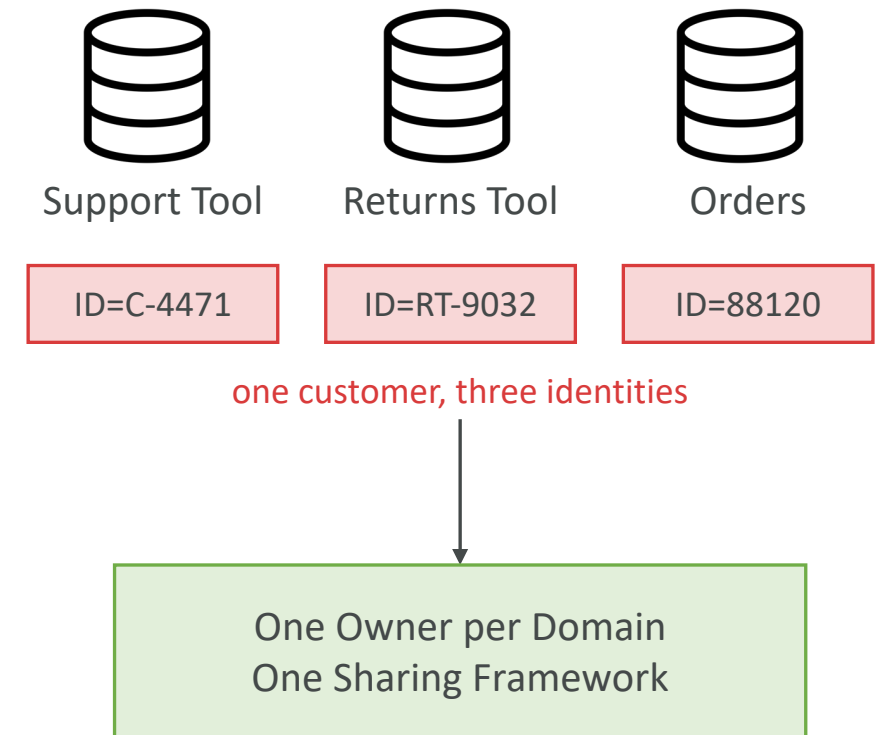
- **Sequence By Dependency:** fix the gap that blocks the others first
 - People before process: a workflow nobody can run exists only on paper
 - Data quality has a lot of impact downstream
- **Align With The Strategy:** an investment serves a business outcome, not gap-filling
- **Progress One Stage At a Time:** recommend the next realistic stage, not the end state
 - From Experimenting today to Launch tomorrow and finally Scale with ready
- **CozyCo:** hire or train the people first, then write the deployment procedure, then deploy to the second team

Is The Data Ready? Three Questions

- **Quality:** data is accurate, complete, consistent, timely?
 - A better model cannot fix bad data
 - Action: clean the data
- **Accessibility:** can the people and systems that need the data reach it?
 - Locked data is invisible to AI
 - Action: open the data with proper access control
- **Integration:** can the data sources combine?
 - Same identifiers, same definitions
 - Action: align the identifiers

How to Remove Data Silos

- **Data Silo:** one system's data, unreachable from the others
- What it costs: the same customer split in three, no cross-team use case, three versions of the truth
- The remedy is a data strategy, not a one-off integration
 - **Ownership:** one owner per data domain, accountable for quality, definitions and access
 - **A Sharing Framework:** purposes and protections agreed in advance, so a request is a decision, not a negotiation
- Workarounds that fail: escalate each request, copy the data, avoid the other teams' data



Technology and Infrastructure Foundations

- Four questions, not an architecture review
 - Can it handle this use case's workload?
 - Can the data reach its target, reliably and in time?
 - Can it be operated once live: monitoring, logging, intervention?
 - Can it scale if this works, without a rebuild?
- Some infrastructure questions are for the strategist and others for the tech team
- Cloud-ready: workloads run in the cloud
- AI-ready adds a layer on top
- A finished migration may be correct for existing workloads and incomplete for AI

Cloud-ready

compute, storage, network, identity

AI-ready

GPU compute, ML pipelines, experiment tracking,
separate experimentation and production,
model monitoring

Roles For a Successful Project

Executive Sponsor

own objectives linked to the success of an initiative



Budget that survives a reprioritization.
Authority to settle disputes between functions.
Their name on it for accountability and ownership

Aligned Leaders

for every function the initiative needs



Consulted and agreed before the start.
An unconsulted leader slows it by not prioritizing what it needs

Champions

many, inside the teams



Respected people who advocate to peers
Report back what is not working
Momentum comes from many champions, not from one sponsor

Champions are empowered when they have

1. time allocated to the role
2. information they are allowed to share
3. a route to raise problems.

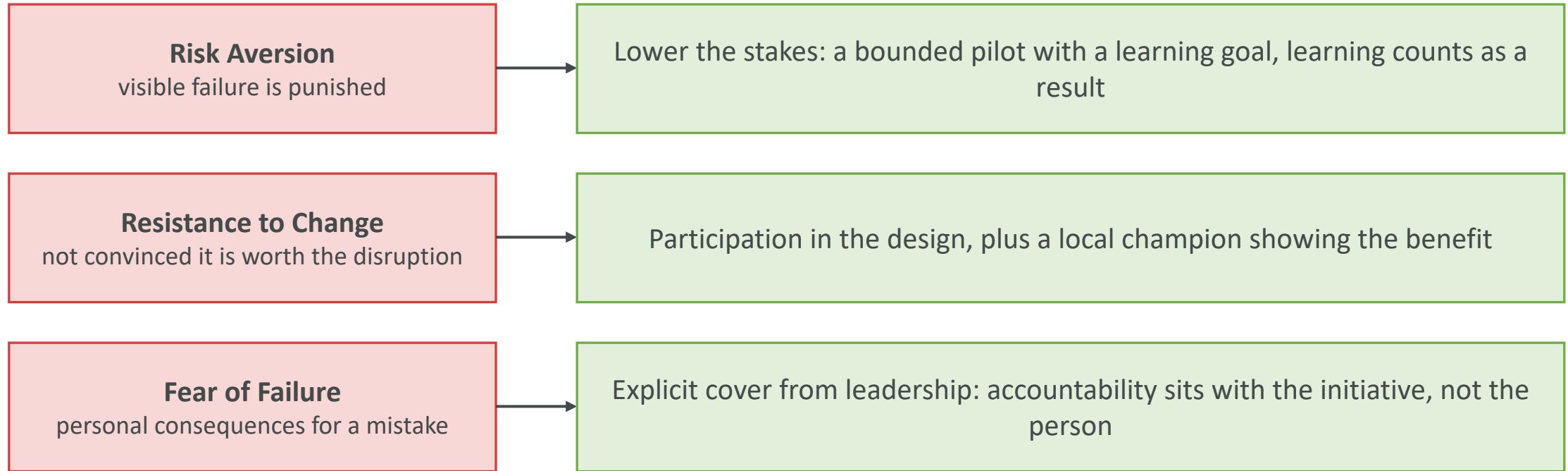
Requirements For The Building Team

- The team that builds **needs to be cross-functional: business, technical, legal and compliance** in it, from day one
- A function that joins late brings constraints and rework
- One owner per initiative, accountable
- Note: a project manager coordinates, they are not the legal review
- CozyCo: general counsel joins the routing rollout team in week one

What To Tell Employees, and When

- Ex: Leadership announces the benefit and stays vague about people.
- Better to provide details to avoid rumors
 - **Timelines:** a date for the next update. “Coming soon” is not a timeline
 - **Outcomes:**
 - overclaim => the next initiative is not believed
 - underclaim => the disruption feels unjustified
 - **Roles:** if unsettled, say so
- Ex: CozyCo, two sentences to the agents: “Routing moves to the model in September, team by team. No agent leaves; your time moves from routing to resolving”

Reactions That Can Block AI Adoption



CozyCo teams 2 and 3: fear of fewer jobs is resistance, not risk aversion. The move: agents help design the queue, one agent per team as champion

Four Ways To Build AI Skills In Your Company

- Match the approach to the goal. Do not train everyone the same way

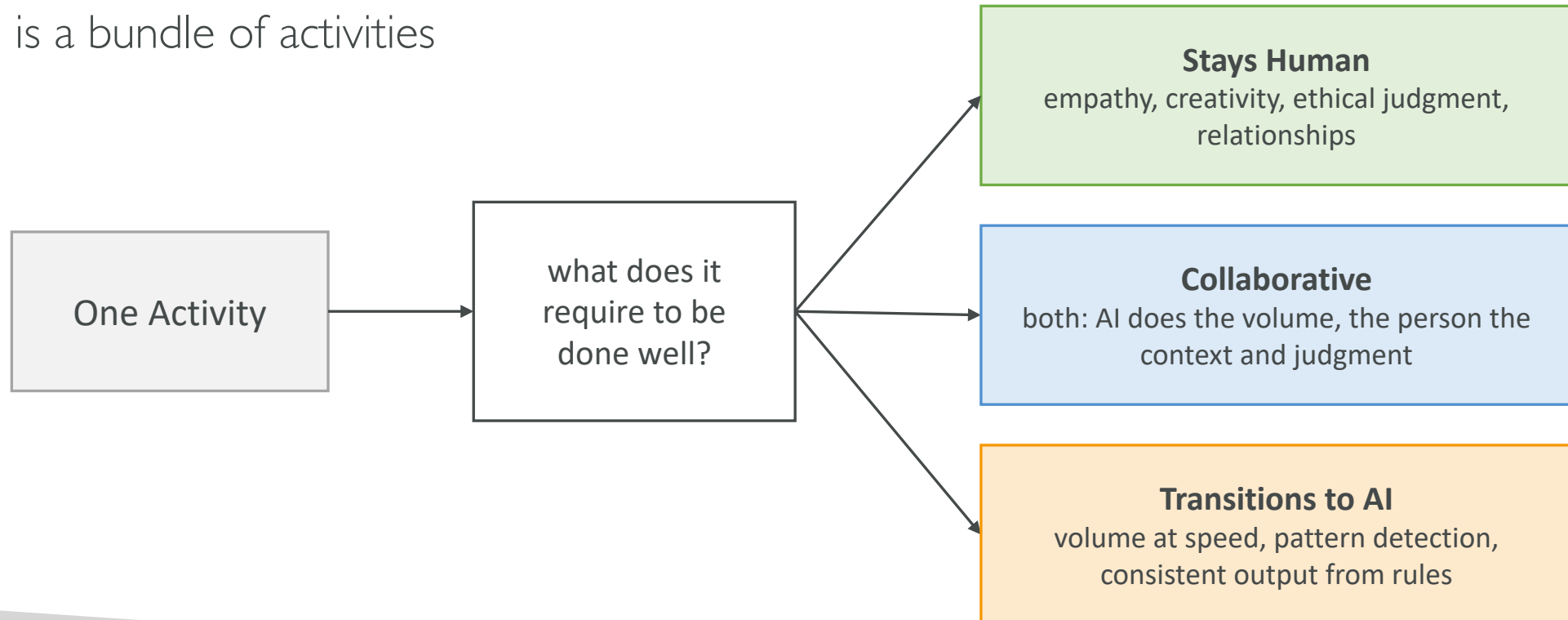
Proof of Concept	Hackathon	Training Program	Responsible AI Training
A bounded project that tests an idea on your own data and context	A short, intensive event, open to people who would never file a proposal	Structured instruction that builds a defined capability at scale	The principles, obligations and judgment calls from the governance domain
<u>Best when:</u> You need evidence before committing. The output is learning	<u>Best when:</u> You need ideas, energy and cross-team relationships, fast	<u>Best when:</u> A known skill gap affects many people. Make it role-specific	<u>Best when:</u> Anyone who builds, approves or relies on AI outputs

Two judgments on top: target literacy by role (a frontline user, an approver and a sponsor need different things), and pair every program with a place to apply it, because unused literacy decays

- Starting from zero, with fear in the workforce: a company-wide awareness program first.
- Hackathons need some awareness to build on; advanced training has no audience yet

Activity Redesign During A Role Transition

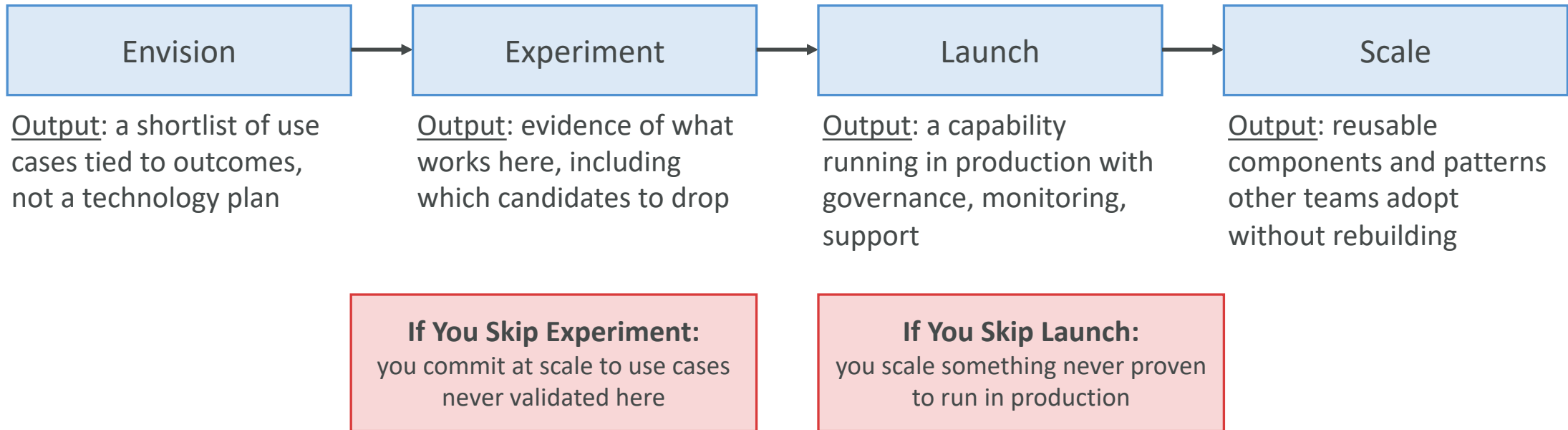
- “Will AI replace me?” Almost always no
- The real question: which parts of my work change?
- A role is a bundle of activities



A CozyCo Support Agent Example: Six Activities, Sorted By What They Require

Stays Human	Collaborative	Transitions to AI
Calm An Angry Customer <i>requires empathy</i>	Check Tickets Routed with Low Confidence <i>AI flags them, the support agent decides</i>	Read and Route Incoming Tickets <i>volume, consistent rules</i>
Decide a Goodwill Refund <i>requires judgment on one case</i>	Spot New Ticket Types <i>AI shows the pattern, the support agent names it</i>	Tag Tickets By Product <i>pattern detection at scale</i>

Transformation Is Iterative: Each Phase Produces What The Next One Needs

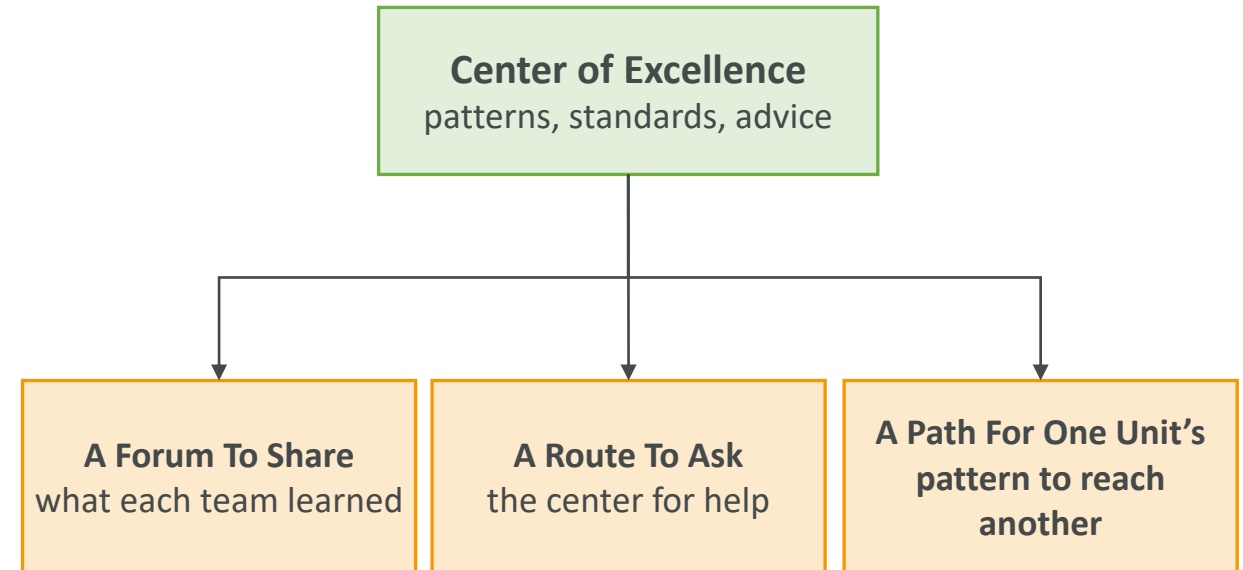


Short-Term Wins build Credibility

- Why a first initiative should be small: an early, visible, genuine win does three things
 - Funds the next stage politically
 - Gives champions something concrete to point at
 - Teaches the organization how it works with AI, when mistakes are recoverable
- The Risk: a win so isolated that nothing generalizes from it
- The Rule: small in scope, representative in kind
- The way it is built, governed and operated becomes the pattern others reuse
- It is not an argument for lighter governance, and not about lacking resources
- CozyCo: routing in team I was the win.
Its owner, monitoring and fallback desk are what every next wave reuses

The Center of Excellence (CoE)

- Group that concentrates AI expertise for the organization
- Reusable patterns and components
- Standards for building and operating AI
- Advice to teams starting out
- One view of everything running
- Fails if...
 - it becomes the only place AI can happen
 - request queue for months, teams route around it
 - ungoverned use of AI returns
- The fix is the operating model
 - enablement, not delivery
 - not more staff, not dropping standards, not dissolving it



Metrics That Keep A Scaling Program Honest

- Domain 2 measured one initiative. Here the question is the whole program

Leading Indicators: on track, before the outcome is visible

Model accuracy week over week
Data pipeline reliability
Adoption of the AI-assisted tools
Time to deploy a new use case

CozyCo: share of tickets routed by the model per week, per team

Lagging Indicators: delivered, once the outcome is visible

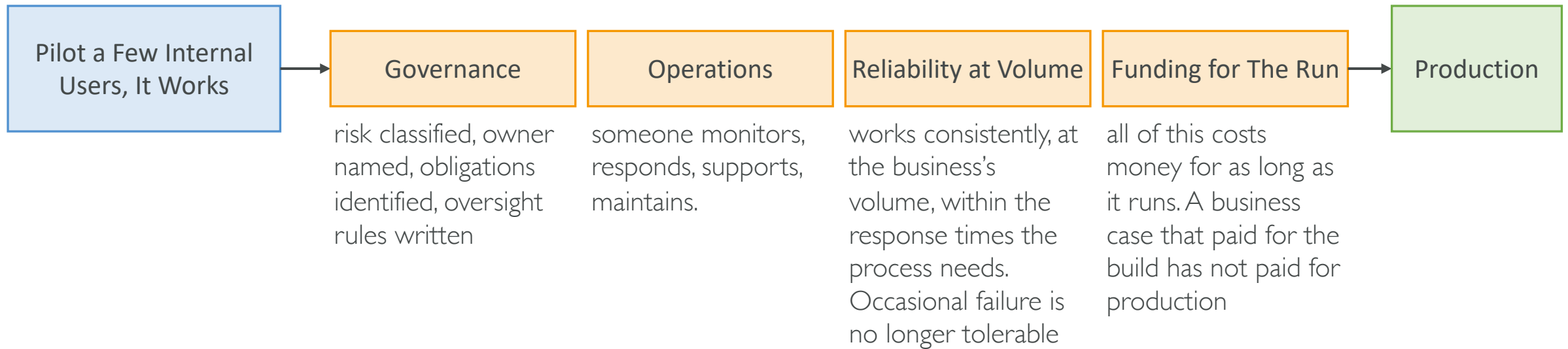
Revenue impact
Cost reduction
Customer satisfaction
Operational efficiency

CozyCo: resolution time, bounced tickets, cost per ticket

Vanity metrics that look good and connect to nothing: pilots started, people trained.

Continuous feedback from users is the qualitative channel: where the system helps, where it gets in the way.
Often the first signal that adoption is failing for a reason no metric shows

From Pilot to Enterprise Grade



- The common and expensive error: treating the move to production as a deployment step
- Production readiness: structured review against **reliability**, **security** and **operations** practices
- Observability for a GenAI system: **model performance**, **usage** and **response quality**

Factors To Consider For Scaling

- A pilot's failure hurts a pilot. A scaled system's failure stops the process it runs
- **Continuity:** fallback, rollback, parallel operation, and the real risk of failing after the old process was retired
- **Performance** at the new volume
- **Dependencies** in both directions
- **Capacity** of the people and support model
- **Cost at enterprise scale** including human oversight
- **Regional variation**
- **Regulatory constraints** per function
- **Technical performance** at enterprise scale

Roll Out In Stages, Keep A Way Back

Wave 1: Team 1

done: 86% routed correctly

Wave 2: Teams 2 and 3

manual desk in parallel, 4 weeks

Wave 3: Returns Team

once its tickets carry the order number

the way back at every wave: the manual routing desk, kept until the wave meets its criteria

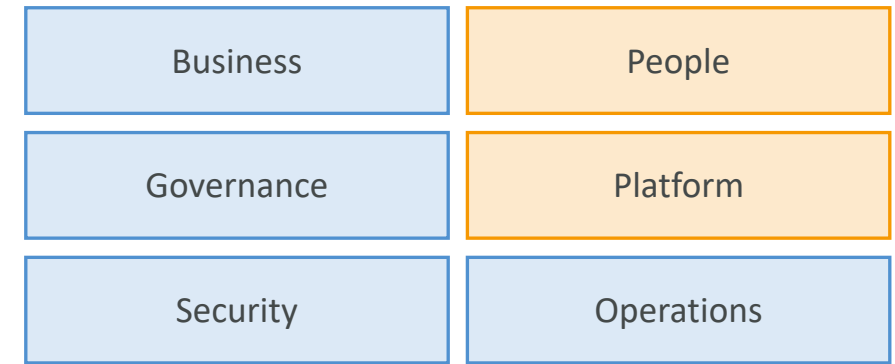
Rejected alternatives:

- switch everyone at once (maximum exposure)
- deploy and react to complaints (users become the detection)
- wait for a guarantee (never comes)

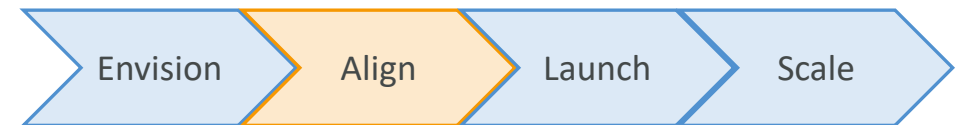
AWS Cloud Adoption Framework (CAF) For Readiness and Scaling



- The answer for an organization-wide readiness assessment: six perspectives at once
- **Platform perspective:** what a sound data and infrastructure foundation covers
- **People perspective:** guidance for leading the change
- Its transformation phases: envision, align (“*experiment*” wording at the exam), launch, scale



six perspectives: one readiness assessment



exam wording: Experiment

Three More AWS Resources, and Where Each Stops

- Recognize what each one gives you, and where it stops

	What it gives you	Where it stops
Well-Architected Framework Generative AI Lens	Best practices for GenAI workloads. Production readiness = a structured review of reliability, security and operations	Not an organization-wide readiness assessment: the tempting distractor against CAF
AWS Prescriptive Guidance	Resources from AWS technology experts to help accelerate cloud adoption and modernization	Not a readiness diagnostic
AWS Training and Certification	Role-based learning paths, digital training (Skill Builder), certifications. Builds the CoE's competency	Closes the skills gap only

AWS Building Blocks For The Data Foundation

- **AWS Lake Formation**: central permissions and catalog for a data lake. Who may read what
- **Amazon DataZone**: share data across teams with the terms attached. The sharing framework, as a service
- **AWS Glue**: pipelines that integrate and transform data between sources



AWS Lake Formation



Amazon DataZone



AWS Glue

Wrap-up: CozyCo Routing, Pilot To Company

- The whole domain on one page, for the routing rollout

Maturity	Experiment. Next stage: Launch in the three support teams
Readiness	Weakest on culture and data. Both are prerequisites, named to leadership
Capability Gaps	People (two builders, no handoff), then process (no deployment procedure). In that order
Foundation	The order number on every returns ticket, with an owner, before wave 3
People	Align the head of support, one champion per team, the agent role redesigned and communicated
Production Grade	Owner, monitoring with thresholds, on-call, run budget
Rollout and Metrics	Waves with the manual desk as fallback. Early metrics first, then resolution time and bounced tickets

Exam Preparation

State of learning checkpoint

- Let's look how far we've gone on our learning journey
- <https://aws.amazon.com/certification/certified-ai-business-strategist/>

How will the exam work?



- You'll have to register online at <https://www.aws.training/>
- Beta exam fee: 50 USD (standard exam price: 100 USD)
- Provide two identity documents (government ID plus a secondary ID, details are in the emails sent to you)
- Pearson testing center or online proctored exam
- No notes are allowed, no pen is allowed, no speaking
- 85 questions in 170 minutes, multiple choice and multiple response
- Multiple response: several correct answers, you must select all of them to get the point
- Results within 5 business days by email, no pass / fail on screen for a beta
- You will not know which answers were right / wrong
- To pass you need a score of at least 700 out of 1000
- A beta exam can be taken only once: no retake until the standard exam is available
- Pass before February 15, 2027 and you also receive the Early Adopter digital badge

Congratulations!

Congratulations!

- Congrats on finishing the course!
- I hope you will pass the exam without a hitch 😊
- If you haven't done so yet, I'd love a review from you!
- If you passed, I'll be more than happy to know I've helped
 - Post it in the Q&A to help & motivate other students. Share your tips!
 - Post it on LinkedIn and tag me!
- Overall, I hope you learned how to use AWS and that you will be a tremendously good AWS AI Business Strategist